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Transposition of nucleic acids into eukaryotic genomes with a transposase from heliothis

Abstract

The present invention provides polynucleotide vectors for high expression of heterologous genes. Some vectors further comprise novel transposons and transposases that further improve expression. Further disclosed are vectors that can be used in a gene transfer system for stably introducing nucleic acids into the DNA of a cell. The gene transfer systems can be used in methods, for example, gene expression, bioprocessing, gene therapy, insertional mutagenesis, or gene discovery.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. application Ser. No. 16/842,707 filed Apr. 7, 2020, which claims priority to U.S. Provisional Application No. 62/831,103 filed Apr. 8, 2019, U.S. Provisional Application No. 62/873,346 filed Jul. 12, 2019 and U.S. Provisional Application No. 62/978,862 filed Feb. 20, 2020, each incorporated by reference in its entirety for all purposes.

REFERENCE TO A SEQUENCE LISTING

(1) The application refers to sequences disclosed in a txt file named 557951SEQLST.TXT, of 3,651,164 bytes, created Jun. 2, 2021, incorporated by reference.

1. FIELD OF THE INVENTION

(2) The field of the present invention relates to configurations of DNA vectors for making stable modifications of the genomes of target cells, and the use of non-natural transposons and transposases.

2. BACKGROUND OF THE INVENTION

(3) The expression levels of genes encoded on a polynucleotide integrated into the genome of a cell depend on the configuration of sequence elements within the polynucleotide. The efficiency of integration and thus the number of copies of the polynucleotide that are integrated into each genome, and the genomic loci where integration occurs also influence the expression levels of genes encoded on the polynucleotide. The efficiency with which a polynucleotide may be integrated into the genome of a target cell can often be increased by placing the polynucleotide into a transposon.

(4) Transposons comprise two ends that are recognized by a transposase. The transposase acts on the transposon to remove it from one DNA molecule and integrate it into another. The DNA between the two transposon ends is transposed by the transposase along with the transposon ends. Heterologous DNA flanked by a pair of transposon ends, such that it is recognized and transposed by a transposase is referred to herein as a synthetic transposon. Introduction of a synthetic transposon and a corresponding transposase into the nucleus of a eukaryotic cell may result in transposition of the transposon into the genome of the cell. These outcomes are useful because they increase transformation efficiencies and because they can increase expression levels from integrated heterologous DNA. There is thus a need in the art for hyperactive transposases and transposons.

(5) Transposition by a piggy Bac-like transposase is perfectly reversible. The transposon is initially

integrated at an integration target sequence in a recipient DNA molecule, during which the target sequence becomes duplicated at each end of the transposon inverted terminal repeats (ITRs). Subsequent transposition removes the transposon and restores the recipient DNA to its former sequence, with the target sequence duplication and the transposon removed. However, this is not sufficient to remove a transposon from a genome into which it has been integrated, as it is highly likely that the transposon will be excised from the first integration target sequence but transposed into a second integration target sequence in the genome. Transposases that are deficient for the integration (or transposition) function, on the other hand, can excise the transposon from the first target sequence, but will be unable to integrate into a second target sequence. Integration-deficient transposases are thus useful for reversing the genomic integration of a transposon.

(6) One application for transposases is for the engineering of eukaryotic genomes. Such engineering may require the integration of more than one different polynucleotide into the genome. These integrations may be simultaneous or sequential. When transposition into a genome of a first transposon comprising a first heterologous polynucleotide by a first transposase is followed by transposition into the same genome of a second transposon comprising a second heterologous polynucleotide by a second transposase, it is advantageous that the second transposase not recognize and transpose the first transposon. This is because the location of a polynucleotide sequence within the genome influences the expressibility of genes encoded on said polynucleotide, so transposition of the first transposon to a different chromosomal location by the second transposase could change the expression properties of any genes encoded on the first heterologous polynucleotide. There is therefore a need for a set of transposons and their corresponding transposases in which the transposases within the set recognize and transpose only their corresponding transposons, but not any other transposons in the set.

(7) Since its discovery in 1983, the piggy Bac transposon and transposase from the looper moth *Trichoplusia ni* has been widely used for inserting heterologous DNA into the genomes of target cells from many different organisms. The piggyBac system is a particularly valuable transposase system because of: “its activity in a wide range of organisms, its ability to integrate multiple large transgenes with high efficiency, the ability to add domains to the transposase without loss of activity, and excision from the genome without leaving a footprint mutation” (Doherty et al., Hum. Gene Ther. 23, 311-320 (2012), at p. 312, LHC, ¶ 12).

(8) The value and versatility of the piggyBac system has inspired significant efforts to identify other active piggyBac-like transposons (commonly referred to as piggyBac-like elements, or PLEs) but these have been largely unsuccessful. “Since piggyBac is one of the most popular transposons used for transgenesis, searching for new active PLEs has attracted lots of attention. However, only a few active PLEs have been reported to date.” (Luo et al., BMC Molecular Biology 15, 28 (2014) world wide web biomedcentral.com/1471-2199/15/28. p. 4 of 12, RHC, ¶ 1 “Discussion”).

(9) Although there are large numbers of homologs of piggy Bac transposons and transposases in sequence databases, few active ones have been identified because the vast majority are inactivated by their hosts to avoid activity deleterious to the hosts as illustrated by the following excerpts: “Related piggy Bac transposable elements have been found in plants, fungi and animals, including humans [125], although they are probably inactive due to mutation.” (Munoz-Lopez & Garcia-Perez, Current Genomics 11, 115-128 (2010) at p. 120, RHC, ¶ 1). “It is believed that transposons invade a genome and subsequently spread throughout it during evolution. The “selfish” mobility of transposons is harmful to the host; hence, they are eliminated or inactivated by the host through natural selection. Even harmless transposons lose the activity eventually because of the absence of conservative selection for them. Thus, in general, transposons have a short life span in a host and they subsequently become fossils in the genome.” (Hikosaka et al., Mol. Biol. Evol. 24, 2648-3656 (2007) at p. 2648, LHC, ¶ 1 “Introduction”). “Frequent movement of transposable elements in a genome is harmful (Belancio et al., 2008; Deininger & Batzer, 1999; Le Rouzic & Capy, 2006; Oliver & Greene, 2009). As a result, most transposable elements are inactivated shortly after they

invade a new host.” (Luo et al., Insect Science 18, 652-662 (2011) at p. 660, LHC, ¶ 1).

(10) Three classes of piggyBac-like elements have been found: (1) those that are very similar to the original piggy Bac from the looper moth (typically >95% identical at the nucleotide level), (2) those that are moderately related (typically 30-50% identical at the amino acid level), and (3) those that are very distantly related (Wu et al., Insect Science 15, 521-528 (2008) at p. 521, RHC. ¶ 2).

(11) PiggyBac-like transposases highly related to the looper moth transposase have been described by several groups. They are extremely highly conserved. Very similar transposase sequences to the original piggyBac (95-98% nucleotide identity) have been reported in three different strains of the fruit fly *Bactrocera dorsalis* (Handler & McCombs, Insect Molecular Biology 9, 605-612, (2000)). Comparably conserved piggyBac sequences have been found in other *Bactrocera* species (Bonizzoni et al., Insect Molecular Biology 16, 645-650 (2007)). Two species of noctuid moth (*Helicoverpa zea* and *Helicoverpa armigera*) and other strains of the looper moth *Trichoplusia ni* had genomic copies of the piggy Bac transposase with 93-100% nucleotide identity to the original piggyBac sequence (Zimowska & Handler, Insect Biochemistry and Molecular Biology, 36, 421-428 (2006)). Zimowska & Handler also found multiple copies of much more significantly mutated (and truncated) versions of the piggy Bac transposase in both *Helicoverpa* species, as well as a homolog in the armyworm *Spodoptera frugiperda*. None of these groups attempted to measure any activity for these transposases. Wu et. al (2008), supra, reported isolating a transposase from *Macdunnoughia crassisigna* with 99.5% sequence identity with the looper moth piggy Bac. They also demonstrated that this transposon and transposase are active, by showing that they could measure both excision and transposition. Their Discussion summarized previous results as follows: “Other reportedly closely related IFP2 class sequences were in various *Bactrocera* species, *T. ni* genome, *Helicoverpa armigera*, and *H. zea* (Handler & McCombs, 2000; Zimowska & Handler, 2006; Bonizzoni et al., 2007). These sequences were partial fragments of piggy Bac-like elements, and most of them were truncated or inactivated by accumulating random mutations.” (Wu et. al., Insect Science 15, 521-528 (2008) at p. 526, LHC, ¶ 3.)

(12) It has proved very difficult to identify active piggy Bac-like transposases that are moderately related to the looper moth enzyme simply by looking at sequence. The presence of features that are known to be necessary: a full-length open reading frame, catalytic aspartate residues and intact ITRs, has not proven to be predictive of activity. “A large diversity of PLEs in eukaryotes has been documented in a computational analysis of genomic sequence data [citations omitted]. However, few elements were isolated with an intact structure consistent with function, and only the original IFP2 piggyBac has been developed into a vector for routine transgenesis.” (Wu et al., Genetica 139, 149-154 (2011), at p. 152, RHC, ¶ 2). Wu et al.'s group from Nanjing University (the “Nanjing group”) published several papers over a 6-year period, each identifying moderately related piggyBac homologs. Although the Nanjing group showed in 2008 that they could measure both excision and transposition of the *Macdunnoughia crassisigna* transposon by its corresponding transposase, and in each subsequent paper they express the desire to identify novel active piggy Bac-like transposases, they only show excision activity and that only for one transposase from *Aphis gossypii*. They conclude that the usefulness of this transposase “remains to be explored with further experiments” (Luo et. al. 2011, p. 660, LHC ¶ 2 “Discussion”). However, none of the other papers published by the Nanjing group in which piggy Bac-like sequences were identified from a variety of other insects, show that any activity was found. Three papers identifying other putative active piggyBac-like transposases were published by a group at Kansas State University. None of these papers reports any activity data. Wang et al., Insect Molecular Biology 15, 435-443 (2006) found multiple copies of piggyBac-like sequences in the genome of the tobacco budworm *Heliothis virescens*. Many of these had obvious mutations or deletions that led the authors not to consider them to be candidate active transposases. Wang et. al., Insect Biochemistry and Molecular Biology 38, 490-498 (2008) reported more than 30 piggyBac-like sequences in the genome of the red flour beetle *Tribolium castaneum*. They concluded “All the TcPLEs identified here, except TcPLE1, were

apparently defective due to the presence of multiple stop codons and/or indels in the putative transposase encoding regions.” Even for TcPLE1 there was “no evidence supporting recent or current mobilization events” (p. 492, section 3.1, ¶ 2&3). Wang et al. (2010) used PCR to identify piggyBac-like sequences from the pink bollworm *Pectinophora gossypiella*. Again, they found many obviously defective copies, as well as one transposase with characteristics the authors believe to be consistent with activity (page 179, RHC, ¶ 2). But no follow up report indicating transposase activity can be found. Other groups have also attempted to identify active piggyBac-like transposases. These reports conclude with statements that the piggy Bac-like elements identified are undergoing testing for activity, but there are no subsequent reports of success. For example, Sarkar et. al. (2003) conclude their Discussion by re-stating the value of novel active piggyBac-like transposons, and describing their ongoing efforts to identify one: “The mobility of the original *T. ni* piggyBac element in various insects suggests that piggy Bac family transposons might prove to be useful genetic tools in organisms other than insects. We are currently isolating an intact piggy Bac element from *An. gambiae* (AgaPB1) to test its mobility in various organisms.” (Mol. Gen. Genomics 270, 173-180 at p. 179, LHC, 11). There appear to be no further published reports of this putative active transposase. Xu et al. analyzed the silkworm genome looking for piggyBac-like sequences (Xu et al., Mol Gen Genomics 276, 31-40 (2006)). They found 98 piggyBac-like sequences and performed various computational analyses of putative transposase sequence and ITR sequences. They conclude: “We have isolated several intact piggyBac-like elements from *B. mori* and are currently testing their activity and the feasibility of using them as transformation vectors.” (p 38, RHC, ¶ 3). There appear to be no further published reports of these putative active transposases.

(13) Four published papers discussing the third class of distantly related piggy Bac-like transposases. The first three of these demonstrate only the excision part of the reaction and acknowledge that this is different from full transposition. Hikosaka et. al., Mol Biol Evol 24, 2648-2656 (2007) reported that “In the present study, we demonstrated that the Xtr-Urib2 Tase has excision activity toward the target transposon, although there is no evidence for the integration of the excised target into the genome thus far.” (page 2654, RHC, ¶ 2). Luo et. al., Insect Science 18, 652-662 (2011) reported “These results demonstrated the activity of the Ago-PLE1.1 transposase in mediating the first step of the cut and-paste movement of the element” (page 658, LHC, ¶ 1). Daimon et. al., Genome 53, 585-593 (2010) discussed the transposase systems yabusabe-1 and yabusabe-W. Although Daimon et al. reported detecting an excision event by PCR, they also report screening approximately 100,000 recovered plasmids for the excision of yabusame-1 and yabusame-W without identifying a single recovered plasmid from which the elements had excised. By contrast Daimon reports the transposition frequency of wildtype piggy Bac enzyme as around 0.3-1.4. Thus, it appears from Daimon et al. that the excision frequency of yabusabe-1 or -W is less than 0.001% (1:100,000). This is at least 2-3 orders of magnitude less than can be achieved with a wild-type piggy Bac enzyme and even less than available genetically engineered variants of piggy Bac transposase, which achieve ten-fold higher transposition than wildtype. The implied transposition frequency for yabasume-1 from Daimon et al. is also two orders of magnitude lower than random integration frequency in mammalian cells (which is of the order of 0.1%). Thus, Daimon et al. show that yabusame-1 was essentially inactive and would not be useful as a genetic engineering tool. Such a view likely underlies Daimon et al.'s own conclusion: “Although we could detect the excision event in the highly sensitive PCR-based assay, our data indicate that both elements have lost their excision activity almost entirely.” This also suggests that the PCR-based excision assay used to show activity of Uribo2 and Ago-PLE1.1 is not predictive of transposition activity that will be useful for inserting heterologous DNA into the genome of a target cell. The only report of a fully active piggy Bac-like transposase (competent for both excision and integration) of the third category of distantly related transposases to the original piggy Bac transposase from *Trichoplusia Ni* is one from the bat *Myotis lucifugus* (Mitra et. al., Proc. Natl.

Acad. Sci. 110, 234-239 (2013)). These authors used a yeast system to demonstrate both excision and transposition activities for the bat transposase. All of the work described here shows that it has been extremely difficult to identify fully active piggyBac-like transposases, even though there are a large number of candidate sequences. There is therefore a need for new piggyBac-like transposons and their corresponding transposases.

3. SUMMARY OF THE INVENTION

(14) Heterologous gene expression from polynucleotide constructs that stably integrate into a target cell genome can be improved by placing the expression polynucleotide between a pair of transposon ends: sequence elements that are recognized and transposed by transposases. DNA sequences inserted between a pair of transposon ends can be excised by a transposase from one DNA molecule and inserted into a second DNA molecule. Three novel piggy Bac-like transposon-transposase systems are disclosed that are not derived from the looper moth *Trichoplusia ni*. They are derived from; the tobacco budworm moth *Heliothis virescens* (the *Heliothis* transposase); the cutworm *Agrotis ipsilon* (the *Agrotis* transposase) and the bollworm *Helicoverpa armigera* (the *Helicoverpa* transposase). Each of these comprise sequences that function as transposon ends and that can be used in conjunction with a corresponding transposase that recognizes and acts on those transposon ends, as gene transfer systems for stably introducing nucleic acids into the DNA of a cell. The gene transfer systems of the invention can be used in methods including but not limited to genomic engineering of eukaryotic cells, heterologous gene expression, gene therapy, cell therapy, insertional mutagenesis, or gene discovery.

(15) Transposition may be effected using a polynucleotide comprising an open reading frame encoding a *Heliothis* transposase, the amino acid sequence of which is at least 90% identical to SEQ ID NO: 39, operably linked to a heterologous promoter. The heterologous promoter may be active in a eukaryotic cell. The heterologous promoter may be active in a mammalian cell. mRNA may be prepared from a polynucleotide comprising an open reading frame encoding a *Heliothis* transposase, the amino acid sequence of which is at least 90% identical to SEQ ID NO: 39, operably linked to a heterologous promoter that is active in an in vitro transcription reaction. The transposase may comprises a mutation as shown in columns C and D in Table 1, relative to the sequence of SEQ ID NO: 39. The transposase may comprise a mutation at an amino acid position selected from 41, 43, 81, 83, 85, 125, 126, 131, 136, 140, 149, 151, 152, 153, 155, 162, 169, 212, 238, 239, 241, 264, 268, 280, 297, 299, 300, 305, 312, 316, 322, 357, 360, 396, 397, 421, 430, 447, 449, 450, 476, 485, 492, 495, 507, 512, 585, 589, 595, and 603, relative to the sequence of SEQ ID NO: 39. The transposase may comprise a mutation selected from S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, 1149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, 1239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, Y421F, H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, Y495F, L507I, H512N, L585K, S589T, T595K and Q603K, relative to the sequence of SEQ ID NO: 39, the transposase optionally including at least 2, 3, 4, or 5 selected from the group. The amino acid sequence of the transposase may be selected from SEQ ID NO: 131-490. The transposase can excise or transpose a transposon from SEQ ID NO: 66. The excision activity or transposition activity of the transposase is at least 2-fold higher than the activity of SEQ ID NO: 39, optionally 2 to 10 fold higher. Codons of the open reading frame of the transposase may be selected for mammalian cell expression. An isolated mRNA may encode a polypeptide, the amino acid sequence of which is at least 90% identical with SEQ ID NO: 39, and wherein the mRNA sequence comprises at least 10 synonymous codon differences relative to SEQ ID NO: 1181 at corresponding positions between the mRNA and SEQ ID NO: 1181, optionally wherein codons in the mRNA at the corresponding positions are selected for mammalian cell expression. The open reading frame encoding the transposase may further encode a heterologous nuclear localization sequence fused to the transposase. The open reading frame encoding the transposase may further encode a

heterologous DNA binding domain (for example derived from a Crispr Cas system, or a zinc finger protein, or a TALE protein) fused to the transposase. A non-naturally occurring polynucleotide may encode a polypeptide, the sequence of which is at least 90% identical to SEQ ID NO: 39.

(16) A *Heliothis* transposon comprises SEQ ID NO: 15 and SEQ ID NO: 16 flanking a heterologous polynucleotide. The transposon may further comprise a sequence at least 90% identical to SEQ ID NO: 19 on one side of the heterologous polynucleotide and a sequence at least 90% identical to SEQ ID NO: 22 on the other. The heterologous polynucleotide may comprise a heterologous promoter that is active in eukaryotic cells. The promoter may be operably linked to at least one or more of: i) an open reading frame; ii) a nucleic acid encoding a selectable marker; iii) a nucleic acid encoding a counter-selectable marker; iii) a nucleic acid encoding a regulatory protein; iv) a nucleic acid encoding an inhibitory RNA. The heterologous promoter may comprise a sequence selected from SEQ ID NOs: 723-812. The heterologous polynucleotide may comprise a heterologous enhancer that is active in eukaryotic cells. The heterologous enhancer may be selected from SEQ ID NOs: 702-722. The heterologous polynucleotide may comprise a heterologous intron that is spliceable in eukaryotic cells. The nucleotide sequence of the heterologous intron may be selected from SEQ ID NO: 813-873. The heterologous polynucleotide may comprise an insulator sequence. The nucleic acid sequence of the insulator may be selected from SEQ ID NO: 684-690. The heterologous polynucleotide may comprise two open reading frames, each operably linked to a separate promoter. The heterologous polynucleotide may comprise a sequence selected from SEQ ID NOs: 997-1180. The heterologous polynucleotide may comprise or encode a selectable marker. The selectable marker may be selected from a glutamine synthetase enzyme, a dihydrofolate reductase enzyme, a puromycin acetyltransferase enzyme, a blasticidin acetyltransferase enzyme, a hygromycin B phosphotransferase enzyme, an aminoglycoside 3'-phosphotransferase enzyme and a fluorescent protein. A eukaryotic cell whose genome comprises SEQ ID NO: 15 and SEQ ID NO: 16 flanking a heterologous polynucleotide is an embodiment of the invention. The cell may be an animal cell, a mammalian cell, a rodent cell or a human cell.

(17) A transposon may be integrated into the genome of a eukaryotic cell by (a) introducing into the cell a transposon comprising SEQ ID NO: 15 and SEQ ID NO: 16 flanking a heterologous polynucleotide, (b) introducing into the cell a transposase, the sequence of which is at least 90% identical with SEQ ID NO: 39 wherein the transposase transposes the transposon to produce a genome comprising SEQ ID NO: 15 and SEQ ID NO: 16 flanking the heterologous polynucleotide. The transposase may be introduced as a polynucleotide encoding the transposase, the polynucleotide may be an mRNA molecule or a DNA molecule. The transposase may be introduced as a protein. The heterologous polynucleotide may also encode a selectable marker, and the method may further comprise selecting a cell comprising the selectable marker. The cell may be an animal cell, a mammalian cell, a rodent cell or a human cell. The human cell may be a human immune cell, for example a B-cell or a T-cell. The heterologous polynucleotide may encode a chimeric antigen receptor. A polypeptide may be expressed from the transposon integrated into the genome of the eukaryotic cell. The polypeptide may be purified. The purified polypeptide may be incorporated into a pharmaceutical composition.

Description

4. BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1. Structure of a *Heliothis* transposon. A *Heliothis* transposon comprises a left transposon end and a right transposon end flanking a heterologous polynucleotide. The left transposon end comprises (i) a left target sequence, which is often 5'-TTAA-3', although a number of other target sequences are used at lower frequency (Li et al., 2013. Proc. Natl. Acad. Sci vol. 110, no. 6, E478-487); (ii) a left ITR (e.g. SEQ ID NO: 15) and (iii) (optionally) additional left transposon end

sequences (e.g. SEQ ID NO: 19). The right transposon end comprises (i) (optionally) additional right transposon end sequences (e.g. SEQ ID NO: 22); (ii) a right ITR (e.g. SEQ ID NO: 16) which is a perfect or imperfect repeat of the left ITR, but in inverted orientation relative to the left ITR and (iii) a right target sequence which is typically the same as the left target sequence.

5. DETAILED DESCRIPTION OF THE INVENTION

5.1 Definitions

(2) Use of the singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise. Thus, for example, reference to “a polynucleotide” includes a plurality of polynucleotides, reference to “a substrate” includes a plurality of such substrates, reference to “a variant” includes a plurality of variants, and the like.

(3) Terms such as “connected,” “attached,” “linked,” and “conjugated” are used interchangeably herein and encompass direct as well as indirect connection, attachment, linkage or conjugation unless the context clearly dictates otherwise. Where a range of values is recited, it is to be understood that each intervening integer value, and each fraction thereof, between the recited upper and lower limits of that range is also specifically disclosed, along with each subrange between such values. The upper and lower limits of any range can independently be included in or excluded from the range, and each range where either, neither or both limits are included is also encompassed within the invention. Where a value being discussed has inherent limits, for example where a component can be present at a concentration of from 0 to 100%, or where the pH of an aqueous solution can range from 1 to 14, those inherent limits are specifically disclosed. Where a value is explicitly recited, it is to be understood that values which are about the same quantity or amount as the recited value are also within the scope of the invention. Where a combination is disclosed, each sub combination of the elements of that combination is also specifically disclosed and is within the scope of the invention. Conversely, where different elements or groups of elements are individually disclosed, combinations thereof are also disclosed. Where any element of an invention is disclosed as having a plurality of alternatives, examples of that invention in which each alternative is excluded singly or in any combination with the other alternatives are also hereby disclosed; more than one element of an invention can have such exclusions, and all combinations of elements having such exclusions are hereby disclosed.

(4) Unless defined otherwise herein, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Singleton, et al., *Dictionary of Microbiology and Molecular Biology*, 2nd Ed., John Wiley and Sons, New York (1994), and Hale & Marham, *The Harper Collins Dictionary of Biology*, Harper Perennial, NY, 1991, provide one of skill with a general dictionary of many of the terms used in this invention. Although any methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present invention, the preferred methods and materials are described. Unless otherwise indicated, nucleic acids are written left to right in 5' to 3' orientation; amino acid sequences are written left to right in amino to carboxy orientation, respectively. The terms defined immediately below are more fully defined by reference to the specification as a whole.

(5) The term “codon usage” or “codon bias” refers to the relative frequencies with which different synonymous codons are used to encode an amino acid within an open reading frame. A nucleic acid sequence having codon preferences for a particular target cell has a balance of synonymous codon choices that result in efficient translation in that cell type. This balance is often not calculable from observed genomic codon frequencies, but must be empirically determined, for example as described in U.S. Pat. Nos. 7,561,972 and 7,561,973 and 8,401,798 and in Welch et. al. (2009) “Design Parameters to Control Synthetic Gene Expression in *Escherichia coli*”. PLOS ONE 4 (9): e7002. <https://doi.org/10.1371/journal.pone.0007002>. A nucleic acid originally isolated from one cell type to be introduced into a target cell of another type can undergo selection of codon preferences for the target site cell such that at least 1 and sometimes, 5, 20, 15, 20, 50, 100 or more

choices among synonymous codons differ between the nucleic acid introduced into the target cell from the original nucleic acid.

(6) The “configuration” of a polynucleotide means the functional sequence elements within the polynucleotide, and the order and direction of those elements.

(7) The terms “corresponding transposon” and “corresponding transposase” are used to indicate an activity relationship between a transposase and a transposon. A transposase transposes its corresponding transposon. Many transposases may correspond with a single transposon, and many transposons may correspond with a single transposase.

(8) The term “counter-selectable marker” means a polynucleotide sequence that confers a selective disadvantage on a host cell. Examples of counter-selectable markers include *sacB*, *rpsL*, *tetAR*, *pheS*, *thyA*, *gata-1*, *ccdB*, *kid* and *barnase* (Bernard, 1995, *Journal/Gene*, 162:159-160; Bernard et al., 1994, *Journal/Gene*, 148:71-74; Gabant et al., 1997, *Journal/Biotechniques*, 23:938-941; Gababt et al., 1998, *Journal/Gene*, 207:87-92; Gababt et al., 2000, *Journal/Biotechniques*, 28:784-788; Galvao and de Lorenzo, 2005, *Journal/Appl Environ Microbiol*, 71:883-892; Hartzog et al., 2005, *Journal/Yeast*, 22:789-798; Knipfer et al., 1997, *Journal/Plasmid*, 37:129-140; Reyrat et al., 1998, *Journal/Infect Immun*, 66:4011-4017; Soderholm et al., 2001, *Journal/Biotechniques*, 31:306-310, 312; Tamura et al., 2005, *Journal/Appl Environ Microbiol*, 71:587-590; Yazynin et al., 1999, *Journal/FEBS Lett*, 452:351-354). Counter-selectable markers often confer their selective disadvantage in specific contexts. For example, they may confer sensitivity to compounds that can be added to the environment of the host cell, or they may kill a host with one genotype but not kill a host with a different genotype. Conditions which do not confer a selective disadvantage on a cell carrying a counter-selectable marker are described as “permissive”. Conditions which do confer a selective disadvantage on a cell carrying a counter-selectable marker are described as “restrictive”.

(9) The term “coupling element” or “translational coupling element” means a DNA sequence that allows the expression of a first polypeptide to be linked to the expression of a second polypeptide. Internal ribosome entry site elements (IRES elements) and cis-acting hydrolase elements (CHYSEL elements) are examples of coupling elements.

(10) The terms “DNA sequence”, “RNA sequence” or “polynucleotide sequence” mean a contiguous nucleic acid sequence. The sequence can be an oligonucleotide of 2 to 20 nucleotides in length to a full length genomic sequence of thousands or hundreds of thousands of base pairs.

(11) The term “expression construct” means any polynucleotide designed to transcribe an RNA. For example, a construct that contains at least one promoter which is or may be operably linked to a downstream gene, coding region, or polynucleotide sequence (for example, a cDNA or genomic DNA fragment that encodes a polypeptide or protein, or an RNA effector molecule, for example, an antisense RNA, triplex-forming RNA, ribozyme, an artificially selected high affinity RNA ligand (aptamer), a double-stranded RNA, for example, an RNA molecule comprising a stem-loop or hairpin dsRNA, or a bi-finger or multi-finger dsRNA or a microRNA, or any RNA). An “expression vector” is a polynucleotide comprising a promoter which can be operably linked to a second polynucleotide. Transfection or transformation of the expression construct into a recipient cell allows the cell to express an RNA effector molecule, polypeptide, or protein encoded by the expression construct. An expression construct may be a genetically engineered plasmid, virus, recombinant virus, or an artificial chromosome derived from, for example, a bacteriophage, adenovirus, adeno-associated virus, retrovirus, lentivirus, poxvirus, or herpesvirus. Such expression vectors can include sequences from bacteria, viruses or phages. Such vectors include chromosomal, episomal and virus-derived vectors, for example, vectors derived from bacterial plasmids, bacteriophages, yeast episomes, yeast chromosomal elements, and viruses, vectors derived from combinations thereof, such as those derived from plasmid and bacteriophage genetic elements, cosmids and phagemids. An expression construct can be replicated in a living cell, or it can be made synthetically. For purposes of this application, the terms “expression construct”, “expression vector”, “vector”, and “plasmid” are used interchangeably to demonstrate the application of the

invention in a general, illustrative sense, and are not intended to limit the invention to a particular type of expression construct.

(12) The term “expression polypeptide” means a polypeptide encoded by a gene on an expression construct.

(13) The term “expression system” means any in vivo or in vitro biological system that is used to produce one or more gene product encoded by a polynucleotide.

(14) A “gene” refers to a transcriptional unit including a promoter and sequence to be expressed from it as an RNA or protein. The sequence to be expressed can be genomic or cDNA among other possibilities. Other elements, such as introns, and other regulatory sequences may or may not be present.

(15) A “gene transfer system” comprises a vector or gene transfer vector, or a polynucleotide comprising the gene to be transferred which is cloned into a vector (a “gene transfer polynucleotide” or “gene transfer construct”). A gene transfer system may also comprise other features to facilitate the process of gene transfer. For example, a gene transfer system may comprise a vector and a lipid or viral packaging mix for enabling a first polynucleotide to enter a cell, or it may comprise a polynucleotide that includes a transposon and a second polynucleotide sequence encoding a corresponding transposase to enhance productive genomic integration of the transposon. The transposases and transposons of a gene transfer system may be on the same nucleic acid molecule or on different nucleic acid molecules. The transposase of a gene transfer system may be provided as a polynucleotide or as a polypeptide.

(16) Two elements are “heterologous” to one another if not naturally associated. For example, a nucleic acid sequence encoding a protein linked to a heterologous promoter means a promoter other than that which naturally drives expression of the protein. A heterologous nucleic acid flanked by transposon ends or ITRs means a heterologous nucleic acid not naturally flanked by those transposon ends or ITRs, such as a nucleic acid encoding a polypeptide other than a transposase, including an antibody heavy or light chain. A nucleic acid is heterologous to a cell if not naturally found in the cell or if naturally found in the cell but in a different location (e.g., episomal or different genomic location) than the location described.

(17) The term “host” means any prokaryotic or eukaryotic organism that can be a recipient of a nucleic acid. A “host,” as the term is used herein, includes prokaryotic or eukaryotic organisms that can be genetically engineered. For examples of such hosts, see Maniatis et al., *Molecular Cloning. A Laboratory Manual*, Cold Spring Harbor Laboratory, Cold Spring Harbor, N.Y. (1982). As used herein, the terms “host,” “host cell,” “host system” and “expression host” can be used interchangeably.

(18) A “hyperactive” transposase is a transposase that is more active than the naturally occurring transposase from which it is derived. “Hyperactive” transposases are thus not naturally occurring sequences.

(19) ‘Integration defective’ or “transposition defective” means a transposase that can excise its corresponding transposon, but that integrates the excised transposon at a lower frequency into the host genome than a corresponding naturally occurring transposase.

(20) An “IRES” or “internal ribosome entry site” means a specialized sequence that directly promotes ribosome binding, independent of a cap structure.

(21) An ‘isolated’ polypeptide or polynucleotide means a polypeptide or polynucleotide that has been either removed from its natural environment, produced using recombinant techniques, or chemically or enzymatically synthesized. Polypeptides or polynucleotides of this invention may be purified, that is, essentially free from any other polypeptide or polynucleotide and associated cellular products or other impurities.

(22) The terms “nucleoside” and “nucleotide” include those moieties which contain not only the known purine and pyrimidine bases, but also other heterocyclic bases which have been modified. Such modifications include methylated purines or pyrimidines, acylated purines or pyrimidines, or

other heterocycles. Modified nucleosides or nucleotides can also include modifications on the sugar moiety, for example, where one or more of the hydroxyl groups are replaced with halogen, aliphatic groups, or is functionalized as ethers, amines, or the like. The term “nucleotidic unit” is intended to encompass nucleosides and nucleotides.

(23) An “Open Reading Frame” or “ORF” means a portion of a polynucleotide that, when translated into amino acids, contains no stop codons. The genetic code reads DNA sequences in groups of three base pairs, which means that a double-stranded DNA molecule can read in any of six possible reading frames—three in the forward direction and three in the reverse. An ORF typically also includes an initiation codon at which translation may start.

(24) The term “operably linked” refers to functional linkage between two sequences such that one sequence modifies the behavior of the other. For example, a first polynucleotide comprising a nucleic acid expression control sequence (such as a promoter, IRES sequence, enhancer or array of transcription factor binding sites) and a second polynucleotide are operably linked if the first polynucleotide affects transcription and/or translation of the second polynucleotide. Similarly, a first amino acid sequence comprising a secretion signal or a subcellular localization signal and a second amino acid sequence are operably linked if the first amino acid sequence causes the second amino acid sequence to be secreted or localized to a subcellular location.

(25) The term “orthogonal” refers to a lack of interaction between two systems. A first transposon and its corresponding first transposase and a second transposon and its corresponding second transposase are orthogonal if the first transposase does not excise or transpose the second transposon and the second transposase does not excise or transpose the first transposon.

(26) The term “overhang” or “DNA overhang” means the single-stranded portion at the end of a double-stranded DNA molecule. Complementary overhangs are those which will base-pair with each other.

(27) A “piggy Bac-like transposase” means a transposase with at least 20% sequence identity as identified using the TBLASTN algorithm to the piggy Bac transposase from *Trichoplusia ni* (SEQ ID NO: 45), and as more fully described in Sakar, A. et. al., (2003). Mol. Gen. Genomics 270:173-180. “Molecular evolutionary analysis of the widespread piggy Bac transposon family and related ‘domesticated’ species”, and further characterized by a DDE-like DDD motif, with aspartate residues at positions corresponding to D268, D346, and D447 of *Trichoplusia ni* piggyBac transposase on maximal alignment. Piggy Bac-like transposases are also characterized by their ability to excise their transposons precisely with a high frequency. A “piggyBac-like transposon” means a transposon having transposon ends which are the same or at least 80% and preferably at least 90, 95, 96, 97, 98 or 99% or 100% identical to the transposon ends of a naturally occurring transposon that encodes a piggy Bac-like transposase. A piggy Bac-like transposon includes an inverted terminal repeat (ITR) sequence of approximately 12-16 bases at each end, and is flanked on each side by a 4 base sequence corresponding to the integration target sequence which is duplicated on transposon integration (the Target Site Duplication or Target Sequence Duplication or TSD). PiggyBac-like transposons and transposases occur naturally in a wide range of organisms including *Argyrogramma agnate* (GU477713), *Anopheles gambiae* (XP_312615; XP_320414; XP_310729), *Aphis gossypii* (GU329918), *Acyrtosiphon pisum* (XP_001948139), *Agrotis ipsilon* (GU477714), *Bombyx mori* (BAD11135), *Ciona intestinalis* (XP_002123602), *Chilo suppressalis* (JX294476), *Drosophila melanogaster* (AAL39784), *Daphnia pulex* (AAM76342), *Helicoverpa armigera* (ABS18391), *Homo sapiens* (NP_689808), *Heliothis virescens* (ABD76335), *Macdunnoughia crassisigna* (EU287451), *Macaca fascicularis* (AB179012), *Mus musculus* (NP_741958), *Pectinophora gossypiella* (GU270322), *Rattus norvegicus* (XP_220453), *Tribolium castaneum* (XP_001814566) and *Trichoplusia ni* (AAA87375) and *Xenopus tropicalis* (BAF82026), although transposition activity has been described for almost none of these.

(28) The terms “polynucleotide,” “oligonucleotide,” “nucleic acid” and “nucleic acid molecule” are used interchangeably to refer to a polymeric form of nucleotides of any length, and may comprise

ribonucleotides, deoxyribonucleotides, analogs thereof, or mixtures thereof. This term refers only to the primary structure of the molecule. Thus, the term includes triple-, double- and single-stranded deoxyribonucleic acid ("DNA"), as well as triple-, double- and single-stranded ribonucleic acid ("RNA"). It also includes modified, for example by alkylation, and/or by capping, and unmodified forms of the polynucleotide. More particularly, the terms "polynucleotide," "oligonucleotide," "nucleic acid" and "nucleic acid molecule" include polydeoxyribonucleotides (containing 2-deoxy-D-ribose), polyribonucleotides (containing D-ribose), including tRNA, IRNA, hRNA, siRNA and mRNA, whether spliced or unspliced, any other type of polynucleotide which is an N- or C-glycoside of a purine or pyrimidine base, and other polymers containing non-nucleotidic backbones, for example, polyamide (for example, peptide nucleic acids ("PNAs")) and polymorpholino (commercially available from the Anti-Virals, Inc., Corvallis, Oreg., as Neugene) polymers, and other synthetic sequence-specific nucleic acid polymers providing that the polymers contain nucleobases in a configuration which allows for base pairing and base stacking, such as is found in DNA and RNA. There is no intended distinction in length between the terms "polynucleotide," "oligonucleotide," "nucleic acid" and "nucleic acid molecule," and these terms are used interchangeably herein. These terms refer only to the primary structure of the molecule. Thus, these terms include, for example, 3'-deoxy-2', 5'-DNA, oligodeoxyribonucleotide N3' P5' phosphoramidates, 2'-O-alkyl-substituted RNA, double- and single-stranded DNA, as well as double- and single-stranded RNA, and hybrids thereof including for example hybrids between DNA and RNA or between PNAs and DNA or RNA, and also include known types of modifications, for example, labels, alkylation, "caps," substitution of one or more of the nucleotides with an analog, internucleotide modifications such as, for example, those with uncharged linkages (for example, methyl phosphonates, phosphotriesters, phosphoramidates, carbamates, or the like) with negatively charged linkages (for example, phosphorothioates, phosphorodithioates, or the like), and with positively charged linkages (for example, aminoalkylphosphoramidates, aminoalkylphosphotriesters), those containing pendant moieties, such as, for example, proteins (including enzymes (for example, nucleases), toxins, antibodies, signal peptides, poly-L-lysine, or the like), those with intercalators (for example, acridine, psoralen, or the like), those containing chelates (of, for example, metals, radioactive metals, boron, oxidative metals, or the like), those containing alkylators, those with modified linkages (for example, alpha anomeric nucleic acids, or the like), as well as unmodified forms of the polynucleotide or oligonucleotide.

(29) A "promoter" means a nucleic acid sequence sufficient to direct transcription of an operably linked nucleic acid molecule. A promoter can be used with or without other transcription control elements (for example, enhancers) that are sufficient to render promoter-dependent gene expression controllable in a cell type-specific, tissue-specific, or temporal-specific manner, or that are inducible by external signals or agents; such elements, may be within the 3' region of a gene or within an intron. Desirably, a promoter is operably linked to a nucleic acid sequence, for example, a cDNA or a gene sequence, or an effector RNA coding sequence, in such a way as to enable expression of the nucleic acid sequence, or a promoter is provided in an expression cassette into which a selected nucleic acid sequence to be transcribed can be conveniently inserted. A regulatory element such as promoter active in a mammalian cells means a regulatory element configurable to result in a level of expression of at least 1 transcript per cell in a mammalian cell into which the regulatory element has been introduced.

(30) The term "selectable marker" means a polynucleotide segment or expression product thereof that allows one to select for or against a molecule or a cell that contains it, often under particular conditions. These markers can encode an activity, such as, but not limited to, production of RNA, peptide, or protein, or can provide a binding site for RNA, peptides, proteins, inorganic and organic compounds or compositions. Examples of selectable markers include but are not limited to: (1) DNA segments that encode products which provide resistance against otherwise toxic compounds

(e.g., antibiotics); (2) DNA segments that encode products which are otherwise lacking in the recipient cell (e.g., tRNA genes, auxotrophic markers); (3) DNA segments that encode products which suppress the activity of a gene product; (4) DNA segments that encode products which can be readily identified (e.g., phenotypic markers such as beta-galactosidase, green fluorescent protein (GFP), and cell surface proteins); (5) DNA segments that bind products which are otherwise detrimental to cell survival and/or function; (6) DNA segments that otherwise inhibit the activity of any of the DNA segments described in Nos. 1-5 above (e.g., antisense oligonucleotides); (7) DNA segments that bind products that modify a substrate (e.g. restriction endonucleases); (8) DNA segments that can be used to isolate a desired molecule (e.g. specific protein binding sites); (9) DNA segments that encode a specific nucleotide sequence which can be otherwise non-functional (e.g., for PCR amplification of subpopulations of molecules); and/or (10) DNA segments, which when absent, directly or indirectly confer sensitivity to particular compounds.

(31) Sequence identity can be determined by aligning sequences using algorithms, such as BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package Release 7.0, Genetics Computer Group, 575 Science Dr., Madison, Wis.), using default gap parameters, or by inspection, and the best alignment (i.e., resulting in the highest percentage of sequence similarity over a comparison window). Percentage of sequence identity is calculated by comparing two optimally aligned sequences over a window of comparison, determining the number of positions at which the identical residues occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of matched and mismatched positions not counting gaps in the window of comparison (i.e., the window size), and multiplying the result by 100 to yield the percentage of sequence identity. Unless otherwise indicated the window of comparison between two sequences is defined by the entire length of the shorter of the two sequences.

(32) A “target nucleic acid” is a nucleic acid into which a transposon is to be inserted. Such a target can be part of a chromosome, episome or vector.

(33) An “integration target sequence” or “target sequence” or “target site” for a transposase is a site or sequence in a target DNA molecule into which a transposon can be inserted by a transposase. The piggyBac transposase from *Trichoplusia ni* inserts its transposon predominantly into the target sequence 5'-TTAA-3'. Other useable target sequences for piggyBac transposons are 5'-CTAA-3', 5'-TTAG-3', 5'-ATAA-3', 5'-TCAA-3', 5'-AGTT-3', 5'-ATTA-3', 5'-GTTA-3', 5'-TTGA-3', 5'-TTTA-3', 5'-TTAC-3', 5'-ACTA-3', 5'-AGGG-3', 5'-CTAG-3', 5'-GTAA-3', 5'-AGGT-3', 5'-ATCA-3', 5'-CTCC-3', 5'-TAAA-3', 5'-TCTC-3', 5'-TGAA-3', 5'-AAAT-3', 5'-AATC-3', 5'-ACAA-3', 5'-ACAT-3', 5'-ACTC-3', 5'-AGTG-3', 5'-ATAG-3', 5'-CAAA-3', 5'-CACA-3', 5'-CATA-3', 5'-CCAG-3', 5'-CCCA-3', 5'-CGTA-3', 5'-CTGA-3', 5'-GTCC-3', 5'-TAAG-3', 5'-TCTA-3', 5'-TGAG-3', 5'-TGTT-3', 5'-TTCA-3', 5'-TTCT-3' and 5'-TTTT-3' (Li et al., 2013. Proc. Natl. Acad. Sci vol. 110, no. 6, E478-487) and 5'-TTAT. PiggyBac-like transposases transpose their transposons using a cut-and-paste mechanism, which results in duplication of their 4 base pair target sequence on insertion into a DNA molecule. The target sequence is thus found on each side of an integrated piggy Bac-like transposon.

(34) The term “translation” refers to the process by which a polypeptide is synthesized by a ribosome ‘reading’ the sequence of a polynucleotide.

(35) A ‘transposase’ is a polypeptide that catalyzes the excision of a corresponding transposon from a donor polynucleotide, for example a vector, and (providing the transposase is not integration-deficient) the subsequent integration of the transposon into a target nucleic acid. A “*Heliothis* transposase” means a transposase with at least 80, 90, 95, 96, 7, 98, 99 or 100% sequence identity to SEQ ID NO: 39, including hyperactive variants of SEQ ID NO: 39, that are able to transposase a corresponding transposon. An “*Agrotis* transposase” means a transposase with at least 80, 90, 95, 96, 97, 98, 99 or 100% sequence identity to SEQ ID NO: 40, including hyperactive variants of SEQ ID NO: 40, that are able to transposase a corresponding transposon. A “*Helicoverpa* transposase”

means a transposase with at least 80, 90, 95, 96, 7, 98, 99 or 100% sequence identity to SEQ ID NO: 41, including hyperactive variants of SEQ ID NO: 41, that are able to transposase a corresponding transposon. A hyperactive transposase is a transposase that is more active than the naturally occurring transposase from which it is derived, for excision activity or transposition activity or both. A hyperactive transposase is preferably at least 1.5-fold more active, or at least 2-fold more active, or at least 5-fold more active, or at least 10-fold more active than the naturally occurring transposase from which it is derived, e.g., 2-5 fold or 2-10 fold. A transposase may or more not be fused to one or more additional domains such as a nuclear localization sequence or DNA binding protein.

(36) The term “transposition” is used herein to mean the action of a transposase in excising a transposon from one polynucleotide and then integrating it, either into a different site in the same polynucleotide, or into a second polynucleotide.

(37) The term “transposon” means a polynucleotide that can be excised from a first polynucleotide, for instance, a vector, and be integrated into a second position in the same polynucleotide, or into a second polynucleotide, for instance, the genomic or extrachromosomal DNA of a cell, by the action of a corresponding trans-acting transposase. A transposon comprises a first transposon end and a second transposon end, which are polynucleotide sequences recognized by and transposed by a transposase. A transposon usually further comprises a first polynucleotide sequence between the two transposon ends, such that the first polynucleotide sequence is transposed along with the two transposon ends by the action of the transposase. This first polynucleotide in natural transposons frequently comprises an open reading frame encoding a corresponding transposase that recognizes and transposes the transposon. Transposons of the present invention are “synthetic transposons” comprising a heterologous polynucleotide sequence which is transposable by virtue of its juxtaposition between two transposon ends. Synthetic transposons may or may not further comprise flanking polynucleotide sequence(s) outside the transposon ends, such as a sequence encoding a transposase, a vector sequence or sequence encoding a selectable marker.

(38) The term “transposon end” means the cis-acting nucleotide sequences that are sufficient for recognition by and transposition by a corresponding transposase. Transposon ends of piggyBac-like transposons comprise perfect or imperfect repeats such that the respective repeats in the two transposon ends are reverse complements of each other. These are referred to as inverted terminal repeats (ITR) or terminal inverted repeats (TIR). A transposon end may or may not include additional sequence proximal to the ITR that promotes or augments transposition.

(39) The term “vector” or “DNA vector” or “gene transfer vector” refers to a polynucleotide that is used to perform a “carrying” function for another polynucleotide. For example, vectors are often used to allow a polynucleotide to be propagated within a living cell, or to allow a polynucleotide to be packaged for delivery into a cell, or to allow a polynucleotide to be integrated into the genomic DNA of a cell. A vector may further comprise additional functional elements, for example it may comprise a transposon.

5.2 Description

(40) 5.2.1 Genomic Integration

(41) Expression of a gene from a heterologous polynucleotide in a eukaryotic host cell can be improved if the heterologous polynucleotide is integrated into the genome of the host cell. Integration of a polynucleotide into the genome of a host cell also generally makes it stably heritable, by subjecting it to the same mechanisms that ensure the replication and division of genomic DNA. Such stable heritability is desirable for achieving good and consistent expression over long growth periods. This is particularly important for cell therapies in which cells are genetically modified and then placed into the body. It is also important for the manufacturing of biomolecules, particularly for therapeutic applications where the stability of the host and consistency of expression levels is also important for regulatory purposes. Cells with gene transfer vectors, including transposon-based gene transfer vectors, integrated into their genomes are thus an

important embodiment of the invention.

(42) Heterologous polynucleotides may be more efficiently integrated into a target genome if they are part of a transposon (i.e., positioned between transposon ITRs), for example so that they may be integrated by a transposase. A particular benefit of a transposon is that the entire polynucleotide between the transposon ITRs is integrated. A transposon comprising target sites flanking ITRs flanking a heterologous polynucleotide integrates at a target site in a genome to result in the genome containing the heterologous polynucleotide flanked by the ITRs, flanked by target sites. This is in contrast to random integration, where a polynucleotide introduced into a eukaryotic cell is often fragmented at random in the cell, and only parts of the polynucleotide become incorporated into the target genome, usually at a low frequency. The piggyBac transposon from the looper moth *Trichoplusia ni* has been shown to be transposed by its transposase in cells from many organisms (see e.g. Keith et al (2008) BMC Molecular Biology 9:72 “Analysis of the piggyBac transposase reveals a functional nuclear targeting signal in the 94 c-terminal residues”). Heterologous polynucleotides incorporated into piggyBac-like transposons may be integrated into eukaryotic cells including animal cells, fungal cells or plant cells. Preferred animal cells can be vertebrate or invertebrate. Preferred vertebrate cells include cells from mammals including rodents such as rats, mice, and hamsters; ungulates, such as cows, goats or sheep; and swine. Preferred vertebrate cells also include cells from human tissues and human stem cells. Target cells types include hepatocytes, neural cells, muscle cells, blood cells, embryonic stem cells, somatic stem cells, hematopoietic cells, embryos, zygotes, sperm cells (some of which are open to be manipulated in an in vitro setting) and immune cells including lymphocytes such as T cells, B cells and natural killer cells, T-helper cells, antigen-presenting cells, dendritic cells, neutrophils and macrophages. Preferred cells can be pluripotent cells (cells whose descendants can differentiate into several restricted cell types, such as hematopoietic stem cells or other stem cells) or totipotent cells (i.e., a cell whose descendants can become any cell type in an organism, e.g., embryonic stem cells). Preferred culture cells are Chinese hamster ovary (CHO) cells or Human embryonic kidney (HEK293) cells. Preferred fungal cells are yeast cells including *Saccharomyces cerevisiae* and *Pichia pastoris*. Preferred plant cells are algae, for example *Chlorella*, tobacco, maize and rice (Nishizawa-Yokoi et al (2014) Plant J. 77:454-63 “Precise marker excision system using an animal derived piggyBac transposon in plants”).

(43) Preferred gene transfer systems comprise a transposon in combination with a corresponding transposase protein that transposases the transposon, or a nucleic acid that encodes the corresponding transposase protein and is expressible in the target cell. A preferred gene transfer system comprises a synthetic *Heliothis* transposon and a corresponding *Heliothis* transposase. A preferred gene transfer system comprises a synthetic *Helicoverpa* transposon and a corresponding *Helicoverpa* transposase. A preferred gene transfer system comprises a synthetic *Agrotis* transposon and a corresponding *Agrotis* transposase.

(44) A transposase protein can be introduced into a cell as a protein or as a nucleic acid encoding the transposase, for example as a ribonucleic acid, including mRNA or any polynucleotide recognized by the translational machinery of a cell; as DNA, e.g. as extrachromosomal DNA including episomal DNA; as plasmid DNA, or as viral nucleic acid. Furthermore, the nucleic acid encoding the transposase protein can be transfected into a cell as a nucleic acid vector such as a plasmid, or as a gene expression vector, including a viral vector. The nucleic acid can be circular or linear. mRNA encoding the transposase may be prepared using DNA in which a gene encoding the transposase is operably linked to a heterologous promoter, such as the bacterial T7 promoter, which is active in vitro. DNA encoding the transposase protein can be stably inserted into the genome of the cell or into a vector for constitutive or inducible expression. Where the transposase protein is transfected into the cell or inserted into the vector as DNA, the transposase encoding sequence is preferably operably linked to a heterologous promoter. There are a variety of promoters that could be used including constitutive promoters, cell-type specific promoters, organism-specific

promoters, tissue-specific promoters, inducible promoters, and the like. Where DNA encoding the transposase is operably linked to a promoter and transfected into a target cell, the promoter should be operable in the target cell. For example if the target cell is a mammalian cell, the promoter should be operable in a mammalian cell; if the target cell is a yeast cell, the promoter should be operable in a yeast cell; if the target cell is an insect cell, the promoter should be operable in an insect cell; if the target cell is a human cell, the promoter should be operable in a human cell; if the target cell is a human immune cell, the promoter should be operable in a human immune cell. All DNA or RNA sequences encoding piggy Bac-like transposase proteins are expressly contemplated. Alternatively, the transposase may be introduced into the cell directly as protein, for example using cell-penetrating peptides (e.g. as described in Ramsey and Flynn (2015) Pharmacol. Ther. 154:78-86 “Cell-penetrating peptides transport therapeutics into cells”); using small molecules including salt plus propanebetaine (e.g. as described in Astolfo et al (2015) Cell 161:674-690); or electroporation (e.g. as described in Morgan and Day (1995) Methods in Molecular Biology 48:63-71 “The introduction of proteins into mammalian cells by electroporation”).

(45) It is possible to insert the transposon into DNA of a cell through non-homologous recombination through a variety of reproducible mechanisms, and even without the activity of a transposase. The transposons described herein can be used for gene transfer regardless of the mechanisms by which the genes are transferred.

(46) 5.2.5 Gene Transfer Systems

(47) Gene transfer systems comprise a polynucleotide to be transferred to a host cell. Preferably the polynucleotide comprises a *Heliothis* transposon or a *Helicoverpa* transposon or an *Agrotis* transposon wherein the polynucleotide is to be integrated into the genome of a target cell.

(48) When there are multiple components of a gene transfer system, for example the one or more polynucleotides comprising genes for expression in the target cell and optionally comprising transposon ends, and a transposase (which may be provided either as a protein or encoded by a nucleic acid), these components can be transfected into a cell at the same time, or sequentially. For example, a transposase protein or its encoding nucleic acid may be transfected into a cell prior to, simultaneously with or subsequent to transfection of a corresponding transposon. Additionally, administration of either component of the gene transfer system may occur repeatedly, for example, by administering at least two doses of this component.

(49) Any of the transposase proteins described herein may be encoded by polynucleotides including RNA or DNA. Similarly, the nucleic acid encoding the transposase protein or the transposon of this invention can be transfected into the cell as a linear fragment or as a circularized fragment, either as a plasmid or as recombinant viral DNA.

(50) A *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase may be provided as a DNA molecule expressible in the target cell. The sequence encoding the *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase should be operably linked to heterologous sequences that enable expression of the transposase in the target cell. A sequence encoding the *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase may be operably linked to a heterologous promoter that is active in the target cell. For example, if the target cell is a mammalian cell, then the promoter should be active in a mammalian cell. If the target is a vertebrate cell, the promoter should be active in a vertebrate cell. If the target cell is a plant cell, the promoter should be active in a plant cell. If the promoter is an insect cell, the promoter should be active in an insect cell. The sequence encoding the *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase may also be operably linked to other sequence elements required for expression in the target cell, for example polyadenylation sequences, terminator sequences etc.

(51) A *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase may be provided as an mRNA expressible in the target cell. mRNA is preferably prepared in an in vitro transcription reaction. For in vitro transcription, a sequence encoding the *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase is operably linked to a promoter that is active in an in vitro

transcription reaction. Exemplary promoters active in an in vitro transcription reaction include a T7 promoter (5'-TAATACGACTCACTATAG-3') which enables transcription by T7 RNA polymerase, a T3 promoter (5'-AATTAACCCTCACTAAAG-3') which enables transcription by T3 RNA polymerase and an SP6 promoter (5'-ATTTAGGTGACACTATAG-3') which enables transcription by SP6 RNA polymerase. Variants of these promoters and other promoters that can be used for in vitro transcription may also be operably linked to a sequence encoding an *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase.

(52) If the *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase is provided as a polynucleotide (either DNA or mRNA) encoding the transposase, then it is advantageous to improve the expressibility of the transposase in the target cell. It is therefore advantageous to use a sequence other than a naturally occurring sequence to encode the transposase, in other words, to use codon-preferences of the cell type in which expression is to be performed. For example, if the target cell is a mammalian cell, then the codons should be biased toward the preferences seen in a mammalian cell. If the target is a vertebrate cell, then the codons should be biased toward the preferences seen in the particular vertebrate cell. If the target cell is a plant cell, then the codons should be biased toward the preferences seen in a plant cell. If the promoter is an insect cell, then the codons should be biased toward the preferences seen in an insect cell.

(53) Preferable RNA molecules include those with appropriate cap structures to enhance translation in a eukaryotic cell, polyadenylic acid and other 3' sequences that enhance mRNA stability in a eukaryotic cell and optionally substitutions to reduce toxicity effects on the cell, for example substitution of uridine with pseudouridine, and substitution of cytosine with 5-methyl cytosine. mRNA encoding the *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase may be prepared such that it has a 5'-cap structure to improve expression in a target cell. Exemplary cap structures are a cap analog (G(5')ppp(5')G), an anti-reverse cap analog (3'-O-Me-m7G(5')ppp(5')G), a clean cap (m7G(5')ppp(5')(2'OMeA)pG), an mCap (m7G(5')ppp(5')G). mRNA encoding the *Heliothis* transposase or *Helicoverpa* transposase or *Agrotis* transposase may be prepared such that some bases are partially or fully substituted, for example uridine may be substituted with pseudouridine, cytosine may be substituted with 5-methyl-cytosine. Any combinations of these caps and substitutions may be made.

(54) The components of the gene transfer system may be transfected into one or more cells by techniques such as particle bombardment, electroporation, microinjection, combining the components with lipid-containing vesicles, such as cationic lipid vesicles, DNA condensing reagents (example, calcium phosphate, polylysine or polyethyleneimine), and inserting the components (that is the nucleic acids thereof into a viral vector and contacting the viral vector with the cell. Where a viral vector is used, the viral vector can include any of a variety of viral vectors known in the art including viral vectors selected from the group consisting of a retroviral vector, an adenovirus vector or an adeno-associated viral vector. The gene transfer system may be formulated in a suitable manner as known in the art, or as a pharmaceutical composition or kit.

(55) 5.2.3 Sequence Elements in Gene Transfer Systems

(56) Expression of genes from a gene transfer polynucleotide such as a piggyBac-like transposon, including a *Heliothis* transposon or a *Helicoverpa* transposon or an *Agrotis* transposon, integrated into a host cell genome is often strongly influenced by the chromatin environment into which it integrates. Polynucleotides that are integrated into euchromatin have higher levels of expression than those that are either integrated into heterochromatin, or which become silenced following their integration. Silencing of a heterologous polynucleotide may be reduced if it comprises a chromatin control element. It is thus advantageous for gene transfer polynucleotides (including any of the transposons described herein) to comprise chromatin control elements such as sequences that prevent the spread of heterochromatin (insulators). Advantageous gene transfer polynucleotides including a *Heliothis* transposon or a *Helicoverpa* transposon or an *Agrotis* transposon comprise an insulator sequence that is at least 95% identical to a sequence selected from one of SEQ ID NOS:

684-690, they may also comprise ubiquitously acting chromatin opening elements (UCOE) or stabilizing and anti-repressor elements (STARs), to increase long-term stable expression from the integrated gene transfer polynucleotide. Advantageous gene transfer polynucleotides may further comprise a matrix attachment region for example a sequence that is at least 95% identical to a sequence selected from one of SEQ ID NOS: 691-701.

(57) In some cases, it is advantageous for a gene transfer polynucleotide to comprise two insulators, one on each side of the heterologous polynucleotide that contains the sequence(s) to be expressed and within the transposon ITRs. The insulators may be the same, or they may be different. Particularly advantageous gene transfer polynucleotides comprise an insulator sequence that is at least 95% identical to a sequence selected from one of SEQ ID NO: 689 or SEQ ID NO: 690 and an insulator sequence that is at least 95% identical to a sequence selected from one of SEQ ID NOS: 684-688. Insulators also shield expression control elements from one another. For example, when a gene transfer polynucleotide comprises genes encoding two open reading frames, each operably linked to a different promoter, one promoter may reduce expression from the other in a phenomenon known as transcriptional interference. Interposing an insulator sequence that is at least 95% identical to a sequence selected from one of SEQ ID NOS: 684-690 between the two transcriptional units can reduce this interference, increasing expression from one or both promoters.

(58) Preferred gene transfer vectors comprise expression elements capable of driving high levels of gene expression. In eukaryotic cells, gene expression is regulated by several different classes of elements, including enhancers, promoters, introns, RNA export elements, polyadenylation sequences and transcriptional terminators.

(59) Advantageous gene transfer polynucleotides for the transfer of genes for expression into eukaryotic cells comprise an enhancer operably linked to a heterologous gene. Advantageous gene transfer polynucleotides for the transfer of genes for expression into mammalian cells comprise an enhancer from immediate early genes 1, 2 or 3 of cytomegalovirus (CMV) from either human, primate or rodent cells (for example sequences at least 95% identical to any of SEQ ID NOS: 702-720), an enhancer from the adenoviral major late protein enhancer (for example sequences at least 95% identical to SEQ ID NO: 721), or an enhancer from SV40 (for example sequences at least 95% identical to SEQ ID NO: 722), operably linked to a heterologous gene.

(60) Advantageous gene transfer polynucleotides for the transfer of genes for expression into eukaryotic cells comprise a promoter operably linked to a heterologous gene. Advantageous gene transfer polynucleotides for the transfer of genes for expression into mammalian cells comprise an EF1a promoter from any mammalian or avian species including human, rat, mice, chicken and Chinese hamster, (for example any of SEQ ID NOS: 723-744); a promoter from the immediate early genes 1, 2 or 3 of cytomegalovirus (CMV) from either human, primate or rodent cells (for example any of SEQ ID NOS: 745-755); a promoter for eukaryotic elongation factor 2 (EEF2) from any mammalian or avian species including human, rat, mice, chicken and Chinese hamster, (for example any of SEQ ID NOS: 756-766); a GAPDH promoter from any mammalian or yeast species (for example any of SEQ ID NOS: 777-793), an actin promoter from any mammalian or avian species including human, rat, mice, chicken and Chinese hamster (for example any of SEQ ID NOS: 767-776); a PGK promoter from any mammalian or avian species including human, rat, mice, chicken and Chinese hamster (for example any of SEQ ID NOS: 794-800), or a ubiquitin promoter (for example SEQ ID NO: 801-804), operably linked to a heterologous gene. The promoter may be operably linked to i) a heterologous open reading frame; ii) a nucleic acid encoding a selectable marker; iii) a nucleic acid encoding a counter-selectable marker; iii) a nucleic acid encoding a regulatory protein; iv) a nucleic acid encoding an inhibitory RNA.

(61) Advantageous gene transfer polynucleotides for the transfer of genes for expression into eukaryotic cells comprise an intron within a heterologous polynucleotide spliceable in a target cell. Advantageous gene transfer polynucleotides for the transfer of genes for expression into mammalian cells comprise an intron from immediate early genes 1, 2 or 3 of cytomegalovirus

(CMV) from either human, primate or rodent cells (for example sequences at least 95% identical to any of SEQ ID NOS: 813-823), an intron from EF1a from any mammalian or avian species including human, rat, mice, chicken and Chinese hamster, (for example sequences at least 95% identical to any of SEQ ID NOS: 833-845), an intron from EEF2 from any mammalian or avian species including human, rat, mice, chicken and Chinese hamster, (for example sequences at least 95% identical to any of SEQ ID NOS: 865-872); an intron from actin from any mammalian or avian species including human, rat, mice, chicken and Chinese hamster (for example sequences at least 95% identical to any of SEQ ID NOS: 846-859), a GAPDH intron from any mammalian or avian species including human, rat, mice, chicken and Chinese hamster (for example sequences at least 95% identical to any of SEQ ID NOS: 860-862); an intron comprising the adenoviral major late protein enhancer for example sequences at least 95% identical to SEQ ID NO: 863-864) or a hybrid/synthetic intron (for example sequences at least 95% identical to any of SEQ ID NOS: 824-832) within a heterologous polynucleotide.

(62) Advantageous gene transfer polynucleotides for the transfer of genes for expression into eukaryotic cells comprise an enhancer and promoter, operably linked to a heterologous coding sequence. Such gene transfer polynucleotides may comprise combinations of enhancers and promoters in which an enhancer from one gene is combined with a promoter from a different gene, that is the enhancer is heterologous to the promoter. For example, for the transfer of genes for expression into mammalian cells, an immediate early CMV enhancer from rodent or human or primate (such as a sequence selected from SEQ ID NOS: 702-720) is advantageously followed by a promoter from an EF1a gene (such as a sequence selected from SEQ ID NOS: 723-744), or a promoter from a heterologous CMV gene (such as a sequence selected from SEQ ID NOS: 745-755), or a promoter from an EEF2 gene (such as a sequence selected from SEQ ID NOS: 756-766), or a promoter from an actin gene (such as a sequence selected from SEQ ID NOS: 767-776), or a promoter from a GAPDH gene (such as a sequence selected from SEQ ID NOS: 777-793) operably linked to a heterologous sequence.

(63) Advantageous gene transfer polynucleotides for the transfer of genes for expression into eukaryotic cells comprise an operably linked promoter and an intron, operably linked to a heterologous open reading frame. Such gene transfer polynucleotides may comprise combinations of promoters and introns in which a promoter from one gene is combined with an intron from a different gene, that is the intron is heterologous to the promoter. For example, for the transfer of genes for expression into mammalian cells, an immediate early CMV promoter from rodent or human or primate (such as a sequence selected from SEQ ID NOS: 745-755) is advantageously followed by an intron from an EF1a gene (such as a sequence that is at least 95% identical to a sequence selected from SEQ ID NOS: 833-845) or an intron from an EEF2 gene (such as a sequence that is at least 95% identical to a sequence selected from SEQ ID NOS: 865-872), or an intron from an actin gene (such as a sequence that is at least 95% identical to a sequence selected from SEQ ID NOS: 846-859) operably linked to a heterologous sequence.

(64) Advantageous gene transfer polynucleotides for the transfer of genes for expression into eukaryotic cells, comprise composite transcriptional initiation regulatory elements comprising promoters that are operably linked to enhancers and/or introns, and the composite transcriptional initiation regulatory element is operably linked to a heterologous sequence. Examples of advantageous composite transcriptional initiation regulatory elements that may be operably linked to a heterologous sequence in gene transfer polynucleotides for the transfer of genes for expression into mammalian cells are sequences selected from SEQ ID NOS: 874-966.

(65) Expression of two open reading frames from a single polynucleotide can be accomplished by operably linking the expression of each open reading frame to a separate promoter, each of which may optionally be operably linked to enhancers and introns as described above. This is particularly useful when expressing two polypeptides that need to interact at specific molar ratios, such as chains of an antibody or chains of a bispecific antibody, or a receptor and its ligand. It is often

advantageous to prevent transcriptional promoter interference by placing a genetic insulator between the two open reading frames, for example to the 3' of the polyadenylation sequence operably linked to the first open reading frame and to the 5' of the promoter operably linked to the second open reading frame encoding the second polypeptide. Transcriptional promoter interference may also be prevented by effectively terminating transcription of the first gene. In many eukaryotic cells the use of strong poly A signal sequences between two open reading frames will reduce transcriptional promoter interference. Examples of poly A signal sequences that can be used to effectively terminate transcription are given as SEQ ID NOs: 967-996. Advantageous gene transfer polynucleotides comprise a sequence that is at least 95% identical to a sequence selected from SEQ ID NOs: 967-996 operably linked to a heterologous open reading frame. Advantageous composite regulatory elements for the termination of transcription of a first gene and the initiation of transcription of a second gene include sequences given as SEQ ID NOs: 997-1180. Particularly advantageous gene transfer polynucleotides for the transfer of a first and a second open reading frame for co-expression into mammalian cells comprise a sequence at least 90% identical or at least 95% identical or at least 99% identical to or 100% identical to a sequence selected from SEQ ID NOs: 997-1180, separating two heterologous open reading frames.

(66) 5.2.4 Selection of Target Cells Comprising Gene Transfer Polynucleotides

(67) A target cell whose genome comprises a stably integrated transfer polynucleotide may be identified, if the gene transfer polynucleotide comprises an open reading frame encoding a selectable marker, by exposing the target cells to conditions that favor cells expressing the selectable marker ("selection conditions"). It is advantageous for a gene transfer polynucleotide to comprise an open reading frame encoding a selectable marker such as an enzyme that confers resistance to antibiotics such as neomycin (resistance conferred by an aminoglycoside 3'-phosphotransferase e.g. a sequence selected from SEQ ID NOs: 513-516), puromycin (resistance conferred by puromycin acetyltransferase e.g. a sequence selected from SEQ ID NOs: 519-521), blasticidin (resistance conferred by a blasticidin acetyltransferase and a blasticidin deaminase e.g. SEQ ID NO: 523), hygromycin B (resistance conferred by hygromycin B phosphotransferase e.g. a sequence selected from SEQ ID NOs: 517-518) and zeocin (resistance conferred by a binding protein encoded by the ble gene, for example SEQ ID NO: 510). Other selectable markers include those that are fluorescent (such as open reading frames encoding GFP, RFP etc.) and can therefore be selected for example using flow cytometry. Other selectable markers include open reading frames encoding transmembrane proteins that are able to bind to a second molecule (protein or small molecule) that can be fluorescently labelled so that the presence of the transmembrane protein can be selected for example using flow cytometry.

(68) A gene transfer polynucleotide may comprise a selectable marker open reading frame encoding glutamine synthetase (GS, for example a sequence selected from SEQ ID NOS: 524-528) which allows selection via glutamine metabolism. Glutamine synthase is the enzyme responsible for the biosynthesis of glutamine from glutamate and ammonia, it is a crucial component of the only pathway for glutamine formation in a mammalian cell. In the absence of glutamine in the growth medium, the GS enzyme is essential for the survival of mammalian cells in culture. Some cell lines, for example mouse myeloma cells do not express sufficient GS enzyme to survive without added glutamine. In these cells a transfected GS open reading frame can function as a selectable marker by permitting growth in a glutamine-free medium. In other cell lines, for example Chinese hamster ovary (CHO) cells express sufficient GS enzyme to survive without exogenously added glutamine. These cell lines can be manipulated by genome editing techniques including CRISPR/Cas9 to reduce or eliminate the activity of the GS enzyme. In all of these cases, GS inhibitors such as methionine sulfoximine (MSX) can be used to inhibit a cell's endogenous GS activity. Selection protocols include introducing a gene transfer polynucleotide comprising sequences encoding a first polypeptide and a glutamine synthase selectable marker, and then treating the cell with inhibitors of glutamine synthase such as methionine sulfoximine. The higher

the levels of methionine sulphoximine that are used, the higher the level of glutamine synthase expression is required to allow the cell to synthesize sufficient glutamine to survive. Some of these cells will also show an increased expression of the first polypeptide.

(69) Preferably the GS open reading frame is operably linked to a weak promoter or other sequence elements that attenuate expression as described herein, such that high levels of expression can only occur if many copies of the gene transfer polynucleotide are present, or if they are integrated in a position in the genome where high levels of expression occur. In such cases it may be unnecessary to use the inhibitor methionine sulphoximine: simply synthesizing sufficient glutamine for cell survival may provide a sufficiently stringent selection if expression of the glutamine synthetase is attenuated.

(70) A gene transfer polynucleotide may comprise a selectable marker open reading frame encoding dihydrofolate reductase (DHFR, for example a sequence selected from SEQ ID NOs: 511-512) which is required for catalyzing the reduction of 5,6-dihydrofolate (DHF) to 5,6,7,8-tetrahydrofolate (THF). Some cell lines do not express sufficient DHFR to survive without added hypoxanthine and thymidine (HT). In these cells a transfected DHFR open reading frame can function as a selectable marker by permitting growth in a hypoxanthine and thymidine-free medium. DHFR-deficient cell lines, for example Chinese hamster ovary (CHO) cells can be produced by genome editing techniques including CRISPR/Cas9 to reduce or eliminate the activity of the endogenous DHFR enzyme. DHFR confers resistance to methotrexate (MTX). DHFR can be inhibited by higher levels of methotrexate. Selection protocols include introducing a construct comprising sequences encoding a first polypeptide and a DHFR selectable marker into a cell with or without a functional endogenous DHFR gene, and then treating the cell with inhibitors of DHFR such as methotrexate. The higher the levels of methotrexate that are used, the higher the level of DHFR expression is required to allow the cell to synthesize sufficient DHFR to survive. Some of these cells will also show an increased expression of the first polypeptide. Preferably the DHFR open reading frame is operably linked to a weak promoter or other sequence elements that attenuate expression as described above, such that high levels of expression can only occur if many copies of the gene transfer polynucleotide are present, or if they are integrated in a position in the genome where high levels of expression occur.

(71) High levels of expression may be obtained from genes encoded on gene transfer polynucleotides that are integrated at regions of the genome that are highly transcriptionally active, or that are integrated into the genome in multiple copies, or that are present extrachromosomally in multiple copies. It is often advantageous to operably link the open reading frame encoding the selectable marker to expression control elements that result in low levels of expression of the selectable polypeptide from the gene transfer polynucleotide and/or to use conditions that provide more stringent selection. Under these conditions, for the expression cell to produce sufficient levels of the selectable polypeptide encoded on the gene transfer polynucleotide to survive the selection conditions, the gene transfer polynucleotide can either be present in a favorable location in the cell's genome for high levels of expression, or a sufficiently high number of copies of the gene transfer polynucleotide can be present, such that these factors compensate for the low levels of expression achievable because of the expression control elements.

(72) Genomic integration of transposons in which a selectable marker is operably linked to regulatory elements that only weakly express the marker usually requires that the transposon be inserted into the target genome by a transposase, see for example Section 6.1.4. By operably linking the selectable marker to elements that result in weak expression, cells are selected which either incorporate multiple copies of the transposon, or in which the transposon is integrated at a favorable genomic location for high expression. Using a gene transfer system that comprises a transposon and a corresponding transposase increases the likelihood that cells will be produced with multiple copies of the transposon, or in which the transposon is integrated at a favorable genomic location for high expression. Gene transfer systems comprising a transposon and a

corresponding transposase are thus particularly advantageous when the transposon comprises a selectable marker operably linked to a weak promoter. A weak promoter in this context is one which does not drive enough transcription of an operably linked selectable marker to permit target cell survival under restrictive conditions if only a single copy of the promoter and selectable marker are integrated into the target cell genome.

(73) A nucleic acid to be expressed as an RNA or protein and a selectable marker may be included on the same gene transfer polynucleotide, but operably linked to different promoters. In this case low expression levels of the selectable marker may be achieved by using a weakly active constitutive promoter such as the phosphoglycerokinase (PGK) promoter (such as a promoter selected from SEQ ID NOs: 794-800), the Herpes Simplex Virus thymidine kinase (HSV-TK) promoter (e.g. a sequence selected from SEQ ID NOs: 805-806), the MC1 promoter (for example SEQ ID NO: 807), a ubiquitin promoter (for example a sequence selected from SEQ ID NOs: 801-804). Other weakly active promoters may be deliberately constructed, for example a promoter attenuated by truncation, such as a truncated SV40 promoter (for example a sequence selected from SEQ ID NO: 808-809), or a promoter attenuated by insertion of a 5'UTR unfavorable for expression (for example a sequence selected from SEQ ID NOS: 811-812) between a promoter and the open reading frame encoding the selectable polypeptide. Particularly advantageous gene transfer polynucleotides comprise a promoter sequence selected from SEQ ID NOS: 794-810, operably linked to an open reading frame encoding a selectable marker.

(74) Expression levels of a selectable marker may also be advantageously reduced by other mechanisms such as the insertion of the SV40 small t antigen intron after the open reading frame for the selectable marker. The SV40 small t intron accepts aberrant 5' splice sites, which can lead to deletions within the preceding open reading frame in a fraction of the spliced mRNAs, thereby reducing expression of the selectable marker. Particularly advantageous gene transfer polynucleotides comprise intron SEQ ID NO: 873, operably linked to an open reading frame encoding a selectable marker. For this mechanism of attenuation to be effective, it is preferable for the open reading frame encoding the selectable marker to comprise an intron donor within its coding region. DNA sequences SEQ ID NOs: 529-532 are exemplary nucleic acid sequences that encode glutamine synthetase sequences with SEQ ID NOs: 524-527 respectively. Each of these nucleic acid sequences comprises an intron donor, and which may be operably linked to the SV40 small t antigen intron by placing the intron into the 3' UTR of the glutamine synthetase open reading frame. Sequence SEQ ID NO: 522 is an exemplary nucleic acid sequence encoding puromycin acetyl transferase SEQ ID NO: 521 which comprises an intron donor, and which may be operably linked to the SV40 small t antigen intron by placing the intron into the 3' UTR of the puromycin open reading frame. Advantageous gene transfer polynucleotides comprise a sequence at least 90% identical or at least 95% identical or at least 99% identical to, or 100% identical to a sequence selected from one of SEQ ID NOs: 529-532 or 522, in combination with SEQ ID NO: 873.

(75) Expression levels of a selectable marker may also be advantageously reduced by other mechanisms such as insertion of an inhibitory 5'-UTR within the transcript, for example SEQ ID NOs: 811-812. Particularly advantageous gene transfer polynucleotides comprise a promoter operably linked to an open reading frame encoding a selectable marker, wherein a sequence that is at least 90% identical or at least 95% identical or at least 99% identical to, or 100% identical to SEQ ID NO: 811-812 is interposed between the promoter and the selectable marker.

(76) Exemplary nucleic acid sequences comprising the glutamine synthetase coding sequence operably linked to regulatory sequences expressible in mammalian cells include SEQ ID NOs: 550-619 or 681-683. A gene transfer polynucleotide comprising a sequence selected from SEQ ID NOs: 550-619 or 681-683, upon integration into the genome of a target cell, expresses glutamine synthetase, thereby helping a cell to grow in the absence of added glutamine or in the presence of MSX. Regulatory elements in these sequences have been balanced to produce low levels of

expression of glutamine synthetase, providing a selective advantage for target cells whose genome comprises either multiple copies of the gene transfer polynucleotide, or for target calls whose genome comprises copies of the gene transfer polynucleotide in regions of the genome that are favorable for expression of encoded genes. Advantageous gene transfer polynucleotides comprise a sequence selected from SEQ ID NOs: 550-619 or 681-683, and they may further comprise a left transposon end and a right transposon end.

(77) Exemplary nucleic acid sequences comprising the blasticidin-S-transferase coding sequence operably linked to regulatory sequences expressible in mammalian cells include SEQ ID NOs: 620-626. A gene transfer polynucleotide comprising a sequence selected from SEQ ID NOs: 620-626, upon integration into the genome of a target cell, expresses blasticidin-S-transferase, thereby helping a cell to grow in the presence of added blasticidin. Regulatory elements in these sequences have been balanced to produce low levels of expression of blasticidin-S-transferase, providing a selective advantage for target cells whose genome comprises either multiple copies of the gene transfer polynucleotide, or for target calls whose genome comprises copies of the gene transfer polynucleotide in regions of the genome that are favorable for expression of encoded genes. Advantageous gene transfer polynucleotides comprise a sequence selected from SEQ ID NOs: 620-626, and they may further comprise a left transposon end and a right transposon end.

(78) Exemplary nucleic acid sequences comprising the hygromycin B phosphotransferase coding sequence operably linked to regulatory sequences expressible in mammalian cells include SEQ ID NOs: 627-628. A gene transfer polynucleotide comprising a sequence selected from SEQ ID NOs: 627-628, upon integration into the genome of a target cell, will express hygromycin B phosphotransferase, thereby helping a cell to grow in the presence of added hygromycin. Regulatory elements in these sequences have been balanced to produce low levels of expression of hygromycin B phosphotransferase, providing a selective advantage for target cells whose genome comprises either multiple copies of the gene transfer polynucleotide, or for target calls whose genome comprises copies of the gene transfer polynucleotide in regions of the genome that are favorable for expression of encoded genes. Advantageous gene transfer polynucleotides comprise a sequence selected from SEQ ID NOs: 627-628, and they may further comprise a left transposon end and a right transposon end.

(79) Exemplary nucleic acid sequences comprising the aminoglycoside 3'-phosphotransferase coding sequence operably linked to regulatory sequences expressible in mammalian cells include SEQ ID NOs: 629-631. A gene transfer polynucleotide comprising a sequence selected from SEQ ID NOs: 629-631, upon integration into the genome of a target cell, expresses aminoglycoside 3'-phosphotransferase, thereby helping a cell to grow in the presence of added neomycin. Regulatory elements in these sequences have been balanced to produce low levels of expression of aminoglycoside 3'-phosphotransferase, providing a selective advantage for target cells whose genome comprises either multiple copies of the gene transfer polynucleotide, or for target calls whose genome comprises copies of the gene transfer polynucleotide in regions of the genome that are favorable for expression of encoded genes. Advantageous gene transfer polynucleotides comprise a sequence selected from SEQ ID NOs: 629-631, and they may further comprise a left transposon end and a right transposon end.

(80) Exemplary nucleic acid sequences comprising the puromycin acetyltransferase coding sequence operably linked to regulatory sequences expressible in mammalian cells include SEQ ID NOs: 632-651 or 659-683. A gene transfer polynucleotide comprising a sequence selected from SEQ ID NOs: 632-651 or 659-683, upon integration into the genome of a target cell, expresses puromycin acetyltransferase, thereby helping a cell to grow in the presence of added puromycin. Regulatory elements in these sequences have been balanced to produce low levels of expression of puromycin acetyltransferase, providing a selective advantage for target cells whose genome comprises either multiple copies of the gene transfer polynucleotide, or for target calls whose genome comprises copies of the gene transfer polynucleotide in regions of the genome that are

favorable for expression of encoded genes. Advantageous gene transfer polynucleotides comprise a sequence selected from SEQ ID NOs: 632-651 or 659-683, and they may further comprise a left transposon end and a right transposon end.

(81) Exemplary nucleic acid sequences comprising the ble gene coding sequence operably linked to regulatory sequences expressible in mammalian cells include SEQ ID NOs: 652-656. A gene transfer polynucleotide comprising a sequence selected from SEQ ID NOs: 652-656, upon integration into the genome of a target cell, expresses the ble gene, thereby helping a cell to grow in the presence of added zeocin. Regulatory elements in these sequences have been balanced to produce low levels of expression of ble gene product, providing a selective advantage for target cells whose genome comprises either multiple copies of the gene transfer polynucleotide, or for target calls whose genome comprises copies of the gene transfer polynucleotide in regions of the genome that are favorable for expression of encoded genes. Advantageous gene transfer polynucleotides comprise a sequence selected from SEQ ID NOs: 652-656, and they may further comprise a left transposon end and a right transposon end.

(82) Exemplary nucleic acid sequences comprising the dihydrofolate reductase coding sequence operably linked to regulatory sequences expressible in mammalian cells include SEQ ID NOs: 533-549 or 657-680. A gene transfer polynucleotide comprising a sequence selected from SEQ ID NOs: 533-549 or 657-680, upon integration into the genome of a target cell, expresses dihydrofolate reductase, thereby helping a cell to grow in the absence of added hypoxanthine and thymidine or in the presence of MTX. Regulatory elements in these sequences have been balanced to produce low levels of expression of dihydrofolate reductase, providing a selective advantage for target cells whose genome comprises either multiple copies of the gene transfer polynucleotide, or for target calls whose genome comprises copies of the gene transfer polynucleotide in regions of the genome that are favorable for expression of encoded genes. Advantageous gene transfer polynucleotides comprise a sequence selected from SEQ ID NOs: 533-549 or 657-680, and they may further comprise a left transposon end and a right transposon end.

(83) The use of transposons and transposases in conjunction with weakly expressed selectable markers has several advantages over non-transposon constructs. One is that linkage between expression of the first polypeptide and the selectable marker is better for transposons, because a transposase integrates the entire sequence that lies between the two transposon ends into the genome. In contrast when heterologous DNA is introduced into the nucleus of a eukaryotic cell, for example a mammalian cell, it is gradually broken into random fragments which may either be integrated into the cell's genome, or degraded. Thus if a gene transfer polynucleotide comprising sequences that encode a first polypeptide and a selectable marker is introduced into a population of cells, some cells will integrate the sequences encoding the selectable marker but not those encoding the first polypeptide, and vice versa. Selection of cells expressing high levels of selectable marker is thus only somewhat correlated with cells that also express high levels of the first polypeptide. In contrast, because the transposase integrates all of the sequences between the transposon ends, cells expressing high levels of selectable marker are highly likely to also express high levels of the first polypeptide.

(84) A second advantage of transposons and transposases is that they are much more efficient at integrating DNA sequences into the genome. A much higher fraction of the cell population is therefore likely to integrate one or more copies of the gene transfer polynucleotide into their genomes, so there will be a correspondingly higher likelihood of good stable expression of both the selectable marker and the first polypeptide.

(85) A third advantage of piggy Bac-like transposons and transposases is that piggy Bac-like transposases are biased toward inserting their corresponding transposons into transcriptionally active chromatin. Each cell is therefore likely to integrate the gene transfer polynucleotide into a region of the genome from which genes are well expressed, so there will be a correspondingly higher likelihood of good stable expression of both the selectable marker and the first polypeptide.

(86) 5.2.5 Novel Piggybac-Like Transposases

(87) Natural DNA transposons undergo a ‘cut and paste’ system of replication in which the transposon is excised from a first DNA molecule and inserted into a second DNA molecule. DNA transposons are characterized by inverted terminal repeats (ITRs) and are mobilized by an element-encoded transposase. The piggy Bac transposon/transposase system is particularly useful because of the precision with which the transposon is integrated and excised (see for example “Fraser, M. J. (2001) The TTAA-Specific Family of Transposable Elements: Identification, Functional Characterization, and Utility for Transformation of Insects. Insect Transgenesis: Methods and Applications. A. M. Handler and A. A. James. Boca Raton, Fla., CRC Press: 249-268”; and “US20070204356 A1: PiggyBac constructs in vertebrates” and references therein).

(88) Many sequences with sequence similarity to the piggy Bac transposase from *Trichoplusia ni* have been found in the genomes of phylogenetically distinct species from fungi to mammals, but very few have been shown to possess transposase activity (see for example Wu M, et al (2011) *Genetica* 139:149-54. “Cloning and characterization of piggy Bac-like elements in lepidopteran insects”, and references therein).

(89) Two properties of transposases that are of particular interest for genomic modifications are their ability to integrate a polynucleotide into a target genome, and their ability to precisely excise a polynucleotide from a target genome. Both of these properties can be measured with a suitable system.

(90) A system for measuring the first step of transposition, which is excision of a transposon from a first polynucleotide, comprises the following components: (i) A first polynucleotide encoding a first selectable marker operably linked to sequences that cause it to be expressed in a selection host and (ii) A transposon comprising transposon ends recognized by a transposase. The transposon is present in, and interrupts the coding sequence of, the first selectable marker, such that the first selectable marker is not active. The transposon is placed in the first selectable marker such that precise excision of the first transposon causes the first selectable marker to be reconstituted. If an active transposase that can excise the first transposon is introduced into a host cell which comprises the first polynucleotide, the host cell will express the active first selectable marker. The activity of the transposase in excising the transposon can be measured as the frequency with which the host cells become able to grow under conditions that require the first selectable marker to be active.

(91) If the transposon comprises a second selectable marker, operably linked to sequences that make the second selectable marker expressible in the selection host, transposition of the second selectable marker into the genome of the host cell will yield a genome comprising active first and second selectable markers. The activity of the transposase in transposing the transposon into a second genomic location can be measured as the frequency with which the host cells become able to grow under conditions that require the first and second selectable markers to be active. In contrast, if the first selectable marker is present, but the second is not, then this indicates that the transposon was excised from the first polynucleotide, but was not subsequently transposed into a second polynucleotide. The selectable markers may, for example, be open reading frames encoding an antibiotic resistance protein, or an auxotrophic marker, or any other selectable marker.

(92) We used such a system to test putative transposase/transposon combinations for activity, as described in Section 6.1. We used computational methods to search publicly available sequenced genomes for open reading frames with homology to known active piggy Bac-like transposases. We selected transposase sequences that appeared to possess the DDDE motif characteristic of active piggyBac-like transposases and searched the DNA sequences flanking these putative transposases for inverted repeat sequences adjacent to a 5'-TTAA-3' target sequence. Amongst those that we identified were putative transposons with intact transposases from: *Spodoptera litura* (Genbank accession number MTZO01002002.1, protein accession number XP_022823959) with an open reading frame encoding a putative transposase with SEQ ID NO: 46 flanked by a putative left end with SEQ ID NO: 91 and a putative right end with SEQ ID NO: 92; *Pieris rapae* (NCBI genomic

reference sequence NW_019093607.1, Genbank protein accession number XP_022123753.1) with an open reading frame encoding a putative transposase with SEQ ID NO: 47 flanked by a putative left end with SEQ ID NO: 93 and a putative right end with SEQ ID NO: 94; *Myzus persicae* (NCBI genomic reference sequence NW_019100532.1, protein accession number XP_022166603) with an open reading frame encoding a putative transposase with SEQ ID NO: 48 flanked by a putative left end with SEQ ID NO: 95 and a putative right end with SEQ ID NO: 96; *Onthophagus taurus* (NCBI genomic reference sequence NW_019280463, protein accession number XP_022900752) with an open reading frame encoding a putative transposase with SEQ ID NO: 49 flanked by a putative left end with SEQ ID NO: 97 and a putative right end with SEQ ID NO: 98; *Temnothorax curvispinosus* (NCBI genomic reference sequence NW_020220783.1, protein accession number XP_024881886) with an open reading frame encoding a putative transposase with SEQ ID NO: 50 flanked by a putative left end with SEQ ID NO: 99 and a putative right end with SEQ ID NO: 100; *Agrius planipenn* (NCBI genomic reference sequence NW_020442437.1, protein accession number XP_025836109) with an open reading frame encoding a putative transposase with SEQ ID NO: 51 flanked by a putative left end with SEQ ID NO: 101 and a putative right end with SEQ ID NO: 102; *Parasteatoda tepidariorum* (NCBI genomic reference sequence NW_018371884.1, protein accession number XP_015905033) with an open reading frame encoding a putative transposase with SEQ ID NO: 52 flanked by a putative left end with SEQ ID NO: 103 and a putative right end with SEQ ID NO: 104; *Pectinophora gossypiella* (Genbank accession number GU270322.1, protein ID ADB45159.1, also described in Wang et al, 2010. Insect Mol. Biol. 19, 177-184. “piggyBac-like elements in the pink bollworm, *Pectinophora gossypiella*”) with an open reading frame encoding a putative transposase with SEQ ID NO: 53 flanked by a putative left end with SEQ ID NO: 105 and a putative right end with SEQ ID NO: 106; *Ctenoplusia agnata* (NCBI accession number GU477713.1, protein accession number ADV17598.1, also described by Wu M, et al (2011) Genetica 139:149-54. “Cloning and characterization of piggyBac-like elements in lepidopteran insects”) with an open reading frame encoding a putative transposase with SEQ ID NO: 54 flanked by a putative left end with SEQ ID NO: 107 and a putative right end with SEQ ID NO: 108; *Macrostomum lignano* (NCBI genomic reference sequence NIVC01003029.1, protein accession number PAA53757) with an open reading frame encoding a putative transposase with SEQ ID NO: 55 flanked by a putative left end with SEQ ID NO: 109 and a putative right end with SEQ ID NO: 110; *Orussus abietinus* (NCBI accession number XM_012421754, protein accession number XP_012277177) with an open reading frame encoding a putative transposase with SEQ ID NO: 56 flanked by a putative left end with SEQ ID NO: 111 and a putative right end with SEQ ID NO: 112; *Eufriesea mexicana* (NCBI genomic reference sequence NIVC01003029.1, protein accession number XP_017759329) with an open reading frame encoding a putative transposase with SEQ ID NO: 57 flanked by a putative left end with SEQ ID NO: 113 and a putative right end with SEQ ID NO: 114; *Spodoptera litura* (NCBI genomic reference sequence NC_036206.1, protein accession number XP_022824855) with an open reading frame encoding a putative transposase with SEQ ID NO: 58 flanked by a putative left end with SEQ ID NO: 115 and a putative right end with SEQ ID NO: 116; *Vanessa tameamea* (NCBI genomic reference sequence NW_020663261.1, protein accession number XP_026490968) with an open reading frame encoding a putative transposase with SEQ ID NO: 59 flanked by a putative left end with SEQ ID NO: 117 and a putative right end with SEQ ID NO: 118; *Blattella germanica* (NCBI genomic reference sequence PYGN01002011.1, protein accession number PSN31819) with an open reading frame encoding a putative transposase with SEQ ID NO: 60 flanked by a putative left end with SEQ ID NO: 119 and a putative right end with SEQ ID NO: 120; *Onthophagus taurus* (NCBI genomic reference sequence NW_019281532.1, protein accession number XP_022910826) with an open reading frame encoding a putative transposase with SEQ ID NO: 61 flanked by a putative left end with SEQ ID NO: 121 and a putative right end with SEQ ID NO: 122; *Onthophagus taurus* (NCBI genomic reference sequence NW_019281689.1, protein accession number XP_022911139)

with an open reading frame encoding a putative transposase with SEQ ID NO: 62 flanked by a putative left end with SEQ ID NO: 123 and a putative right end with SEQ ID NO: 124; *Onthophagus taurus* (NCBI genomic reference sequence NW_019286114.1, protein accession number XP_022913435) with an open reading frame encoding a putative transposase with SEQ ID NO: 63 flanked by a putative left end with SEQ ID NO: 125 and a putative right end with SEQ ID NO: 126; *Megachile rotundata* (NCBI genomic reference sequence NW_003797295, protein accession number XP_012145925) with an open reading frame encoding a putative transposase with SEQ ID NO: 64 flanked by a putative left end with SEQ ID NO: 127 and a putative right end with SEQ ID NO: 128; *Xiphophorus maculatus* (NCBI genomic reference sequence NC_036460.1, protein accession number XP_023207869) with an open reading frame encoding a putative transposase with SEQ ID NO: 65 flanked by a putative left end with SEQ ID NO: 129 and a putative right end with SEQ ID NO: 130; *Heliothis virescens* (NCBI accession number NWSH01000300.1, protein accession number PCG77621) with an open reading frame encoding a putative transposase with SEQ ID NO: 39 flanked by a putative left end with SEQ ID NO: 1 and a putative right end with SEQ ID NO: 3; *Agrotis ipsilon* (NCBI accession number GU477714.1, protein accession number ADV17599.1) with an open reading frame encoding a putative transposase with SEQ ID NO: 40 flanked by a putative left end with SEQ ID NO: 23 and a putative right end with SEQ ID NO: 24 and *Helicoverpa armigera* (NCBI accession number ABS18391.1, protein accession number ABS18391) with an open reading frame encoding a putative transposase with SEQ ID NO: 41 flanked by a putative left end with SEQ ID NO: 31 and a putative right end with SEQ ID NO: 32.

(93) 5.2.5.1 the *Heliothis* Transposase and its Corresponding Transposon

(94) One active transposase and its corresponding transposon identified by transposition activity in yeast was a *Heliothis* transposase, as described in Section 6.1.2. A *Heliothis* transposase comprises a polypeptide sequence that is at least 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% identical to the sequence given by SEQ ID NO: 39, and which is capable of transposing the transposon from transposase reporter construct SEQ ID NO: 66, as described in Section 6.1.2. Exemplary *Heliothis* transposases include sequences given as SEQ ID NOs: 131-465.

(95) A *Heliothis* transposase may be provided as a part of a gene transfer system as a protein, or as a polynucleotide encoding the *Heliothis* transposase, wherein the polynucleotide is expressible in the target cell. When provided as a polynucleotide, the *Heliothis* transposase may be provided as DNA or mRNA. If provided as DNA, the open reading frame encoding the *Heliothis* transposase is preferably operably linked to heterologous regulatory elements including a promoter that is active in the target cell such that the transposase is expressible in the target cell, for example a promoter that is active in a eukaryotic cell or a vertebrate cell or a mammalian cell. If provided as mRNA, the mRNA may be prepared in vitro from a DNA molecule in which the open reading frame encoding the *Heliothis* transposase is preferably operably linked to a heterologous promoter active in the invitro transcription system used to prepare the mRNA, for example a T7 promoter.

(96) A *Heliothis* transposon comprises a heterologous polynucleotide flanked by a left transposon end comprising a left ITR with sequence given by SEQ ID NO: 15 and a right transposon end comprising a right ITR with sequence given by SEQ ID NO: 16, and wherein the distal end of each ITR is immediately adjacent to a target sequence. Here and elsewhere when inverted repeats are defined by a sequence including a nucleotide defined by an ambiguity code, the identity of that nucleotide can be selected independently in the two repeats. A preferred target sequence is 5'-TTAA-3', although other useable target sequences may be used; preferably the target sequence on one side of the transposon is a direct repeat of the target sequence on the other side of the transposon. The left transposon end may further comprise additional sequences proximal to the ITR, for example a sequence at least 90% identical to, or 100% identical to SEQ ID NO: 13 or SEQ ID NO: 19. The right transposon end may further comprise additional sequences proximal to the ITR, for example a sequence at least 90% identical to, or 100% identical to SEQ ID NO: 14, or

SEQ ID NO: 20, or SEQ ID NO: 21, or SEQ ID NO: 22. The structure of a representative *Heliothis* transposon is shown in FIG. 1. A *Heliothis* transposon can be transposed by a transposase with a polypeptide sequence given by SEQ ID NO: 39, for example as encoded by a polynucleotide with sequence given by SEQ ID NO: 42 operably linked to a Gall promoter.

(97) Transposon ends, including ITRs and target sequences may be added to the ends of a heterologous polynucleotide sequence to create a synthetic *Heliothis* transposon which may be efficiently transposed into a target eukaryotic genome by an *Heliothis* transposase. For example, SEQ ID NO: 1 and SEQ ID NO: 2 each comprise a left 5'-TTAA-3' target sequence followed by a left transposon ITR followed by additional end sequences that may be added to one side of a heterologous polynucleotide, with the target sequence distal relative to the heterologous polynucleotide, to generate a synthetic *Heliothis* transposon. SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5 and SEQ ID NO: 6 each comprise additional end sequences followed by a right transposon ITR sequence followed by a right 5'-TTAA-3' target sequence that may be added to the other side of a heterologous polynucleotide, with the target sequence distal relative to the heterologous polynucleotide, to generate a synthetic *Heliothis* transposon. The preceding transposon end sequences comprise 5'-TTAA-3' as the target sequence, but SEQ ID NO: 7 and SEQ ID NO: 8 each comprise a left transposon ITR followed by additional end sequences that may be added to one side of a heterologous polynucleotide, with the ITR sequence distal relative to the heterologous polynucleotide, and SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11 and SEQ ID NO: 12 each comprise additional end sequences followed by a right transposon ITR sequence that may be added to the other side of a heterologous polynucleotide, with the target sequence distal relative to the heterologous polynucleotide, to generate a synthetic *Heliothis* transposon which may then be flanked by alternative target sequences.

(98) *Heliothis* transposases recognize synthetic *Heliothis* transposons. They excise the transposon from a first DNA molecule, by cutting the DNA at the target sequence at the left end of one transposon end and the target sequence at the right end of the second transposon end, re-join the cut ends of the first DNA molecule to leave a single copy of the target sequence. The excised transposon sequence, including any heterologous DNA that is between the transposon ends, is integrated by the transposase into a target sequence of a second DNA molecule, such as the genome of a target cell. A cell whose genome comprises a synthetic *Heliothis* transposon is an embodiment of the invention.

(99) 5.2.5.2 the *Heliothis* Transposase is Active in Mammalian Cells

(100) The looper moth piggyBac transposase has been shown to be active in a very wide variety of eukaryotic cells. In Section 6.1.2 we show that the *Heliothis* transposase can transpose its corresponding transposon into the genome of the yeast *Saccharomyces cerevisiae*. In Section 6.1.4 we show that the *Heliothis* transposase can transpose its corresponding transposon into the genome of a mammalian CHO cell. These results provide evidence that, like the other known active piggyBac-like transposases, the *Heliothis* transposase is also active in transposing its corresponding transposon into the genomes of most eukaryotic cells. Although the *Heliothis* transposase is active in a wide range of eukaryotic cells, the naturally occurring open reading frame encoding the *Heliothis* transposase (given as SEQ ID NO: 1181) is unlikely to express well in a similarly wide range of cells, as optimal codon usage differs significantly between cell types. It is therefore advantageous to use a sequence other than a naturally occurring sequence to encode the transposase, in other words, to use codon-preferences of the cell type in which expression is to be performed. Likewise, the promoter and other regulatory sequences are selected so as to be active in the cell type in which expression is to be performed. An advantageous polynucleotide for expression of a *Heliothis* transposase comprises at least 2, 5, 10, 20, 30, 40 or 50 synonymous codon differences relative to SEQ ID NO: 1181 at corresponding positions between the polynucleotide and SEQ ID NO: 1181, optionally wherein codons in the polynucleotide at the corresponding positions are selected for mammalian cell expression. The polynucleotide may be

DNA or mRNA. Exemplary polynucleotide sequences encoding *Heliothis* transposases with polypeptide sequences given by SEQ ID NOs: 39, 134 and 144, wherein codons have been selected for expression in mammalian cells, are given as SEQ ID NOs: 42, 1182 and 1183 respectively.

(101) 5.2.6 Cross-Reacting Piggybac-Like Transposases

(102) 5.2.6.1 the *Helicoverpa* Transposase and its Corresponding Transposon

(103) A putative piggyBac-like transposase and its corresponding transposon has been identified from *Helicoverpa armigera* (Sun et al. (2008). Insect Molecular Biology 17, 9-18 “piggyBac like elements in cotton bollworm, *Helicoverpa armigera* (Hübner)”). The authors state that the transposase “is currently being tested for function and the feasibility of its being reconstructed as a transformation vector”. However there have been no further reports in the past 10 years regarding these experiments, nor any filed patents, although positive identification of a novel active piggyBac-like transposase would be both publishable and patentable. We therefore infer that the authors have failed to identify an active *Helicoverpa* transposase. Nor do the authors suggest that the transposon may be useful in other eukaryotic cells including mammalian cells. Nor do the authors teach how a synthetic *Helicoverpa* transposon may be constructed or used.

(104) Here we show that a synthetic *Helicoverpa* transposon can be constructed by adding *Helicoverpa* transposon ends to either side of a heterologous polynucleotide. The synthetic *Helicoverpa* transposon can be transposed into the genome of a target eukaryotic cell, including cells other than an insect cell, by a *Helicoverpa* transposase.

(105) A *Helicoverpa* transposase comprises a polypeptide sequence that is at least 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% identical to the sequence given by SEQ ID NO: 41, and is capable of transposing the transposon from transposase reporter construct SEQ ID NO: 68, as described in Section 6.1.3. Although the *Helicoverpa* transposase is active in a wide range of eukaryotic cells, the naturally occurring open reading frame encoding the *Helicoverpa* transposase (given as SEQ ID NO: 1181) is unlikely to express well in a similarly wide range of cells, as optimal codon usage differs significantly between cell types. It is therefore advantageous to use a sequence other than a naturally occurring sequence to encode the transposase, in other words, to use codon-preferences of the cell type in which expression is to be performed. Likewise, the promoter and other regulatory sequences are selected so as to be active in the cell type in which expression is to be performed. An advantageous polynucleotide for expression of a *Helicoverpa* transposase comprises at least 2, 5, 10, 20, 30, 40 or 50 synonymous codon differences relative to SEQ ID NO: 1184 at corresponding positions between the polynucleotide and SEQ ID NO: 1184, optionally wherein codons in the polynucleotide at the corresponding positions are selected for mammalian cell expression. The polynucleotide may be DNA or mRNA. An exemplary polynucleotide sequence encoding a *Helicoverpa* transposase with polypeptide sequence given by SEQ ID NO: 41, wherein codons have been selected for expression in mammalian cells, is given as SEQ ID NO: 44.

(106) A *Helicoverpa* transposon comprises a heterologous polynucleotide flanked by a left transposon end comprising a left ITR with sequence given by SEQ ID NO: 37 and a right transposon end comprising a right ITR with sequence given by SEQ ID NO: 38, and wherein the distal end of each ITR is immediately adjacent to a target sequence. A preferred target sequence is 5'-TTAA-3', although other useable target sequences may be used; preferably the target sequence on one side of the transposon is a direct repeat of the target sequence on the other side of the transposon. The left transposon end may further comprise additional sequences proximal to the ITR, for example a sequence at least 90% identical to, or 100% identical to SEQ ID NO: 35. The right transposon end may further comprise additional sequences proximal to the ITR, for example a sequence at least 90% identical to, or 100% identical to SEQ ID NO: 36. A *Helicoverpa* transposon can be transposed by a transposase with a polypeptide sequence given by 41, for example as encoded by a polynucleotide with sequence given by SEQ ID NO: 44 operably linked to a Gall promoter.

(107) Transposon ends, including ITRs and target sequences may be added to the ends of a

heterologous polynucleotide sequence to create a synthetic *Helicoverpa* transposon which may be efficiently transposed into a target eukaryotic genome by an *Helicoverpa* transposase. For example, SEQ ID NO: 31 comprises a left target sequence followed by a left transposon end sequence that may be added to one side of a heterologous polynucleotide to generate a synthetic *Helicoverpa* transposon. SEQ ID NO: 32 comprises a right transposon end sequence followed by a right target sequence that may be added to the other side of a heterologous polynucleotide to generate a synthetic *Helicoverpa* transposon. Preferably a synthetic *Helicoverpa* transposon comprises a promoter that is active in a eukaryotic cell other than an insect cell. Preferably a synthetic *Helicoverpa* transposon comprises a promoter that is active in a mammalian cell.

(108) *Helicoverpa* transposases recognize synthetic *Helicoverpa* transposons. They excise the transposon from a first DNA molecule, by cutting the DNA at the target sequence at the left end of one transposon end and the target sequence at the right end of the second transposon end, re-join the cut ends of the first DNA molecule to leave a single copy of the target sequence. The excised transposon sequence, including any heterologous DNA that is between the transposon ends, is integrated by the transposase into a target sequence of a second DNA molecule, such as the genome of a target cell. A cell whose genome comprises a synthetic *Helicoverpa* transposon is an embodiment of the invention.

(109) 5.2.6.2 the *Agrotis* Transposase and its Corresponding Transposon

(110) Similar sequences to the putative piggy Bac-like transposase and its corresponding putative transposon from *Helicoverpa armigera* were identified from *Agrotis ypsilon* (Wu et. al. (2011). *Genetica* 139, 149-154. “Cloning and characterization of piggy Bac-like elements in lepidopteran insects”) by the same group that identified the putative piggyBac-like transposase and its corresponding putative transposon from *Helicoverpa armigera*. The authors state that “These elements are currently being tested for function, and preliminary results indicate that the transposases support transposition in cell culture, though at varying frequencies. (M. Wu, G. H. Luo, X. F. Wang and Z. J. Han, unpublished data). If HaPLE1.1, Ay-PLE1.1 and AaPLE1.1 are found to be transpositionally active, then comparative and functional assays should allow an assessment of domains critical to their mobility, in addition to the known “DDD” domains. This may allow the development of new vectors for insect transformation, and possibly improvements in the efficiency of IFP2 piggyBac by modifications to critical domains.” Although this text suggests initial encouraging results, there have been no further published reports in the past 8 years regarding these experiments, nor any filed patents, although positive identification of a novel active piggy Bac-like transposase would be both publishable and patentable. We therefore infer that the authors have failed to identify an active *Agrotis* transposase. Nor do the authors suggest that the transposon or transposase may be useful in eukaryotic cells other than insect cells. Nor do the authors teach how a synthetic *Agrotis* transposon may be constructed.

(111) Here we show that a synthetic *Agrotis* transposon can be constructed by adding *Agrotis* transposon ends to either side of a heterologous polynucleotide. The synthetic *Agrotis* transposon can be transposed into the genome of a target eukaryotic cell, including cells other than an insect cell, by an *Agrotis* transposase.

(112) An *Agrotis* transposase comprises a polypeptide sequence that is at least 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% identical to a sequence given by SEQ ID NO: 40, and is capable of transposing the transposon from transposase reporter construct SEQ ID NO: 67, as described in Section 6.1.3. Although the *Agrotis* transposase is active in a wide range of eukaryotic cells, the naturally occurring open reading frame encoding the *Agrotis* transposase (given as SEQ ID NO: 1185) is unlikely to express well in a similarly wide range of cells, as optimal codon usage differs significantly between cell types. It is therefore advantageous to use a sequence other than a naturally occurring sequence to encode the transposase, in other words, to use codon-preferences of the cell type in which expression is to be performed. Likewise, the promoter and other regulatory sequences are selected so as to be active in the cell type in which expression is to be performed. An

advantageous polynucleotide for expression of an *Agrotis* transposase comprises at least 2, 5, 10, 20, 30, 40 or 50 synonymous codon differences relative to SEQ ID NO: 1185 at corresponding positions between the polynucleotide and SEQ ID NO: 1185, optionally wherein codons in the polynucleotide at the corresponding positions are selected for mammalian cell expression. The polynucleotide may be DNA or mRNA. An exemplary polynucleotide sequence encoding an *Agrotis* transposase with polypeptide sequence given by SEQ ID NO: 40, wherein codons have been selected for expression in mammalian cells, is given as SEQ ID NO: 43.

(113) An *Agrotis* transposon comprises a heterologous polynucleotide flanked by a left transposon end comprising a left ITR with sequence given by SEQ ID NO: 29 and a right transposon end comprising a right ITR with sequence given by SEQ ID NO: 30, and wherein the distal end of each ITR is immediately adjacent to a target sequence. A preferred target sequence is 5'-TTAA-3', although other useable target sequences may be used; preferably the target sequence on one side of the transposon is a direct repeat of the target sequence on the other side of the transposon. The left transposon end may further comprise additional sequences proximal to the ITR, for example a sequence at least 90% identical to, or 100% identical to SEQ ID NO: 27. The right transposon end may further comprise additional sequences proximal to the ITR, for example a sequence at least 90% identical to, or 100% identical to SEQ ID NO: 28. An *Agrotis* transposon can be transposed by a transposase with a polypeptide sequence given by SEQ ID NO: 40, for example as encoded by a polynucleotide with sequence given by SEQ ID NO: 43 operably linked to a Gall promoter.

(114) Transposon ends, including ITRs and target sequences may be added to the ends of a heterologous polynucleotide sequence to create a synthetic *Agrotis* transposon which may be efficiently transposed into a target eukaryotic genome by an *Agrotis* transposase. For example, SEQ ID NO: 23 comprises a left target sequence followed by a left transposon end sequence that may be added to one side of a heterologous polynucleotide to generate a synthetic *Agrotis* transposon. SEQ ID NO: 24 comprises a right transposon end sequence followed by a right target sequence that may be added to the other side of a heterologous polynucleotide to generate a synthetic *Agrotis* transposon. Preferably a synthetic *Agrotis* transposon comprises a promoter that is active in a eukaryotic cell other than an insect cell. Preferably a synthetic *Agrotis* transposon comprises a promoter that is active in a mammalian cell.

(115) *Agrotis* transposases recognize synthetic *Agrotis* transposons. They excise the transposon from a first DNA molecule, by cutting the DNA at the target sequence at the left end of one transposon end and the target sequence at the right end of the second transposon end, re-join the cut ends of the first DNA molecule to leave a single copy of the target sequence. The excised transposon sequence, including any heterologous DNA that is between the transposon ends, is integrated by the transposase into a target sequence of a second DNA molecule, such as the genome of a target cell. A cell whose genome comprises a synthetic *Agrotis* transposon is an embodiment of the invention.

(116) 5.2.6.3 Cross-Reactivity of the *Agrotis* and *Helicoverpa* Transposases

(117) As described in Section 6.1.3, the *Agrotis* and *Helicoverpa* transposases are more than 92% identical. The ITRs for the corresponding transposons for these two transposases are identical, and the *Agrotis* transposase can transpose a *Helicoverpa* transposon and the *Helicoverpa* transposase can transpose an *Agrotis* transposon. The sequences adjacent to the ITRs differ between the two transposons: the sequences adjacent to the *Agrotis* left ITR (SEQ ID NO: 27) is about 90% identical with the sequences adjacent to the *Helicoverpa* left ITR (SEQ ID NO: 35), similarly the sequences adjacent to the *Agrotis* right ITR (SEQ ID NO: 28) is about 90% identical with the sequences adjacent to the *Helicoverpa* right ITR (SEQ ID NO: 36). As described in Section 6.1.3, these differences do not significantly affect the relative excision or transposition frequencies of the transposons with the different transposases. The ITRs thus appear to be the critical element for specificity of piggy Bac-like transposases.

(118) 5.2.7 Hyperactive *Heliiothis* Transposases

(119) Individual favorable mutations may be combined in a variety of different ways, for example by “DNA shuffling” or by methods described in U.S. Pat. No. 8,635,029 B2 and Liao et al (2007, BMC Biotechnology 2007, 7:16 doi: 10.1186/1472-6750-7-16 “Engineering proteinase K using machine learning and synthetic genes”). A transposase with modified activity, either for activity on a new target sequence, or increased activity on an existing target sequence may be obtained by using variations of the selection scheme described herein (for example Section 6.1.6) with an appropriate corresponding transposon.

(120) An alignment of known active piggyBac-like transposases may be used to identify amino acid changes likely to result in enhanced activity. Transposases are often deleterious to their hosts, so tend to accumulate mutations that inactivate them. However the mutations that accumulate in different transposases are different, as each occurs by random chance. A consensus sequence can be obtained from an alignment of sequences, and this can be used to improve activity (Ivics et al, 1997. Cell 91:501-510. “Molecular reconstruction of Sleeping Beauty, a Tc1-like transposon from fish, and its transposition in human cells.”). We aligned known active piggy Bac-like transposases using the CLUSTAL algorithm, and enumerated the amino acids found at each position. This diversity is shown in Table 1 relative to a *Heliothis* transposase (relative to SEQ ID NO: 39), the amino acids shown in column C are found in known active piggyBac-like transposases at the equivalent position in an alignment, and are thus likely to be acceptable changes in a *Heliothis* transposase. Column D shows amino acid changes found in known active piggy Bac-like transposases other than the *Heliothis* transposase at positions where there is good conservation within the rest of the transposase set, but the amino acid in the *Heliothis* transposase sequence is an outlier. Mutation of the position shown in column A to an amino acid shown in column D is particularly likely to result in enhanced transposase activity. Table 2 shows this diversity relative to a *Helicoverpa* transposase (relative to SEQ ID NO:41), the amino acids shown in column C are found in known active piggy Bac-like transposases at the equivalent position in an alignment, and are thus likely to be acceptable changes in a *Helicoverpa* transposase. Column D shows amino acid changes found in known active piggy Bac-like transposases other than the *Helicoverpa* transposase at positions where there is good conservation within the rest of the transposase set, but the amino acid in the *Helicoverpa* transposase sequence is an outlier. Mutation of the position shown in column A to an amino acid shown in column D is particularly likely to result in enhanced transposase activity. Table 3 shows this diversity relative to an *Agrotis* transposase (relative to SEQ ID NO: 40), the amino acids shown in column C are found in known active piggy Bac-like transposases at the equivalent position in an alignment, and are thus likely to be acceptable changes in a *Agrotis* transposase. Column D shows amino acid changes found in known active piggy Bac-like transposases other than the *Agrotis* transposase at positions where there is good conservation within the rest of the transposase set, but the amino acid in the *Agrotis* transposase sequence is an outlier. Mutation of the position shown in column A to an amino acid shown in column D is particularly likely to result in enhanced transposase activity.

(121) We selected amino acid substitutions to make in *Heliothis* transposase SEQ ID NO: 39 from column D in Table 1. Genes encoding *Heliothis* transposase variants comprising combinations of these substitutions were synthesized and tested for transposase activity as described in Section 6.1.7. As described in Section 6.1.7, we identified 27 *Heliothis* transposases with transposition activity over 10-fold greater than SEQ ID NO: 39 (sequences given by SEQ ID NOs: 131-157), 211 *Heliothis* transposases with transposition activity between 2 and 10-fold greater than SEQ ID NO:39 (sequences given by SEQ ID NOs: 158-368), and 33 *Heliothis* transposases with transposition activity less than 2-fold greater than SEQ ID NO: 39 (sequences given by SEQ ID NOs: 369-400). Exemplary sequences of hyperactive *Heliothis* transposases comprises a polypeptide with the sequence of one of these transposases.

(122) *Heliothis* transposases can thus be created that are not naturally occurring sequences, but that are at least 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 99.5% identical to SEQ ID NO: 39.

Such variants can retain partial activity of the transposase of SEQ ID NO: 39 (as determined by either or both of transposition and/or excision activity), can be functionally equivalent of the transposase of SEQ ID NO: 39 in either or both of transposition and excision, or can have enhanced activity relative to the transposase of SEQ ID NO: 39 in transposition, excision activity or both. Such variants can include mutations shown herein to increase transposition and/or excision, mutations shown herein to be neutral as to transposition and/or excision, and mutations detrimental to transposition and/or integration. Preferred variants include mutations shown to be neutral or to enhance transposition/and or excision. Some such variants lack mutations shown to be detrimental to transposition and/or excision. Some such variants include only mutations shown to enhance transposition, only mutations shown to enhance excision, or mutations shown to enhance both transposition and excision.

(123) Enhanced activity means activity (e.g., transposition or excision activity) that is greater beyond experimental error than that of a reference transposase from which a variant was derived. The activity can be greater by a factor of e.g., 1.1, 2, 5, 10, 15, 20, 50 or 100 fold of the reference transposase. The enhanced activity can lie within a range of for example 1.1-100 fold, 2-50 fold, 5-50 fold or 1.1-10 fold of the reference transposase. Here and elsewhere activities can be measured as demonstrated in the examples. Functional equivalence means a variant transposase can mediate transposition and/or excision of the same transposon with a comparable efficiency (within experimental error) to a reference transposase.

(124) Furthermore, variant sequences of SEQ ID NO: 39 can be created by combining two, three, four, or five or more substitutions selected from Table 1 column D. Combining beneficial substitutions, for example those shown in column D of Table 1 can result in hyperactive variants of SEQ ID NO: 39. We identified more than 220 *Heliothis* transposases with transposition or excision frequencies more than about 2-fold greater than that of naturally occurring *Heliothis* transposase SEQ ID NO: 39. These preferred hyperactive *Heliothis* transposases comprised one or more of the following substitutions (relative to SEQ ID NO: 39): S41V, L43S, V81E, S136V, K152L, L153F, D155T, T162I, I239L, H241N, S264R, T268D, V280P, T297C, L299F, M300R, M305N, L312I, Y421F, V450I, R485K, Y495F, L507I, H512N, Q603E, D83S, V85L, P125S, Q126S, Q131T, Q131R, E140C, E140A, N151Q, K169E, N212S, C316A, N322T, P351G, H357R, K360Q, E379P, K397S, A447N, A447D, A449V, A449S, K476L, V492A, 1500M, L585K and T595K. A preferred hyperactive *Heliothis* transposase comprises an amino acid substitution selected from this set. Some hyperactive *Heliothis* transposases may further comprise a heterologous nuclear localization sequence.

(125) We used machine learning methods as described in Liao et al (2007, BMC Biotechnology 2007, 7:16 doi: 10.1186/1472-6750-7-16 “Engineering proteinase K using machine learning and synthetic genes”) to determine the effect of various amino acid substitutions on the excision and transposition activities of a *Heliothis* transposase. Each substitution was empirically tested in a minimum of 5 different sequence contexts (i.e. in the presence of different other amino acid substitutions). As described in Liao et. al., the mean value for the regression weight for a substitution is a measure of the average effect of that substitution within multiple different *Heliothis* transposases. A substitution with a positive mean regression weight is one that on average has a positive effect on the transposition activity of *Heliothis* transposases. Addition of a substitution with a positive mean regression weight to an active *Heliothis* transposase that does not already comprise such a substitution is thus expected to improve the transposition activity of that *Heliothis* transposase. Tables 11 and 14 identify 56 substitutions within a *Heliothis* transposase with positive mean regression weights for transposition: S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, I149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, L507I, H512N, L585K, S589T,

T595K and Q603K.

(126) Preferred hyperactive *Heliothis* transposases comprise an amino acid sequence, other than a naturally occurring protein (e.g., not a transposase whose amino acid sequence comprises SEQ ID NO: 39), that is at least 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% identical to the amino acid sequence of any of SEQ ID NOs: 39 or 131-400 and which comprise a substitution at a position selected from amino acid 41, 43, 81, 83, 85, 125, 126, 131, 136, 140, 149, 151, 152, 153, 155, 162, 169, 212, 238, 239, 241, 264, 268, 280, 297, 299, 300, 305, 312, 316, 322, 357, 360, 396, 397, 421, 430, 447, 449, 450, 476, 485, 492, 495, 507, 512, 585, 589, 595, and 603, relative to SEQ ID NO: 39 and numbered according to SEQ ID NO: 39. Preferably the substitution is one shown in Table 1 columns C or D. Preferably the hyperactive *Heliothis* transposase comprises an amino acid substitution, relative to the sequence of SQ ID NO: 39, selected from S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, I149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, L507I, H512N, L585K, S589T, T595K and Q603K, or any combination of substitutions thereof including at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or all of these substitutions. The hyperactive *Heliothis* transposase may also comprise substitutions relative to SEQ ID NO: 39 at many other positions that are not associated with enhanced transposition, for example conservative substitutions that have a neutral effect on transposition.

(127) Methods of creating transgenic cells using naturally occurring or hyperactive *Heliothis* transposases are an aspect of the invention. A method of creating a transgenic cell comprises (i) introducing into a eukaryotic cell a naturally occurring or hyperactive *Heliothis* transposase (as a protein or as a polynucleotide encoding the transposase) and a corresponding *Heliothis* transposon. Creating the transgenic cell may further comprise (ii) identifying a eukaryotic cell whose genome comprises a *Heliothis* transposon. Identifying the cell in which an *Heliothis* transposon is incorporated into the genome of the eukaryotic cell may comprise selecting the eukaryotic cell for a selectable marker encoded on the *Heliothis* transposon. The selectable marker may be any selectable polypeptide, including any described herein. The eukaryotic cell into which the *Heliothis* transposon is introduced, and whose genome comprises a *Heliothis* transposon may be an animal cell, a fungal cell or a plant cell. The animal cell may be vertebrate or invertebrate. Preferred vertebrate cells include cells from mammals including rodents such as rats, mice, and hamsters; ungulates, such as cows, goats or sheep; swine and cells from primates including humans. Types of cells whose genomes may comprise a *Heliothis* transposon include hepatocytes, neural cells, muscle cells, blood cells, embryonic stem cells, somatic stem cells, hematopoietic cells, and immune cells including lymphocytes such as T cells, B cells and natural killer cells, T-helper cells, antigen-presenting cells, dendritic cells, neutrophils and macrophages. Preferred culture cells are Chinese hamster ovary (CHO) cells or Human embryonic kidney (HEK293) cells. Preferred fungal cells are yeast cells including *Saccharomyces cerevisiae* and *Pichia pastoris*.

(128) Activity of transposases may also be increased by fusion of nuclear localization signal (NLS) at the N-terminus, C-terminus, both at the N- and C-termini or internal regions of the transposase protein, as long as transposase activity is retained. A nuclear localization signal or sequence (NLS) is an amino acid sequence that 'tags' or facilitates interaction of a protein, either directly or indirectly with nuclear transport proteins for import into the cell nucleus. Nuclear localization signals (NLS) used can include consensus NLS sequences, viral NLS sequences, cellular NLS sequences, and combinations thereof.

(129) Transposases may also be fused to other protein functional domains. Such protein functional domains can include DNA binding domains, flexible hinge regions that can facilitate one or more domain fusions, and combinations thereof. Fusions can be made either to the N-terminus, C-terminus, or internal regions of the transposase protein so long as transposase activity is retained.

Fusions to DNA binding domains can be used to direct the *Heliothis* transposase or the *Agrotis* transposase or the *Helicoverpa* transposase to a specific genomic locus or loci. DNA binding domains may include a helix-turn-helix domain, a zinc-finger domain, a leucine zipper domain, a TALE (transcription activator-like effector) domain, a CRISPR-Cas protein or a helix-loop-helix domain. Specific DNA binding domains used can include a Gal4 DNA binding domain, a LexA DNA binding domain, or a Zif268 DNA binding domain. Flexible hinge regions used can include glycine/serine linkers and variants thereof.

5.3 Kits

(130) The present invention also features kits comprising a *Heliothis* transposase as a protein or encoded by a nucleic acid, and/or a *Heliothis* transposon; or a gene transfer system as described herein comprising a *Heliothis* transposase as a protein or encoded by a nucleic acid as described herein, in combination with a *Heliothis* transposon; optionally together with a pharmaceutically acceptable carrier, adjuvant or vehicle, and optionally with instructions for use. Any of the components of the inventive kit may be administered and/or transfected into cells in a subsequent order or in parallel, e.g. a *Heliothis* transposase protein or its encoding nucleic acid may be administered and/or transfected into a cell as defined above prior to, simultaneously with or subsequent to administration and/or transfection of a *Heliothis* transposon. Alternatively, a *Heliothis* transposon may be transfected into a cell as defined above prior to, simultaneously with or subsequent to transfection of a *Heliothis* transposase protein or its encoding nucleic acid. If transfected in parallel, preferably both components are provided in a separated formulation and/or mixed with each other directly prior to administration to avoid transposition prior to transfection. Additionally, administration and/or transfection of at least one component of the kit may occur in a time staggered mode, e.g. by administering multiple doses of this component.

(131) In addition, the present invention also features kits comprising a *Helicoverpa* transposase as a protein or encoded by a nucleic acid, and/or a *Helicoverpa* transposon; or a gene transfer system as described herein comprising a *Helicoverpa* transposase as a protein or encoded by a nucleic acid as described herein, in combination with a *Helicoverpa* transposon; optionally together with a pharmaceutically acceptable carrier, adjuvant or vehicle, and optionally with instructions for use. Any of the components of the inventive kit may be administered and/or transfected into cells in a subsequent order or in parallel, e.g. a *Helicoverpa* transposase protein or its encoding nucleic acid may be administered and/or transfected into a cell as defined above prior to, simultaneously with or subsequent to administration and/or transfection of a *Helicoverpa* transposon. Alternatively, a *Helicoverpa* transposon may be transfected into a cell as defined above prior to, simultaneously with or subsequent to transfection of a *Helicoverpa* transposase protein or its encoding nucleic acid. If transfected in parallel, preferably both components are provided in a separated formulation and/or mixed with each other directly prior to administration to avoid transposition prior to transfection. Additionally, administration and/or transfection of at least one component of the kit may occur in a time staggered mode, e.g. by administering multiple doses of this component.

(132) In addition, the present invention also features kits comprising a *Agrotis* transposase as a protein or encoded by a nucleic acid, and/or a *Agrotis* transposon; or a gene transfer system as described herein comprising a *Agrotis* transposase as a protein or encoded by a nucleic acid as described herein, in combination with a *Agrotis* transposon; optionally together with a pharmaceutically acceptable carrier, adjuvant or vehicle, and optionally with instructions for use. Any of the components of the inventive kit may be administered and/or transfected into cells in a subsequent order or in parallel, e.g. a *Agrotis* transposase protein or its encoding nucleic acid may be administered and/or transfected into a cell as defined above prior to, simultaneously with or subsequent to administration and/or transfection of a *Agrotis* transposon. Alternatively, a *Agrotis* transposon may be transfected into a cell as defined above prior to, simultaneously with or subsequent to transfection of a *Agrotis* transposase protein or its encoding nucleic acid. If transfected in parallel, preferably both components are provided in a separated formulation and/or

mixed with each other directly prior to administration the to avoid transposition prior to transfection. Additionally, administration and/or transfection of at least one component of the kit may occur in a time staggered mode, e.g. by administering multiple doses of this component.

6. EXAMPLES

(133) The following examples illustrate the methods, compositions and kits disclosed herein and should not be construed as limiting in any way. Various equivalents will be apparent from the following examples; such equivalents are also contemplated to be part of the invention disclosed herein.

(134) 6.1 New Transposases

(135) 6.1.1 Measuring Transposase Activity

(136) As described in Section 5.2.5, transposition frequencies for active transposases may be measured using a system in which a transposon interrupts a selectable marker. Transposase reporter polynucleotides were constructed in which the open reading frame of the yeast *Saccharomyces cerevisiae* URA3 open reading frame was interrupted by a yeast TRP 1 open reading frame operably linked to a promoter and terminator such that it was expressible in the yeast *Saccharomyces cerevisiae*. The TRP1 gene was flanked by putative transposon ends with 5'-TTAA-3' target sites, such that excision of the putative transposon would leave a single copy of the 5'-TTAA-3' target site and exactly reconstitute the URA3 open reading frame. A yeast transposase reporter strain was constructed by integrating the transposase reporter polynucleotide into the URA3 gene of a haploid yeast strain auxotrophic for LEU2 and TRP1, such that the strain became LEU2⁻, URA3⁻ and TRP1⁺.

(137) Transposases were tested for their ability to transposase the TRP1 gene-containing transposons from within the URA3 open reading frame. Each open reading frame encoding a putative transposase was cloned into a *Saccharomyces cerevisiae* expression vector comprising a 2 micron origin of replication and a LEU2 gene expressible in *Saccharomyces*. Each transposase open reading frame was operably linked to a Gal promoter. Each cloned transposase open reading frame was transformed into a yeast transposase reporter strain and plated on minimal media lacking leucine. After 2 days, all LEU⁺ colonies were harvested by scraping the plates. The Gal promoter was induced by growing in galactose for 4 hours, and cells were then plated onto 3 different plates: plates lacking only leucine, plates lacking leucine and uracil, and plates lacking leucine, uracil and tryptophan. These plates were incubated for 2-4 days, and the colonies on each plate were counted, measuring the number of live cells, the number of transposon excision events and the number of transposon excision and re-integration (i.e. transposition events) respectively.

(138) 6.1.2 Identification of a New Piggybac-Like Transposase

(139) As described in Section 5.2.5, thirteen putative piggyBac-like transposases were identified from Genbank as being at least 20% identical to the piggyBac transposase from *Trichoplusia ni*. The flanking gene sequences were analyzed for the presence of inverted repeat sequences including the 5'-TTAA-3' target sequence characteristic of piggyBac transposition. Putative left and right transposon end sequences were taken from these flanking sequences. These transposon ends were incorporated into transposase reporter constructs configured as described in Section 6.1.1 and integrated into the genome of *Saccharomyces cerevisiae* thereby generating transposase reporter strains. The corresponding transposase sequence for each reporter strain was back-translated, cloned into a *Saccharomyces cerevisiae* expression vector and transformed into the reporter strain. Transposase activities were measured as described in Section 6.1.1.

(140) The following twenty combinations showed no excision or transposition: reporter construct SEQ ID NO: 71 (comprising putative left transposon end SEQ ID NO: 91, and putative right transposon end SEQ ID NO: 92) with transposase SEQ ID NO: 46, reporter construct SEQ ID NO: 72 (comprising putative left transposon end SEQ ID NO: 93, and putative right transposon end SEQ ID NO: 94) with transposase SEQ ID NO: 47, reporter construct SEQ ID NO: 73 (comprising putative left transposon end SEQ ID NO: 95, and putative right transposon end SEQ ID NO: 96)

with transposase SEQ ID NO: 48, reporter construct SEQ ID NO: 74 (comprising putative left transposon end SEQ ID NO: 97, and putative right transposon end SEQ ID NO: 98) with transposase SEQ ID NO: 49, reporter construct SEQ ID NO: 75 (comprising putative left transposon end SEQ ID NO: 99, and putative right transposon end SEQ ID NO: 100) with transposase SEQ ID NO: 50, reporter construct SEQ ID NO: 76 (comprising putative left transposon end SEQ ID NO: 101, and putative right transposon end SEQ ID NO: 102) with transposase SEQ ID NO: 51, reporter construct SEQ ID NO: 77 (comprising putative left transposon end SEQ ID NO: 103, and putative right transposon end SEQ ID NO: 104) with transposase SEQ ID NO: 52, reporter construct SEQ ID NO: 78 (comprising putative left transposon end SEQ ID NO: 105, and putative right transposon end SEQ ID NO: 106) with transposase SEQ ID NO: 53, reporter construct SEQ ID NO: 79 (comprising putative left transposon end SEQ ID NO: 107, and putative right transposon end SEQ ID NO: 108) with transposase SEQ ID NO: 54, reporter construct SEQ ID NO: 80 (comprising putative left transposon end SEQ ID NO: 109, and putative right transposon end SEQ ID NO: 110) with transposase SEQ ID NO: 55, reporter construct SEQ ID NO: 81 (comprising putative left transposon end SEQ ID NO: 111, and putative right transposon end SEQ ID NO: 112) with transposase SEQ ID NO: 56, reporter construct SEQ ID NO: 82 (comprising putative left transposon end SEQ ID NO: 113, and putative right transposon end SEQ ID NO: 114) with transposase SEQ ID NO: 57, reporter construct SEQ ID NO: 83 (comprising putative left transposon end SEQ ID NO: 115, and putative right transposon end SEQ ID NO: 116) with transposase SEQ ID NO: 58, reporter construct SEQ ID NO: 84 (comprising putative left transposon end SEQ ID NO: 117, and putative right transposon end SEQ ID NO: 118) with transposase SEQ ID NO: 59, reporter construct SEQ ID NO: 85 (comprising putative left transposon end SEQ ID NO: 119, and putative right transposon end SEQ ID NO: 120) with transposase SEQ ID NO: 60, reporter construct SEQ ID NO: 86 (comprising putative left transposon end SEQ ID NO: 121, and putative right transposon end SEQ ID NO: 122) with transposase SEQ ID NO: 61, reporter construct SEQ ID NO: 87 (comprising putative left transposon end SEQ ID NO: 123, and putative right transposon end SEQ ID NO: 124) with transposase SEQ ID NO: 62, reporter construct SEQ ID NO: 88 (comprising putative left transposon end SEQ ID NO: 125, and putative right transposon end SEQ ID NO: 126) with transposase SEQ ID NO: 63, reporter construct SEQ ID NO: 89 (comprising putative left transposon end SEQ ID NO: 127, and putative right transposon end SEQ ID NO: 128) with transposase SEQ ID NO: 64, reporter construct SEQ ID NO: 90 (comprising putative left transposon end SEQ ID NO: 129, and putative right transposon end SEQ ID NO: 130) with transposase SEQ ID NO: 65. This is consistent with reports in the literature that while computational recognition of sequences that are homologous to the piggyBac transposase from *Trichoplusia ni* is straightforward, most of these sequences are transpositionally inactive, even when they appear to have intact terminal repeats and the transposases appear to comprise the DDDE motif found in active piggy Bac-like transposases. It is therefore necessary to measure excision and transposition activity, in order to identify novel active piggyBac-like transposases and transposons.

(141) One transposase that showed good activity in excising its corresponding transposon from the reporter construct (shown by the appearance of URA⁺ colonies) and transposing the TRP gene in the transposon into another genomic location in the *Saccharomyces cerevisiae* reporter strain was transposase SEQ ID NO: 39. Transposase SEQ ID NO: 39 was able to transpose the transposon from reporter construct SEQ ID NO: 66. This is shown in Table 4: the number of excision events, measured by the appearance of URA⁺ colonies, is shown in column G; the number of full transposition events, measured by the appearance of URA⁺ TRP⁺ colonies, is shown in column H.

(142) 6.1.3 Identification of a Pair of Cross-Reacting Piggybac-Like Transposases

(143) Two putative piggyBac-like transposases, one from a *Helicoverpa* species and another from

an *Agrotis* species were identified from Genbank as being at least 20% identical to the piggy Bac transposase from *Trichoplusia ni*. The flanking gene sequences were analyzed for the presence of inverted repeat sequences including the 5'-TTAA-3' target sequence characteristic of piggyBac transposition. Putative left and right transposon end sequences were taken from these flanking sequences. These transposon ends were incorporated into transposase reporter constructs configured as described in Section 6.1.1 and integrated into the genome of *Saccharomyces cerevisiae* thereby generating transposase reporter strains.

(144) Unlike other known piggy Bac-like transposons, the putative transposons from *Helicoverpa* and *Agrotis* had identical 16 bp ITR sequences given by SEQ ID NO: 37 (5'-CCCTAGAAGCCCAATC-3') immediately adjacent to a 5'-TTAA-3' target sequence at each end of the putative transposon. We hypothesized that if these ITR sequences are what a corresponding transposase recognizes, the *Agrotis* transposase would be able to transpose the *Helicoverpa* transposon, and vice versa. The *Agrotis* and *Helicoverpa* transposases were back-translated, cloned into a *Saccharomyces cerevisiae* expression vector and each was transformed separately into the *Agrotis* and *Helicoverpa* reporter strains. Activities for each transposase in each reporter strain were measured as described in Section 6.1.1.

(145) Table 5 shows the number of excision events (measured by the appearance of URA⁺ colonies, shown in column G) and the number of full transposition events (measured by the appearance of URA⁺ TRP⁺ colonies, shown in column H). The relative excision frequency, determined by dividing the number of live cells (colonies on leu plates) by the number of URA⁺ colonies, is shown in column I. The relative transposition frequency, determined by dividing the number of live cells (colonies on leu plates) by the number of URA⁺ colonies, is shown in column J.

(146) The two transposases showed comparable activity in excising their corresponding transposons from the reporter construct (shown by the appearance of URA⁺ colonies) and transposing the TRP gene in their corresponding transposons into another genomic location in the *Saccharomyces cerevisiae* reporter strain (shown by the appearance of URA⁺ TRP⁺ colonies). These activities were also very comparable with the activities of the *Agrotis* transposase in excising or transposing the *Helicoverpa* transposon (compare rows 1 and 4 for the activity of the *Agrotis* transposase on the *Helicoverpa* (row 4) and *Agrotis* (row 1) transposon), and of the *Helicoverpa* transposase in excising or transposing the *Agrotis* transposon (compare rows 2 and 3 for the activity of the *Helicoverpa* transposase on the *Helicoverpa* (row 3) and *Agrotis* (row 2) transposon). The relative excision and relative transposition frequencies for the two transposases acting on their corresponding transposons differed by less than 2-fold from their activities on the heterologous transposon.

(147) The *Agrotis* and *Helicoverpa* transposase sequences (SEQ ID NOs 40 and 41 respectively) differ by 45 amino acids over their total length of 599 amino acids, so they are 92.48% identical. The sequences between the transposon ITRs and the open reading frame encoding the transposase in the naturally occurring transposon are slightly shorter in the *Agrotis* transposon than the *Helicoverpa* transposon. Over the length of the 303 bp of the left sequence (i.e. to the 5' of the transposase when considering the direction of translation) but excluding the ITR, there are 24 differences between the *Agrotis* sequence (SEQ ID NO: 27) and the *Helicoverpa* sequence (SEQ ID NO: 35), so these two left end sequences are 92.1% identical. Over the length of the 347 bp of the right sequence but excluding the ITRs, there are 37 differences between the *Agrotis* sequence (SEQ ID NO: 28) and the *Helicoverpa* sequence (SEQ ID NO: 36), so the two right end sequences are 89.3% identical. This suggests the degree of identity between transposons and transposases that may be required for the transposase to act on a heterologous transposon. If a first and second transposon share identical ITR sequences, and the first transposon left transposon end is more than 93% identical with the second transposon left transposon end, and the first transposon right transposon end is more than 93% identical with the second transposon right transposon end, if their

corresponding transposases are also at least 93% identical, then the first transposase may be expected to transpose the second transposon and the second transposase may be expected to transpose the first transposon.

(148) 6.1.4 the *Heliothis* Transposase is Active in Mammalian Cells

(149) PiggyBac-like transposases can transpose their corresponding transposons into the genomes of eukaryotic cells including yeast cells such as *Pichia pastoris* and *Saccharomyces cerevisiae*, and mammalian cells such as human embryonic kidney (HEK) and Chinese hamster ovary (CHO) cells. To determine the activity of the *Heliothis* transposase in mammalian cells, we constructed a gene transfer polynucleotide comprising transposon ends, and further comprising a selectable marker encoding glutamine synthetase with a polypeptide sequence given by SEQ ID NO: 527, encoded by DNA sequence given by SEQ ID NO: 532 and operably linked to regulatory elements that give weak glutamine synthetase expression, the sequence of the glutamine synthetase and its associated regulatory elements given by SEQ ID NO: 570. The gene transfer polynucleotides further comprised open reading frames encoding the heavy and light chains of an antibody, each operably linked to a promoter and polyadenylation signal sequence. The gene transfer polynucleotide (with SEQ ID NO: 507) comprised a left transposon end comprising a 5'-TTAA-3' target integration sequence immediately followed by a *Heliothis* left transposon end with ITR sequence given by SEQ ID NO: 17, which is an embodiment of SEQ ID NO: 15. The gene transfer polynucleotide further comprised a *Heliothis* right transposon end with ITR sequence given by SEQ ID NO: 18, which is an embodiment of SEQ ID NO: 16. The two *Heliothis* transposon ends were placed on either side of the heterologous polynucleotide comprising the glutamine synthetase selectable marker and the open reading frames encoding the heavy and light chains of the antibody. The left transposon end further comprised a sequence given by SEQ ID NO: 13 immediately adjacent to the left ITR. The right transposon end further comprised a sequence given by SEQ ID NO: 14 immediately adjacent to the right ITR.

(150) Gene transfer polynucleotides were transfected into CHO cells which lacked a functional glutamine synthetase gene. Cells were transfected by electroporation: 25 µg of gene transfer polynucleotide DNA with SEQ ID NO: 507 was co-transfected with 3 µg of DNA comprising an open reading frame encoding *Heliothis* transposase SEQ ID NO: 39 operably linked to a human CMV promoter and a polyadenylation signal sequence. The cells were incubated in media containing 4 mM glutamine for 48 hours following electroporation, and subsequently diluted to 300,000 cells per ml in media lacking glutamine. Cells were exchanged into fresh glutamine-free media every 5 days. The viability of the cells from each transfection were measured at various times following transfection using a Beckman-Coulter Vi-Cell. The total number of viable cells were also measured with the same instrument. The results are shown in Table 6.

(151) As shown in Table 6, the viability of cells transfected with the gene transfer polynucleotide but no transposase fell to about 27% by 12 days post-transfection (column B). The total number of live cells fell to fewer than 50,000 per ml within 7 days (column C). At or below this density of live cells, viability measurements become inaccurate. The culture never recovered. In contrast when the gene transfer polynucleotide was co-transfected with the *Heliothis* transposase, cells recovered to greater than 90% viability within 21 days (Table 6 column D), by which time the density of live cells exceeded 1 million per ml (Table 6 column E). This shows that a gene transfer polynucleotide comprising a left and right *Heliothis* transposon end can be efficiently transposed into the genome of a mammalian target cell by a corresponding *Heliothis* transposase.

(152) The recovered pools of CHO cells comprising piggy Bac-like transposons integrated into their genomes were grown in a 14 day fed-batch using Sigma Advanced Fed Batch media. Antibody titers were measured in culture supernatant using an Octet. Table 7 shows the titers measured at 7, 10, 12 and 14 days of the fed batch culture. The titer of antibody from cells comprising gene transfer polynucleotide with SEQ ID NO: 507, that had been integrated by co-transfection with the *Heliothis* transposase SEQ ID NO: 39 reached approximately 2 g/L after 14

days. This shows that the *Heliothis* transposon and its corresponding transposase, as described in Section 5.2.5, is a novel, piggyBac-like transposon/transposase system that is active in mammalian cells.

(153) 6.1.5 Messenger RNA Encoding the *Heliothis* Transposase is Active in Mammalian Cells

(154) We further tested gene transfer polynucleotide with SEQ ID NO: 507, whose configuration is described in Section 6.1.4, to determine whether the synthetic *Heliothis* transposon could be integrated into the genome of a mammalian cell if the corresponding transposase was provided in the form of mRNA.

(155) mRNA encoding transposases was prepared by in vitro transcription using T7 RNA polymerase. The mRNA comprised a 5' sequence SEQ ID NO: 508 preceding the sequence encoding the open reading frame, and a 3' sequence SEQ ID NO: 509 following the stop codon at the end of the open reading frame. The mRNA had an anti-reverse cap analog (3'-O-Me-m7G (5')ppp(5') G. DNA molecules comprising a sequence encoding a transposase operably linked to a heterologous promoter that is active in vitro are useful for the preparation of transposase mRNA. Isolated mRNA molecules comprising a sequence encoding a transposase are useful for integration of a corresponding transposon into a target genome.

(156) Gene transfer polynucleotide SEQ ID NO: 507 was transfected into CHO cells which lacked a functional glutamine synthetase gene. Cells were transfected by electroporation: 25 µg of gene transfer polynucleotide DNA was co-transfected with 3 µg of mRNA comprising an open reading frame encoding a corresponding transposase (amino acid sequence SEQ ID NO: 39, nucleotide sequence SEQ ID NO: 42). The cells were incubated in media containing 4 mM glutamine for 48 hours following electroporation, and subsequently diluted to 300,000 cells per ml in media lacking glutamine. Cells were exchanged into fresh glutamine-free media every 5 days. The viability of the cells from each transfection were measured at various times following transfection using a Beckman-Coulter Vi-Cell. The total number of viable cells were also measured with the same instrument. The results are shown in Table 8.

(157) When gene transfer polynucleotide with SEQ ID NO: 507 was co-transfected with mRNA encoding *Heliothis* transposase SEQ ID NO: 39, viability fell to around 33% by 14 days post-transfection (Table 8 column B), by which time the density of live cells was around 30,000 per ml (Table 8 column C). Cell viability and the density of live cells then increased until by 30 days post-transfection viability was above 95% and there were over 2 million live cells per ml. This shows that a gene transfer polynucleotide comprising a left and right *Heliothis* transposon end can be efficiently transposed into the genome of a mammalian target cell when co-transfected with mRNA encoding a corresponding *Heliothis* transposase.

(158) 6.1.6 *Heliothis* Transposon End Sequences Active in Mammalian Cells

(159) When we originally tested the *Heliothis* transposon, we used the entire sequence between the 5'-TTAA-3' target sequences and the transposase open reading frame as transposon ends. We have found that for other piggy Bac-like sequences this full sequence is generally not required for transposition activity. We therefore constructed synthetic *Heliothis* transposons with truncated ends to determine whether these were transposable by a *Heliothis* transposase. A heterologous polynucleotide with SEQ ID NO: 70 encoded glutamine synthetase with a polypeptide sequence given by SEQ ID NO: 528, operably linked to regulatory elements that give weak glutamine synthetase expression as a selectable marker. On one side of the heterologous polynucleotide was a left *Heliothis* transposon end comprising a 5'-TTAA-3' integration target sequence immediately followed by a transposon ITR sequence with SEQ ID NO: 17 (which is an embodiment of SEQ ID NO: 15). On the other side of the heterologous polynucleotide was a right *Heliothis* transposon end comprising a transposon ITR sequence with SEQ ID NO: 18 (which is an embodiment of SEQ ID NO: 16) immediately followed by a 5'-TTAA-3' integration target sequence. The transposon further comprised an additional sequence selected from SEQ ID NOs: 13 or 19 immediately adjacent to the left transposon ITR sequence. The transposon further comprised an additional sequence selected

from SEQ ID NOs: 14, or 20-22 immediately adjacent to the right transposon ITR sequence. Transposons were transfected into CHO cells which lacked a functional glutamine synthetase gene. Cells were transfected by electroporation: 25 µg of gene transfer polynucleotide DNA were transfected, optionally the cells were co-transfected with 3 µg of a second plasmid comprising an open reading frame encoding a corresponding transposase (amino acid sequence SEQ ID NO: 144) operably linked to a CMV promoter and expressible in a mammalian cell. The cells were incubated in media containing 4 mM glutamine for 48 hours following electroporation, and subsequently diluted to 300,000 cells per ml in media lacking glutamine. Cells were exchanged into fresh glutamine-free media every 5 days. The viability of the cells from each transfection were measured at various times following transfection using a Beckman-Coulter Vi-Cell. The total number of viable cells were also measured with the same instrument. The results are shown in Table 9. (160) Table 9 columns A and B show the reduction in cell viability and viable cell density when cells were transfected with a transposon comprising full length transposon ends in the absence of transposase. Cell viability and viable cell density can both be seen to fall throughout the experiment. In contrast when any the same transposon was co-transfected with DNA encoding a *Heliothis* transposase, the cell viability and viable cell density fell initially, but had begun to recover by day 12 and was fully recovered between day 17 and 21 (Table 9 columns C and D). A comparable result was obtained when the left transposon end was truncated from the sequence given by SEQ ID NO: 13, to the sequence given by SEQ ID NO: 19 (compare Table 9 columns E and F with columns G and H respectively). A comparable result was also obtained when the right transposon end was truncated from the sequence given by SEQ ID NO: 14, to the sequence given by SEQ ID NO: 20 (compare Table 9 columns I and J with columns K and L respectively); or to the sequence given by SEQ ID NO: 21 (compare Table 9 columns M and N with columns O and P respectively); or to the sequence given by SEQ ID NO: 22 (compare Table 9 columns Q and R with columns S and T respectively). This shows that in addition to an integration target sequence immediately adjacent to a transposon ITR sequence with SEQ ID NO: 15, a *Heliothis* synthetic transposon left transposon end may further comprise an additional sequence selected from SEQ ID NOs: 13 and 19 immediately adjacent to the left transposon ITR sequence; and a *Heliothis* synthetic transposon right transposon end may comprise an additional sequence selected from SEQ ID NOs: 14 or 20-22, immediately adjacent to an ITR with sequence SEQ ID NO: 16, immediately adjacent to an integration target sequence.

(161) 6.1.7 Identifying Hyperactive *Heliothis* Transposases

(162) To identify *Heliothis* transposase mutations that led to either increased transposition activity, or increased excision activity, relative to the naturally occurring *Heliothis* transposase sequence given by SEQ ID NO: 39, we analyzed a CLUSTAL alignment of active piggy Bac-like transposases. Table 1 column C shows the amino acids found in active piggyBac-like transposases relative to each position in the *Heliothis* transposase (position shown in Table 1 column A). The amino acid present in *Heliothis* transposase given by SEQ ID NO: 39 is shown in column B of Table 1. Because transposases are often deleterious to their hosts, they tend to accumulate mutations that inactivate them. The mutations that accumulate in different transposases are different, as each occurs by random chance. A consensus sequence can therefore be used to approximate an ancestral sequence that pre-dates the accumulation of deleterious mutations. It is difficult to accurately calculate an ancestral sequence from a small number of extant sequences, so we chose to focus on positions where active transposases were more highly conserved, and where the consensus amino acid(s) differed from the one in the *Heliothis* transposase. We considered that mutating these amino acids to the consensus amino acids found in other active transposases would be likely to increase the activity of the *Heliothis* transposase. These candidate beneficial amino acid substitutions are shown in Table 1 column D.

(163) 6.1.7.1 First Set of *Heliothis* Transposase Variants

(164) A set of 95 open reading frames encoding variant *Heliothis* transposases comprised one or

more substitutions selected from S41V, L43S, Q74S, V81E, V81P, V135G, S136V, S136P, N137K, N137T, I149C, W150F, K152L, L153F, D155T, T162I, L188T, K233R, K233L, V238F, I239L, H241N, H241R, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M303I, M305N, L312I, L314F, L314I, R334K, I378Y, K385L, K396R, Y419F, Y421F, L428V, H430Y, V450I, I459M, R485K, R485N, Y495F, D499N, S501A, L507I, H512N, L552I, Q586D, Q586R, S589T, V600I, Q603K, Q603E and C604H. Each substitution was represented at least 5 times within the set of 96 variants, and the number of different pairwise combinations of substitutions was maximized so that each substitution was tested in as many different sequence contexts as possible. Each variant open reading frame was cloned into a vector comprising a leucine selectable marker; each open reading frame encoding a transposase variant was operably linked to the *Saccharomyces cerevisiae* Gal-1 promoter. Each of these variants was then individually transformed into a *Saccharomyces cerevisiae* strain comprising a chromosomally integrated copy of SEQ ID NO: 66, as described above. After 48 hours cells were scraped from the plate into minimal media lacking leucine and with galactose as the carbon source. The A600 for each culture was adjusted to 2. Cultures were grown for 4 hours in galactose to induce expression of the transposases, then a 100×-diluted aliquot was plated on media lacking leucine, uracil and tryptophan (to count transposition), a 100×-diluted aliquot was plated on media lacking leucine and uracil (to count excision) and a 25,000×-diluted aliquot was plated on media lacking leucine (to count total live cells). Two days later, colonies were counted to determine transposition (=number of cells on -leu -ura -trp media divided by (250× number of cells on -leu media)) and excision (=number of cells on-leu-ura media divided by (250×number of cells on -leu media)) frequencies. The results are shown in Table 10. We identified 2 *Heliothis* transposase variants (with sequences given by SEQ ID NO: 158 and 159) with excision or transposition activities that were between 2 and 4-fold higher than the activities measured for the naturally occurring *Heliothis* transposase. We also identified 8 *Heliothis* transposase variants (with sequences given by SEQ ID NO: 369-376) with excision or transposition activities that were improved by less than 2-fold relative to the naturally occurring *Heliothis* transposase. We also identified 45 *Heliothis* transposase variants (with sequences given by SEQ ID NO: 401-445) whose excision or transposition activities ranged from equal to the naturally occurring *Heliothis* transposase to around one fifth as active, and we identified 24 variants that were less than one fifth as active as the naturally occurring *Heliothis* transposase, but whose activity was still measurable (with sequences given by SEQ ID NO: 466-489). Only 16 of the 95 variants (with sequences given by SEQ ID NO: 491-506) possessed activities that were so low that they were essentially inactive.

(165) The effects of sequence changes on excision and transposition activities were modelled as described in U.S. Pat. No. 8,635,029 and Liao et al (2007, BMC Biotechnology 2007, 7:16 doi: 10.1186/1472-6750-7-16 “Engineering proteinase K using machine learning and synthetic genes”). Mean values and standard deviations for the regression weights were calculated for each substitution, these are shown in Table 11. The effect of an individual substitution upon transposase activity may vary depending on the context (i.e. the other substitutions present). A positive mean regression weight indicates that on average, considering all of the different sequence contexts in which it has been tested, the substitution has a positive influence on the measured property. Incorporation of substitutions with positive mean regression weights into a sequence generally results in variants with improved activity (Liao et. al., *ibid*). A measure of this context-dependent variability is the standard deviation of the regression weight. If the mean regression weight plus the standard deviation of the regression weight for a substitution is zero or greater, then there are contexts within which the substitution has a positive effect. Thirty-one of the sixty substitutions we selected by looking for changes toward the consensus in other active piggyBac-like transposases had a mean regression weight plus the standard deviation of the regression weight of zero or greater: S41V, L43S, V81E, V81P, S136V, I149C, K152L, L153F, D155T, T162I, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, K396R, Y421F, H430Y, V450I, Y495F, L507I, H512N, S589T, Q603K. In addition to identifying specific

substitutions with a beneficial effect, this also provides an indication of positions at which analogous substitutions may be beneficial. For example, replacement of the nucleophilic threonine at position 268 with either the acidic residue aspartate or the acidic residue glutamate has a positive effect. This provides evidence that multiple different analogous substitutions at a position can be beneficial. Analogous substitutions are those in which properties of the amino acids are conserved. For example: glycine and alanine are in the “small” amino acid group; valine, leucine, isoleucine and methionine are in the “hydrophobic” amino acid group; phenylalanine, tyrosine and tryptophan are in the “aromatic” amino acid group; aspartate and glutamate are in the “acidic” amino acid group; asparagine and glutamine are in the “amide” amino acid group; histidine, lysine and arginine are in the “basic” amino acid group; cysteine, serine and threonine are in the “nucleophilic” amino acid group. If a substitution at an amino acid position within the *Heliothis* transposase is beneficial for excision or transposition activity, other substitutions at the same position drawn from the same amino acid group are likely to be beneficial. For example, since replacing the hydrophobic residue valine at position 81 with the acidic residue glutamate (V81E) is beneficial, replacing with the acidic residue aspartate (i.e. V81D) is likely also to be beneficial. Similarly, since replacing the hydrophobic residue methionine at position 300 with the basic residue arginine (M300R) is beneficial, replacing with the basic residues histidine or lysine (i.e. M300H or M300K) are likely also to be beneficial. An advantageous hyperactive *Heliothis* transposase comprises a substitution at one or more positions selected from amino acids 41, 43, 81, 81, 136, 149, 152, 153, 155, 162, 238, 239, 241, 264, 268, 268, 280, 297, 299, 300, 305, 312, 396, 421, 430, 450, 495, 507, 512, 589 and 603, for example one or more substitutions selected from S41V, L43S, V81E, V81P, S136V, I149C, K152L, L153F, D155T, T162I, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, K396R, Y421F, H430Y, V450I, Y495F, L507I, H512N, S589T and Q603K, or analogous changes at the same positions. (166) Of the 60 amino acid substitutions selected, only one (R485N) was found exclusively in essentially inactive transposases. This indicates that amino acid changes that make the *Heliothis* transposase sequence closer to a consensus sequence for active piggy Bac-like transposases are substitutions that can be incorporated the *Heliothis* transposase to create variants that retain transposition and excision activity. These are the changes listed in Table 1 column D. The positive results from the initial set of substitutions provide evidence that other changes in column D of Table 1 can also be incorporated into an *Heliothis* transposase to improve its activity.

(167) 6.1.7.2 Second Set of *Heliothis* Transposase Variants

(168) As described in Liao et al (2007, BMC Biotechnology 2007, 7:16 doi: 10.1186/1472-6750-7-16 “Engineering proteinase K using machine learning and synthetic genes”) and U.S. Pat. No. 8,635,029, Sections 5.4.2 and 5.4.3, substitutions that have been tested several times in the contexts of different combinations of other substitutions and that have “a positive regression coefficient, weight or other value describing its relative or absolute contribution to one or more activity” of a protein are usefully incorporated into a protein to obtain a protein that is “improved for one or more property, activity or function of interest”. Based on the substitution weights shown in Table 11, we designed a set of open reading frames encoding 95 new variant transposases combining some of the most positive substitutions (S41V, L43S, V81E, S136V, K152L, L153F, D155T, T162I, I239L, H241N, S264R, T268D, V280P, T297C, L299F, M300R, M305N, L312I, Y421F, V450I, Y495F, L507I and H512N). Each substitution was represented at least 10 times within the set of 95 variants, and the number of pairwise combinations was maximized so that each substitution was tested in as many different sequence contexts as possible. Each variant open reading frame was cloned into a vector comprising a leucine selectable marker; each open reading frame encoding a transposase variant was operably linked to the *Saccharomyces cerevisiae* Gal-1 promoter. Each of these variants was then individually transformed into a *Saccharomyces cerevisiae* strain comprising a chromosomally integrated copy of SEQ ID NO: 66, as described in Section 6.1.7.1. After 48 hours cells were scraped from the plate into minimal media lacking leucine and with galactose as

the carbon source. The A600 for each culture was adjusted to 2. Cultures were grown for 4 hours in galactose to induce expression of the transposases, then a 100×-diluted aliquot was plated on media lacking leucine, uracil and tryptophan (to count transposition), a 100×-diluted aliquot was plated on media lacking leucine and uracil (to count excision) and a 25,000×-diluted aliquot was plated on media lacking leucine (to count total live cells). Two days later, colonies were counted to determine transposition (=number of cells on -leu -ura -trp media divided by (250×number of cells on -leu media)) and excision (=number of cells on -leu -ura media divided by (250×number of cells on -leu media)) frequencies. The results are shown in Table 12.

(169) In addition to the activities of the 95 new *Heliothis* transposase variants, Table 12 also shows the activities of 6 variants from the first set that were among the most active variants in that set (transposases with polypeptide sequences given by SEQ ID NOs: 158, 159, 369, 372, 375 and 376). More than 60 of the new set of variants had greater transposition activity than the best variants from the first set. Every variant in the new set of variants, which all comprised various combinations of amino acid substitutions selected from S41V, L43S, V81E, S136V, K152L, L153F, D155T, T162I, I239L, H241N, S264R, T268D, V280P, T297C, L299F, M300R, M305N, L312I, Y421F, V450I, Y495F, L507I and H512N (all of which had positive mean regression weights) was increased in either excision or transposition activity relative to the naturally occurring transposase. A preferred hyperactive *Heliothis* transposase comprises one or more amino acid substitution selected from S41V, L43S, V81E, S136V, K152L, L153F, D155T, T162I, I239L, H241N, S264R, T268D, V280P, T297C, L299F, M300R, M305N, L312I, Y421F, V450I, Y495F, L507I and H512N, or analogous changes at the same positions.

(170) 6.1.7.3 Third Set of *Heliothis* Transposase Variants

(171) Using the activities shown in Table 12, we modelled the effects of substitutions on transposition and excision as described in Section 6.1.7.2 We selected the fourteen most positive substitutions (S41V, L43S, V81E, S136V, I239L, H241N, T268D, T297C, M300R, M305N, L312I, Y421F, V450I, Y495F) and a new set of 29 substitutions identified from Table 1 column D (D83S, V85L, P125S, Q126S, Q131R, Q131T, E140C, E140A, N151Q, K169E, N212S, C316A, C316M, N322T, P351G, H357R, H357D, K360Q, E379P, K397S, A447N, A447D, A449S, A449V, K476L, V492A, I500M, L585K and T595K). We designed a set of open reading frames encoding 96 new variants combining these substitutions. Each substitution was represented at least 5 times within the set of 96 variants, and the number of different pairwise combinations of substitutions was maximized so that each substitution was tested in as many different sequence contexts as possible. Each variant open reading frame was cloned into a vector comprising a leucine selectable marker; each open reading frame encoding a transposase variant was operably linked to the *Saccharomyces cerevisiae* Gal-1 promoter. Each of these variants was then individually transformed into a *Saccharomyces cerevisiae* strain comprising a chromosomally integrated copy of SEQ ID NO: 66, as described in Section 6.1.7.1. After 48 hours cells were scraped from the plate into minimal media lacking leucine and with galactose as the carbon source. The A600 for each culture was adjusted to 2. Cultures were grown for 4 hours in galactose to induce expression of the transposases, then a 1,000×-diluted aliquot was plated on media lacking leucine, uracil and tryptophan (to count transposition), a 1,000×-diluted aliquot was plated on media lacking leucine and uracil (to count excision) and a 25,000×-diluted aliquot was plated on media lacking leucine (to count total live cells). Two days later, colonies were counted to determine transposition (=number of cells on -leu -ura -trp media divided by (25×number of cells on -leu media)) and excision (=number of cells on -leu -ura media divided by (25×number of cells on -leu media)) frequencies. The results are shown in Table 13.

(172) In addition to the activities of the 95 new *Heliothis* transposase variants, Table 13 also shows the activities of 2 variants from previous sets that were among the most active variants in those sets (transposases with polypeptide sequences given by SEQ ID NO: 158 and 174). More than 60 of the new set of variants had greater transposition activity than the best of these. A preferred hyperactive

Heliothis transposase comprises one or more amino acid substitution selected from S41V, L43S, V81E, S136V, I239L, H241N, T268D, T297C, M300R, M305N, L312I, Y421F, V450I, Y495F, D83S, V85L, P125S, Q126S, Q131R, Q131T, E140C, E140A, N151Q, K169E, N212S, C316A, C316M, N322T, P351G, H357R, H357D, K360Q, E379P, K397S, A447N, A447D, A449S, A449V, K476L, V492A, I500M, L585K and T595K, or analogous changes at the same positions.

(173) The effects of sequence changes on excision and transposition frequencies were modelled as described in Liao et al (2007, BMC Biotechnology 2007, 7:16 doi: 10.1186/1472-6750-7-16 “Engineering proteinase K using machine learning and synthetic genes”) and U.S. Pat. No. 8,635,029. Mean values and standard deviations for the regression weights were calculated for each substitution, these are shown in Table 14. A positive regression weight indicates that the substitution has a positive influence on the measured property, but this is relative to the mean of the set of variants. In the third set of variants, several substitutions that were identified as positive in rounds 1 and 2 were incorporated into every variant in set 3, for example L43S, V81E, T268D, M300R and M305N. These all had positive regression weights when their effects on activity were modelled in round 1 (see Table 11). However in Table 14 it can be seen that all of these substitutions have regression weights of 0 within the third set of variants. This does not mean that these substitutions are no longer have a positive effect on transposase activity, it means that they are now having a neutral effect relative to the activity of the entire set. Since they are present in every variant in the set, they cannot be increasing the activities of some members of the set but not others. Instead they are now increasing the average activity of the entire set of variants. From Table 14, the following substitutions are seen to have positive regression weights for transposition, or were present in every variant: S41V, L43S, V81E, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, N151Q, K169E, N212S, I239L, T268D, T297C, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K397S, Y421F, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, L585K and T595K. A preferred hyperactive *Heliothis* transposase comprises one or more of these amino acid substitutions, or analogous changes at the same positions.

(174) 6.1.7.4 Fourth Set of *Heliothis* Transposase Variants

(175) Using the activities shown in Table 13, we modelled the effects of substitutions on transposition and excision as described in Section 6.1.7.2 We selected some of the most positive substitutions (S41V, L43S, V81E, D83S, Q126S, Q131R, Q131T, S136V, E140C, E140A, K169E, T268D, T297C, M300R, M305N, L312I, C316A, N322T, H357R, K397S, Y421F, A449S, V450I, V492A, Y495F and L585K). We designed a set of open reading frames encoding 90 new variants combining these substitutions. Each substitution was represented at least 5 times within the set of 90 variants, and the number of different pairwise combinations of substitutions was maximized so that each substitution was tested in as many different sequence contexts as possible. Each variant open reading frame was cloned into a vector comprising a leucine selectable marker; each open reading frame encoding a transposase variant was operably linked to the *Saccharomyces cerevisiae* Gal-1 promoter. Each of these variants was then individually transformed into a *Saccharomyces cerevisiae* strain comprising a chromosomally integrated copy of SEQ ID NO: 66, as described in Section 6.1.7.1. After 48 hours cells were scraped from the plate into minimal media lacking leucine and with galactose as the carbon source. The A600 for each culture was adjusted to 2. Cultures were grown for 4 hours in galactose to induce expression of the transposases, then a 2,000×-diluted aliquot was plated on media lacking leucine, uracil and tryptophan (to count transposition), a 2,000×-diluted aliquot was plated on media lacking leucine and uracil (to count excision) and a 25,000×-diluted aliquot was plated on media lacking leucine (to count total live cells). Two days later, colonies were counted to determine transposition (=number of cells on -leu -ura -trp media divided by (12.5×number of cells on -leu media)) and excision (=number of cells on -leu -ura media divided by (12.5×number of cells on -leu media)) frequencies. The results are shown in Table 15.

(176) In addition to the activities of the 90 new *Heliothis* transposase variants, Table 15 also shows the activities of 3 variants from previous sets that were among the most active variants in that set. More than 50 of the new set of variants had greater transposition activity than the best of these. A preferred hyperactive *Heliothis* transposase comprises one or more amino acid substitution selected from S41V, L43S, V81E, D83S, Q126S, Q131R, Q131T, S136V, E140C, E140A, K169E, T268D, T297C, M300R, M305N, L312I, C316A, N322T, H357R, K397S, Y421F, A449S, V450I, V492A, Y495F and L585K.

(177) We also analyzed the variants with transposition activities at least twice as high as the naturally occurring transposase with sequence given by SEQ ID NO: 39. Twenty-seven of these were more than 10-fold more active (sequences given by SEQ ID NO: 131-157), and 211 were between 2 and 10-fold more active (sequences given by SEQ ID NO: 158-368). The following substitutions occurred at least once within this set: S41V, L43S, V81E, S136V, K152L, L153F, D155T, T162I, 1239L, H241N, S264R, T268D, V280P, T297C, L299F, M300R, M305N, L312I, Y421F, V450I, R485K, Y495F, L507I, H512N, Q603E, D83S, V85L, P125S, Q126S, Q131T, Q131R, E140C, E140A, N151Q, K169E, N212S, C316A, N322T, P351G, H357R, K360Q, E379P, K397S, A447N, A447D, A449V, A449S, K476L, V492A, 1500M, L585K and T595K. A preferred hyperactive *Heliothis* transposase comprises one or more amino acid substitution selected from this set, or analogous changes at the same positions.

(178) We also analyzed the mean regression weights for each set of variants. A preferred hyperactive *Heliothis* transposase comprises an amino acid substitution at one or more of the positions at which we observed positive mean regression weights for transposition, that is at one of the following amino acid positions relative to SEQ ID NO: 39: 41, 43, 81, 81, 83, 85, 125, 126, 131, 131, 136, 140, 140, 149, 151, 152, 153, 155, 162, 169, 212, 238, 239, 241, 264, 268, 268, 280, 297, 299, 300, 305, 312, 316, 322, 357, 360, 396, 397, 421, 421, 430, 447, 447, 449, 449, 450, 476, 485, 492, 495, 495, 507, 512, 585, 589, 595, 603. A preferred hyperactive *Heliothis* transposase comprises one or more amino acid substitution selected from S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, 1149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, 1239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, Y421F, H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, Y495F, L507I, H512N, L585K, S589T, T595K, Q603K, or an analogous substitution at the same position.

Brief Description of Tables

(179) Table 1. Amino Acid Changes Likely to Result in Enhanced Activity of the *Heliothis* Transposase.

(180) Amino acid substitutions with the potential to improve transposase activity were identified as described in Section 5.2.7. Column A shows the position in a *Heliothis* transposase (relative to SEQ ID NO: 39), column B shows the amino acid in the native protein, column C shows the amino acids found in known active piggy Bac-like transposases at the equivalent position in an alignment, column D shows amino acid changes found in known active piggyBac-like transposases other than the *Heliothis* transposase at positions where there is good conservation within the rest of the transposase set, but the amino acid in the *Heliothis* transposase sequence is an outlier. Mutation to these amino acids are particularly likely to result in enhanced transposase activity.

(181) Table 2. Amino Acid Changes Likely to Result in Enhanced Activity of the *Helicoverpa* Transposase.

(182) Amino acid substitutions with the potential to improve transposase activity were identified as described in Section 5.2.7. Column A shows the position in a *Helicoverpa* transposase (relative to SEQ ID NO: 41), column B shows the amino acid in the native protein, column C shows the amino acids found in known active piggy Bac-like transposases at the equivalent position in an alignment, column D shows amino acid changes found in known active piggyBac-like transposases other than the *Helicoverpa* transposase at positions where there is good conservation within the rest of the

transposase set, but the amino acid in the *Helicoverpa* transposase sequence is an outlier. Mutation to these amino acids are particularly likely to result in enhanced transposase activity.

(183) Table 3. Amino Acid Changes Likely to Result in Enhanced Activity of the *Agrotis* Transposase.

(184) Amino acid substitutions with the potential to improve transposase activity were identified as described in Section 5.2.7. Column A shows the position in a *Agrotis* transposase (relative to SEQ ID NO: 40), column B shows the amino acid in the native protein, column C shows the amino acids found in known active piggy Bac-like transposases at the equivalent position in an alignment, column D shows amino acid changes found in known active piggyBac-like transposases other than the *Agrotis* transposase at positions where there is good conservation within the rest of the transposase set, but the amino acid in the *Agrotis* transposase sequence is an outlier. Mutation to these amino acids are particularly likely to result in enhanced transposase activity.

(185) Table 4. Excision and Transposition of Transposons in Yeast.

(186) Transposon and transposase sources are listed in column A. The left sequence with SEQ ID NO shown in column B and the right sequence with SEQ ID NO shown in column C were used to construct reporter plasmids as described in Section 6.1.2. The reporter plasmids have insert sequence given by the SEQ ID NO listed in column D. These reporter plasmids were integrated into the Ura3 gene of a Trp-strain of *Saccharomyces cerevisiae*. The amino acid sequence given by the SEQ ID NO shown in column E was back translated, synthesized and cloned into a plasmid comprising a Leu2 gene expressible in *Saccharomyces cerevisiae* and 2 micron origin of replication. The transposase open reading frame was operably linked to a Gall promoter. The plasmid comprising the transposase was transformed into the reporter strain, expression was induced, and cells were plated as described in Section 6.1.1. Induced culture was diluted 25,000-fold prior to plating 100 μ l on leu dropout plates, and 100-fold prior to plating 100 μ l on leu ura or leu ura trp dropout plates. Column F shows the number of colonies on the leu dropout plates; column G shows the number of colonies on the leu ura dropout plates (indicating excision of the transposon from the middle of the ura open reading frame in the reporter); column H shows the number of colonies on the leu ura trp dropout plates (indicating excision of the transposon from the middle of the ura open reading frame in the reporter and transposition to another site in the genome).

(187) Table 5. Transposition of Transposons by Related Transposases.

(188) Putative transposons were identified from the sources listed in column A. The ends from these transposons were used to construct reporter plasmids as described in Section 6.1.3. The reporter plasmids have sequence given by the SEQ ID NO listed in column B. These reporter plasmids were integrated into the Ura3 gene of a Trp-strain of *Saccharomyces cerevisiae*. Putative transposases were identified from the sources listed in column C. The amino acid sequence given by the SEQ ID NO shown in column D was back translated to the DNA sequence given by the SEQ ID NO shown in column E. This DNA sequence was cloned into a plasmid comprising a Leu2 gene expressible in *Saccharomyces cerevisiae* and 2 micron origin of replication. The transposase open reading frame was operably linked to a Gall promoter. The plasmid comprising the transposase was transformed into the reporter strain, expression was induced, and cells were plated as described in Section 6.1.1. Induced culture was diluted 25,000-fold prior to plating 100 μ l on leu dropout plates, and 100 μ l of culture was directly plated on leu ura or leu ura trp dropout plates. Column F shows the number of colonies on the leu dropout plates; column G shows the number of colonies on the leu ura dropout plates (indicating excision of the transposon from the middle of the ura open reading frame in the reporter); column H shows the number of colonies on the leu ura trp dropout plates (indicating excision of the transposon from the middle of the ura open reading frame in the reporter and transposition to another site in the genome). The relative excision frequency, determined by dividing the number of live cells (colonies on leu plates) by the number of URA+ colonies, is shown in column I. The relative transposition frequency, determined by dividing the

number of live cells (colonies on leu plates) by the number of URA⁺ colonies, is shown in column J.

(189) Table 6. Transposition of Transposons into the Genome of CHO Target Cells.

(190) Cells were transfected with transposon SEQ ID NO: 507 and transposases as described in Section 6.1.4. The transposase SEQ ID NO is shown in row 1. For each transfection, viability (the percentage of cells that are viable) and the total viable cell density (in millions of cells per ml) are shown in adjacent columns, as indicated in row 2. Rows 3-15 show these measurements at various times post-transfection, the days elapsed are shown in column A.

(191) Table 7. Antibody Production from Transposons Integrated into the Genome of CHO Target Cells.

(192) Cells were transfected with transposon SEQ ID NO: 507 and DNA encoding a transposase with polypeptide sequence given by SEQ ID NO: 39 as described in Section 6.1.4. The transposon SEQ ID NO is shown in column A, the co-transfected transposase SEQ ID NO is shown in column B. Recovery is shown in Table 6. During a 14 day fed batch antibody production run, the culture supernatant contained the concentration of antibody shown: column C shows the titer on Day 7; column D shows the titer on Day 10; column E shows the titer on Day 12; column F shows the titer on Day 14.

(193) Table 8. Transposition of Transposons into the Genome of CHO Target Cells by mRNA-Encoded Transposase.

(194) Cells were transfected with a transposon and mRNA-encoded transposase as described in Section 6.1.5. The transposon with sequence given by SEQ ID NO: 507 was co-transfected with mRNA encoding a transposase with polypeptide sequence given by SEQ ID NO: 39. The viability (the percentage of cells that are viable) and the total viable cell density (in millions of cells per ml) are shown in adjacent columns, as indicated in row 1. Rows 4-14 show these measurements at various times post-transfection, the days elapsed since transfection are shown in column A.

(195) Table 9. Transposition of Transposons with Truncated End Sequences into the Genome of CHO Target Cells.

(196) Cells were transfected with a transposon as described in Section 6.1.6. The transposon comprised a left transposon end comprising a 5'TTAA-3' integration target sequence immediately followed by a transposon ITR sequence with SEQ ID NO: 17 which was immediately followed by a left end sequence with SEQ ID NO shown in row 2. The transposon further comprised a heterologous polynucleotide with sequence SEQ ID NO: 70 which comprised a glutamine synthetase open reading frame operably linked to regulatory sequences to make it expressible in a mammalian cell. The transposon further comprised a right transposon end comprising a right end sequence with SEQ ID NO shown in row 3 immediately followed by a transposon ITR sequence with SEQ ID NO: 18 immediately followed by a 5'TTAA-3' integration target sequence. Row 1 shows transfections in which a plasmid encoding a transposase with amino acid sequence given by SEQ ID NO: 144 was co-transfected with the transposon. The viability (the percentage of cells that are viable) is indicated in columns labelled "V" in row 5 and the total viable cell density (in millions of cells per ml) is indicated in columns labelled "VCD" in row 4. Rows 5-14 show these measurements at various times post-transfection, the days elapsed since transfection are shown in column U.

(197) Table 10. Transposition and Excision Activities of *Heliothis* Transposase Variants (First Set).

(198) Open reading frames encoding *Heliothis* transposase variants were designed, synthesized and cloned as described in Section 6.1.7.1. SEQ ID NOs of each variant are given in column A. Genes were transformed into a *Saccharomyces cerevisiae* strain whose genome comprised a single copy of transposase reporter transposon with SEQ ID NO: 66, and plated on media lacking leucine. After 48 hours cells were scraped from the plate into minimal media lacking leucine and with galactose as the carbon source. The A600 for each culture was adjusted to 2. Cultures were grown for 4 hours in galactose to induce expression of the transposases. Cultures were diluted 100-fold into minimal

media lacking leucine; one 100 µl aliquot was plated onto minimal media agar plates lacking leucine and uracil (to measure transposon excision) another 100 µl aliquot was plated onto minimal media agar plates lacking leucine, tryptophan and uracil (to measure transposon transposition). Each culture was also diluted 25,000-fold and a 100 µl aliquot was plated onto minimal media agar plates lacking leucine (to measure live cells). After 48 hours colonies on each plate were counted, the number of colonies on plates lacking leucine are shown in column B, the number of colonies on plates lacking leucine, uracil and tryptophan are shown in column C, the number of colonies on plates lacking leucine and uracil are shown in column D. Column E shows the transposition frequency (calculated as the number in column C, divided by the number in column B, and further divided by 250) Column F shows the excision frequency (calculated as the number in column D, divided by the number in column B, and further divided by 250).

(199) Table 11. Model Weights for Amino Acid Substitutions in *Heliothis* Transposase Variants.

(200) The effects of sequence changes on *Heliothis* transposase excision and transposition activities for the first set of variants were modelled as described in Liao et al (2007, BMC Biotechnology 2007, 7:16 doi: 10.1186/1472-6750-7-16 “Engineering proteinase K using machine learning and synthetic genes”) and U.S. Pat. No. 8,635,029, see also Section 6.1.7.1. The mean values and standard deviations for the regression weights were calculated for each substitution. The position (relative to SEQ ID NO: 39) is shown in column A, the amino acid found at this position in SEQ ID NO: 39 is shown in column B. The tested amino acid substitution is shown in column C. The mean regression weight for the substitution on transposition activity is shown in column D, the standard deviation for this regression weight is shown in column E. The mean regression weight for the substitution on excision activity is shown in column F, the standard deviation for this regression weight is shown in column G.

(201) Table 12. Transposition and Excision Activities of *Heliothis* Transposase Variants (Second Set).

(202) Open reading frames encoding *Heliothis* transposase variants were designed, synthesized and cloned as described in Section 6.1.7.2. SEQ ID NOs of each variant are given in column A. Genes were transformed into a *Saccharomyces cerevisiae* strain whose genome comprised a single copy of transposase reporter SEQ ID NO: 66, and plated on media lacking leucine. After 48 hours cells were scraped from the plate into minimal media lacking leucine and with galactose as the carbon source. The A600 for each culture was adjusted to 2. Cultures were grown for 4 hours in galactose to induce expression of the transposases. Cultures were diluted 100-fold into minimal media lacking leucine; one 100 µl aliquot was plated onto minimal media agar plates lacking leucine and uracil (to measure transposon excision) another 100 µl aliquot was plated onto minimal media agar plates lacking leucine, tryptophan and uracil (to measure transposon transposition). Each culture was also diluted 25,000-fold and a 100 µl aliquot was plated onto minimal media agar plates lacking leucine (to measure live cells). After 48 hours colonies on each plate were counted, the number of colonies on plates lacking leucine are shown in column B, the number of colonies on plates lacking leucine, uracil and tryptophan are shown in column C, the number of colonies on plates lacking leucine and uracil are shown in column D. Column E shows the transposition frequency (calculated as the number in column C, divided by the number in column B, and further divided by 250) Column F shows the excision frequency (calculated as the number in column D, divided by the number in column B, and further divided by 250).

(203) Table 13. Transposition and Excision Activities of *Heliothis* Transposase Variants (Third Set).

(204) Open reading frames encoding *Heliothis* transposase variants were designed, synthesized and cloned as described in Section 6.1.7.3. SEQ ID NOs of each variant are given in column A. Genes were transformed into a *Saccharomyces cerevisiae* strain whose genome comprised a single copy of transposase reporter SEQ ID NO: 66, and plated on media lacking leucine. After 48 hours cells were scraped from the plate into minimal media lacking leucine and with galactose as the carbon source. The A600 for each culture was adjusted to 2. Cultures were grown for 4 hours in galactose

to induce expression of the transposases. Cultures were diluted 1,000-fold into minimal media lacking leucine; one 100 µl aliquot was plated onto minimal media agar plates lacking leucine and uracil (to measure transposon excision) another 100 µl aliquot was plated onto minimal media agar plates lacking leucine, tryptophan and uracil (to measure transposon transposition). Each culture was also diluted 25,000-fold and a 100 µl aliquot was plated onto minimal media agar plates lacking leucine (to measure live cells). After 48 hours colonies on each plate were counted, the number of colonies on plates lacking leucine are shown in column B, the number of colonies on plates lacking leucine, uracil and tryptophan are shown in column C, the number of colonies on plates lacking leucine and uracil are shown in column D. Column E shows the transposition frequency (calculated as the number in column C, divided by the number in column B, and further divided by 25) Column F shows the excision frequency (calculated as the number in column D, divided by the number in column B, and further divided by 25).

(205) Table 14. Model Weights for Amino Acid Substitutions in *Heliothis* Transposase Variants.

(206) The effects of sequence changes on *Heliothis* transposase excision and transposition activities were modelled for the third set of variants as described in Liao et al (2007, BMC Biotechnology 2007, 7:16 doi: 10.1186/1472-6750-7-16 “Engineering proteinase K using machine learning and synthetic genes”) and U.S. Pat. No. 8,635,029 and in Section 6.1.7.3. The mean values and standard deviations for the regression weights were calculated for each substitution. The position (relative to SEQ ID NO: 39) is shown in column A, the amino acid found at this position in SEQ ID NO: 39 is shown in column B. The tested amino acid substitution is shown in column C. The mean regression weight for the substitution on transposition activity is shown in column D, the standard deviation for this regression weight is shown in column E. The mean regression weight for the substitution on excision activity is shown in column F, the standard deviation for this regression weight is shown in column G. The number of times a substitution occurred within the set of 96 variants is shown in column H.

(207) Table 15. Transposition and Excision Activities of *Heliothis* Transposase Variants.

(208) Open reading frames encoding *Heliothis* transposase variants were designed, synthesized and cloned as described in Section 6.1.7.4. SEQ ID NOs of each variant are given in column A. Genes were transformed into a *Saccharomyces cerevisiae* strain whose genome comprised a single copy of transposase reporter SEQ ID NO: 66, and plated on media lacking leucine. After 48 hours cells were scraped from the plate into minimal media lacking leucine and with galactose as the carbon source. The A600 for each culture was adjusted to 2. Cultures were grown for 4 hours in galactose to induce expression of the transposases. Cultures were diluted 2,000-fold into minimal media lacking leucine; one 100 µl aliquot was plated onto minimal media agar plates lacking leucine and uracil (to measure transposon excision) another 100 µl aliquot was plated onto minimal media agar plates lacking leucine, tryptophan and uracil (to measure transposon transposition). Each culture was also diluted 25,000-fold and a 100 µl aliquot was plated onto minimal media agar plates lacking leucine (to measure live cells). After 48 hours colonies on each plate were counted, the number of colonies on plates lacking leucine are shown in column B, the number of colonies on plates lacking leucine, uracil and tryptophan are shown in column C, the number of colonies on plates lacking leucine and uracil are shown in column D. Column E shows the transposition frequency (calculated as the number in column C, divided by the number in column B, and further divided by 12.5) Column F shows the excision frequency (calculated as the number in column D, divided by the number in column B, and further divided by 12.5).

(209) TABLE-US-00001 TABLE 1 A B C D heliothis_position heliothis Acceptable Beneficial 1 M M 2 E EM 3 P PSEAG 4 S SMK 5 T TARDN 6 S SRQFI 7 S SGFRLYE 8 G GLTDSRA 9 R RTANDQ 10 K KDEQH 11 R RLEDHN 12 S SEAIR 13 I ILA 14 G GNLASR 15 N NQLTAH 16 V VIFLCM 17 H HLFM LFM 18 N NEDQA 19 Q QLSNE 20 R REDSV 21 A ADSLE 22 A AVEDST 23 K KEYLD 24 N NESVFY 25 R RDIPGS 26 R RVSLEGDY 27 A AIEDS 28 V VFISD 29 V VDES 30 P PGSVE 31 G GDEP 32 T TEAKVP 33 R RSEAT 34 D DS 35 F

FHNRC 36 G GVDSC 37 T TSIVD 38 T TIDES 39 L LRDSH 40 T TPDEN 41 S SVDE VDE 42
W WEFQN 43 L LSQY SQY 44 D DFW 45 N NTSC 46 E EDS 47 D DESQ 48 S SVEA 49 S
SEMIFTRA 50 G GIDV 51 S SPD 52 E ETYFASP 53 V VLD 54 E EHD 55 D DPSEVL 56 I
ITVPL 57 G GSQ 58 D DRAPN 59 N NQTS 60 F FESRG 61 T TDSQ 62 P PGE 63 E EDS 64 R
RA 65 H HN 66 E EAT 67 I IV 68 E EIV 69 S SA 70 D DQVNE 71 T TQVNIME 72 I IGSLET 73
S SPADVY 74 Q QSED SED 75 S SDQRE 76 E ESNDP 77 S SENVD 78 E ENLAPID 79 E
EDGN 80 Q QMLV 81 V VPE PE 82 A ALQVGD 83 D DSTQ 84 H HNSELA 85 V VLSA 86 T
TAGR 87 E ERSDQ 88 E ERGHN 89 H HSTRAM 90 N NWIDSF 91 M MAILC 92 S STA 93 S
SKLAR 94 D DPGQ 95 D D 96 D D 97 A A 98 P P 99 L L 100 S SV 101 T TSG 102 R RQG 103
R RPS 104 S SFTY 105 F FYI 106 Y YKCRS 107 G GS 108 K K 109 N NDG 110 R RNKEG
111 Y YTIHP 112 K KIVCA 113 W W 114 A ANSGY 115 C CRTPK 116 Q QASTPN 117 P PKC
118 L LPHGSNQ 119 S SNQRTF 120 R RPST 121 A ARTN 122 V VGSI 123 R R 124 V VTL 125
P PRS 126 Q QSAE 127 H HESIL 128 N NP 129 I IP 130 I IVF 131 Q QTKR 132 R RGMTSE
133 T TNQVR 134 N NPRA 135 V VGQL GQL 136 S SVP VP 137 N NKT KT 138 L LRVNT
139 T TQFMDIG 140 E EACT 141 D DKVRS 142 D DNT 143 P PAVI 144 K KLDVYFS 145 D
DLET 146 P PEI 147 F FLQIYS 148 S SDLNEK 149 I ICAF CAF 150 W WF 151 N NHQK 152
K KLI LI 153 L LF F 154 M MVIF 155 D DNST NST 156 D DEQS 157 E ESPAD 158 I IM 159 L
LEI 160 Q QSRHD 161 E EVID 162 T TIM 163 L LV 164 K KEDLT 165 W WHMY 166 T T 167
N N 168 E EHLVAS 169 K KEYS 170 I IGMA 171 I IRSE 172 Q QSRVLH 173 Y YEVKRS 174
R RFLQ 175 S SQRTV 176 K KGNQ 177 F FKLENTS 178 S SNIKPA 179 D DTMLA 180 K
KSITPE 181 D DNSR 182 R RYS 183 P PYEAHFV 184 E EAKSTYH 185 L LYWFMK 186 R
RSKHQ 187 N NESDP 188 L LTI 189 D DTN 190 M MLQTEI 191 V VTMDACS 192 E ED 193
L LMI 194 H HRNWYK 195 A AR 196 F FVYL 197 I IVF 198 G GA 199 L LI 200 L LT 201 L
LYITV 202 F FLMAI 203 T TAM 204 A AG 205 V VL 206 F FYMRIT 207 K KR 208 S SDA 209
N NGK 210 H HRG 211 E EQLMS 212 N NASL 213 V VLTE 214 N NQSDK 215 Y YDSE 216 L
LW 217 F FWD 218 A ANDTR 219 T TSAR 220 D DETS 221 G GEFLV 222 T TNSL 223 G GS
224 R RIV 225 E EPTMD 226 I IRV 227 F FY 228 R RPVS 229 C CMAST 230 V VT 231 M M
232 S S 233 K KLR LR 234 N NREDQ 235 R RT 236 F FY 237 L LAHEDQY 238 V VFML FML
239 I IL 240 L LVSIQ 241 H HNR RN 242 C CVFNS 243 L LIM 244 R RH 245 F FM 246 D DN
247 N ND 248 P PSRKT 249 D DTSA 250 D DTLIV 251 R RP 252 E EVPD 253 E EGTD 254 R
RLQ 255 R RAPK 256 E EASGQK 257 S SIDNHT 258 D D 259 K KRAVN 260 I ILFM 261 A
AILTH 262 A APK 263 I IVLF 264 S SR R 265 Y YQDKPS 266 I IVLM 267 F FYWI 268 T TED
ED 269 K KEILSQ 270 F FWL 271 V VIS 272 G GKENHQ 273 N NIQRC 274 C CLF 275 Q
QKPRIA 276 K KDLQAN 277 I IVLNA 278 Y YH 279 N NTVS 280 V VP P 281 C CYGS 282 E
EPGSAQ 283 Y YNHF 284 A ALVI LVI 285 T TC 286 V VI 287 D D 288 E E 289 M MERQS
290 L L 291 V VL 292 P PAGLS 293 F F 294 R RK 295 G G 296 R R 297 T TCL CL 298 H
HKPQL 299 L LF F 300 M MR R 301 I IQMV 302 Y Y 303 M MLI 304 P P 305 M MNS NS 306
K K 307 P PR 308 A ADS 309 K KR 310 Y Y 311 G G 312 L LI 313 K KR 314 L LIF IF 315 M
MIWPLYF 316 C CAMK 317 L LAM 318 C CV 319 D DAE 320 A AS 321 N NYKAGS 322 N
NTS 323 G GYSKF 324 Y Y 325 F FSAMTV 326 Y YLWISV 327 N NKDY 328 C CMAGFL 329
Y YEQIML 330 I IVP 331 Y Y 332 T TALE 333 G G 334 R RDK 335 G GQSD 336 S SPT 337 D
DGKQSL 338 G GTL 339 A APND 340 G GYP 341 L LKEVPA 342 T TVKPG 343 E ESNC 344
E ENQP 345 E EP 346 K K 347 K KD 348 F FL 349 M MT 350 V VKLRT 351 P PGSA 352 T
THMEGF 353 Q QDRFYKE 354 S SVYI 355 V V 356 I IDLKWE 357 H HRED 358 L LIM 359
A AVSTI 360 K KQES 361 P PGT 362 L LIV 363 F FSQHLA 364 G GQR 365 S STQ 366 N
NGCH 367 R RHF 368 N NH 369 I IVL 370 T TY 371 C CMVF 372 D D 373 N N 374 W WF
375 F FY 376 T TS 377 S SG 378 I IY 379 E EPRT 380 L LT 381 I IYGAFM 382 E EAKTL 383 Y
YHENA 384 L LM 385 K KLQY LQY 386 K KQCN 387 K KNRAEL 388 G GKNDR 389 L LT
390 T TP 391 C CAMIS 392 V VLCT 393 G G 394 T T 395 M MVI 396 K KRN 397 K KSR 398
N N 399 K KR 400 R RTPK 401 E ECGQ 402 I ILM 403 P P 404 K KPSERD 405 E EKVAS 406

F FIL 1 L LRKIT 208 P PEKND 409 S SRIK 410 K KQRDG 411 Q QGSL 412 R RN 413 D
DEPRQ 414 V VIMGP 415 G GNHEA 416 S ST 417 S SY 418 L LIMAV 419 Y YFL FL 420 G
GACR 421 Y YFK FK 422 A AQTDN 423 G GDEK 424 Q QDKPL 425 N NFALI 426 T TA 427 I
IVL 428 L LVK VK 429 S SF 430 H HYF YF 431 V VICKDA 432 P P 433 K K 434 K KRP 435 N
NSAK 436 K KR 437 A ANMV 438 V V 439 I IFLYV 440 L LMVA 441 L LM 442 S ST 443 S ST
444 M MLCI 445 H HD 446 H HTED 447 A ADNE 448 E ESAN 449 A AESV 450 V VIL IL 451
D DSNR 452 E ESTQ 453 T TERSQ 454 T TDNR 455 G GDV 456 K K 457 P P 458 E ESQDL
459 I IMC 460 I IVS 461 G GTLMK 462 F FDYE 463 Y Y 464 N NS 465 K KSQ 466 T TY 467
K KM 468 G GSA 469 G G 470 V V 471 D D 472 E ENSTRV 473 I IVLFT 474 D D 475 K KQE
476 K KLVM 477 C CITQS 478 A ARKSH 479 I ITSVDN 480 Y YM 481 T TDSN 482 S SVCA
483 S SQNT 484 R R 485 R RNK NK 486 T TS 487 R RANK 488 R RA 489 W W 490 P PY 491
M MLK 492 V VTAK 493 V VIL 494 F FLG 495 Y YFI 496 R RWNGY 497 M MILV 498 L LVI
499 D DNQ 500 I ITVM 501 S SA 502 T TGAFLS 503 V VIYR 504 N N 505 S SA 506 H
HKYFC 507 L LIV IV 508 I IVL 509 Y YQW 510 D DMSCKRQ 511 I ILEHAT 512 H HNIA NIA
513 H HSNKV 514 D DGKE 515 K KNVA 516 T TLPV 517 T TPYIQSV 518 E ETRNSYK 519
R RY 520 G GRKT 521 M MALEKY 522 F FQ 523 L LIM 524 K KERQ 525 Q QKENIS 526 L L
527 A AGSYP 528 R RMKTIAL 529 T TSADQL 530 L LM 531 V VITF 532 L LATSGY 533 P
PSGE 534 Q QHKWFV 535 M MQIEL 536 K KAQREH 537 R REKQS 538 R RT 539 A AKLVN
540 L LTRKQEP 541 N NPAEK 542 E ESAPMK 543 R RKTPN 544 L LISP 545 P PKS 546 R
RVADTF 547 E ESATYNH 548 L LAVI 549 R RA 550 L LKVDRQ 551 S SRINL 552 L LIE IE
553 A AGKTSE 554 R RSINK 555 V VHKEQ 556 L LF 557 G GRKPI 558 P PETNKD 559 D
DSVETP 560 M MSNVLT 561 P PAQ 562 V VSATR 563 P PMSH 564 D DAEGV 565 P PKND
566 Q QIVNSM 567 E EPDSTR 568 V VNTE 569 D DE 570 E EP 571 T TGEV 572 F FVPM 573
K KG 574 T TVKPR 575 R RKQY 576 R RKSTV 577 R RYG 578 C C 579 H HYQGTRK 580 T
TIVFYDE 581 C C 582 P PSR 583 L LVSYN 584 K KR 585 L LKDI 586 Q QDR DR 587 R RS
588 K KMD 589 S STA 590 T TKSND 591 H HYTR 592 T TISQY 593 C CF 594 Y YIVCKPN
595 T TSKA 596 C C 597 K KTPA 598 K KSNR 599 H HFAVNP 600 V VIL LI 601 C C 602 L
LRGFM 603 Q QEK KE 604 C CHP HP 605 A ATNC 606 K KNVIF 607 Q QFTDE 608 V
VFMIL 609 C CY 610 A AEPQH 611 D DNST 612 C CQ 613 V VRGIFLA

(210) TABLE-US-00002 TABLE 2 A B C D helicopterpa_position helicopterpa Acceptable
Beneficial_Ha 1 M M 2 A PSEAG 3 S SMK 4 R TARDN 5 Q SRQFI 6 R SGFRLYE 7 L
GLTDSRA 8 N RTANDQ 9 H KDEQH DE 10 D RLEDHN 11 E SEAIR 12 I ILA 13 A GNLASR
14 T NQLTAH 15 I VIFLCM 16 L HLFM 17 E NEDQA 18 N QLSNE 19 D REDSV 20 D ADSLE
21 D AVEDST 22 Y KEYLD 23 S NESVYF 24 P RDIPGS 25 L RVSLEGDY 26 D AIEDS 27 S
VFISD 28 E VDES 29 S PGSVE 30 E GDEP 31 K TEAKVP 32 E RSEAT 33 D DS 34 C
FHNRC 35 V GVDSC 36 V TSIVD 37 E TIDES 38 D LRDSH 39 D TPDEN 40 V SVDE 41 W
WEFQN 42 S LSQY 43 D DFW 44 N NTSC 45 E EDS 46 D DESQ 47 A SVEA 48 I SEMIFTRA
49 V GIDV 50 D SPD 51 F ETYFASP 52 V VLD 53 E EHD 54 D DPSEVL 55 T ITVPL 56 S
GSQ 57 A DRAPN 58 Q NQTS 59 E FESRG 60 D TDSQ 61 P PGE 62 D EDS 63 N DQVNE 64
N TQVNIME 65 I IGSLET 66 A SPADVY 67 S QSED 68 R SDQRE 69 E ESNDP 70 S SENVD
71 P ENLAPID 72 N EDGN 73 L QMLV 74 E VPE 75 V ALQVGD 76 T DSTQ 77 S HNSELA 78
L VLAS 79 T TAGR 80 S ERSDQ 81 H ERGHN 82 R HSTRAM 83 I NWIDSF 84 I MAILC 85 T
STA 86 L SKLAR 87 P DPGQ 88 Q RQG 89 R RPS 90 S SFTY 91 I FYI 92 R YKCRS 93 G GS
94 K K 95 N NDG 96 N RNKEG 97 H YTIHP 98 V KIVCA 99 W W 100 S ANSGY 101 T
CRTPK 102 T QASTPN 103 K PKC 104 G LPHGSNQ 105 R SNQRTF 106 T RPST 107 T ARTN
108 G VGS 109 R R 110 T VTL 111 S PRS 112 A QSAE 113 I HESIL 114 N NP 115 I IP 116 I
IVF 117 R QTKR 118 T RGMTSE 119 N TNQVR 120 R NPRA 121 G VGQL 122 P SVP 123 T
NKT 124 R LRVNT 125 M TQFMDIG 126 C EACT 127 R DKVRS 128 N DNT 129 I PAVI 130
V KLDVYFS 131 D DLET 132 P PEI 133 L FLQIYS 134 L SDLNEK 135 C ICAF 136 F WF 137
Q NHQK 138 L KLI 139 F LF 140 I MVIF 141 T DNST 142 D DEQS 143 E ESPAD 144 I IM 145

I LEI 146 H QSRH 147 E EVID 148 I TIM 149 V LV 150 K WEDLT 151 W WHMY 152 T T
153 N N 154 V EHLVAS 155 E KEYS 156 I IGMA GA 157 I IRSE 158 V QSRVLH 159 K
YEVKRS 160 R RFLQ 161 Q SQRTV 162 N KGNQ 163 L FKLENTS 164 K SNIKPA 165 D
DTMLA 166 I KSITPE 167 S DNSR 168 A PYEAHFV 169 S EAKSTYH 170 Y LYWFMK 171 R
RSKHQ 172 D NESDP 173 T LTI 174 N DTN 175 T MLQTEI 176 M VTMDACS 177 E ED 178
I LMI LM 179 W HRNWKY 180 A AR 181 L FVYL 182 V IVF 183 G GA 184 I LI 185 L LT 186
T LYITV 187 L FLMAI 188 T TAM 189 A AG 190 V VL 191 M FYMRIT 192 K KR 193 D SDA
SA 194 N NGK 195 H HRG 196 L EQLMS 197 S NASL 198 T VLTE 199 D NQSDK 200 E
YDSE 201 L LW 202 F FWD 203 D ANDTR 204 A TSAR 205 T DETS 206 F GEFLV 207 S
TNSL 208 G GS 209 T EPTMD 210 R IRV 211 Y FY F 212 V RPTS 213 S CMAST 214 V VT
215 M M 216 S S 217 R KLR 218 E NREDQ 219 R RT 220 F FY 221 E LAHEDQY 222 F VFML
223 L IL 224 I LVSIQ 225 R HNR 226 C CVFNS 227 I LIM LA 228 R RH 229 M FM 230 D DN
231 D ND 232 K PSRKT 233 T DTSA 234 L DTLIV 235 R RP 236 P EVPD 237 T EGTD 238 L
RLQ 239 R RAPK 240 S EASGQK 241 D SIDNHT 242 D D 243 A KRAVN 244 F ILFM 245 L
AILTH 246 P APK 247 V IVLF 248 R SR 249 K YQDKPS 250 I IVLM 251 W FYWI 252 E TED
253 I KEILSQ 254 F FWI 255 I VIS 256 N GKENHQ 257 Q NIQRC 258 C CLF 259 R QKPRIA
260 Q KDLQAN 261 N IVLNA 262 H YH Y 263 V NTVS 264 P VP 265 G CYGS 266 S
EPGSAQ 267 N YNHF 268 L ALVI 269 T TC 270 V VI 271 D D 272 E E 273 Q MERQS 274 L L
275 L VL 276 G PAGLS 277 F F 278 R RK 279 G G 280 R R 281 C TCL 282 P HKPQL 283 F LF
284 R MR 285 M IQMV 286 Y Y 287 I MLI 288 P P 289 N MNS 290 K K 291 P PR 292 D ADS
AS 293 K KR 294 Y Y 295 G G 296 I LI 297 K KR 298 F LIF LI 299 P MIWPLYF 300 M CAMK
301 M LAM LA 302 C CV 303 A DAE DE 304 A AS 305 A NYKAGS 306 T NTS 307 K GYSKF
308 Y Y 309 M FSAMTV 310 I YLWISV 311 D NKDY 312 A CMAGFL 313 I YEQIML 314 P
IVP 315 Y Y 316 L TALE 317 G G 318 K RDK 319 S GQSD 320 T SPT 321 K DGKQSL 322 T
GTL 323 N APND 324 G GYP 325 L LKEVPA 326 P TVKPG 327 L VKLRT 328 G PGSA 329 E
THMEGF 330 F QDRFYKE 331 Y SVYI 332 V V 333 K IDLKWE 334 D HRED E 335 L LIM
336 T AVSTI 337 K KQES 338 T PGT PG 339 V LIV 340 H FSQHLA 341 G GQR 342 T STQ
343 N NGCH 344 R RHF 345 N NH 346 I IVL 347 T TY 348 C CMVF 349 D D 350 N N 351 W
WF 352 F FY 353 T TS 354 S SG 355 I IY 356 P EPRT 357 L LT 358 A IYGAFM 359 K EAKTL
360 N YHENA 361 M LM L 362 L KLQY 363 Q KQCN 364 A KNRAEL 365 P P 366 Y HY 367
N GKNDR 368 L LT 369 T TP 370 I CAMIS 371 V VLCT 372 G G 373 T T 374 I MVI 375 R
KRN 376 S KSR 377 N N 378 K KR 379 R RTPK 380 E ECGQ 381 M ILM IL 382 P P 383 E
KPSEED 384 E EKVAS 385 I FIL 386 K LRKIT 387 N PEKNDR 388 S SRIKT 389 R KQRDG
390 S QGSL 391 R RN 392 P DEPRQ 393 V VIMGP 394 G GNHEA 395 S ST 396 S SY 397 M
LIMAV 398 F YFL 399 C GACR 400 F YFK 401 D AQTDN 402 G GDEK 403 P QDKPL 404 L
NFALI 405 T TA 406 L IVL 407 V LVK 408 S SF 409 Y HYF 410 K VICKDA 411 P P 412 K K
413 P KRP 414 S NSAK 415 K KR 416 M ANMV 417 V V 418 F IFLYV 419 L LMVA 420 L LM
421 S ST 422 S ST 423 C MLCI 424 D HD H 425 E HTED 426 N ADNE 427 A ESAN 428 V
AESV 429 I VIL 430 N DSNR 431 E ESTQ 432 S TERSQ 433 N TDNR 434 G GDV 435 K K
436 P P 437 D ESQDL 438 M IMC 439 I IVS 440 L GTLMK 441 F FDYE 442 Y Y 443 N NS 444
Q KSQ 445 T TY 446 K KM 447 G GSA 448 G G 449 V V 450 D D 451 S ENSTRV 452 F IVLFT
453 D D 454 Q KQE 455 M KLVM 456 C CITQS 457 K ARKSH 458 S ITSVMY 459 M YM Y
460 S TDSN 461 A SVCA 462 N SQNT 463 R R 464 K RNK 465 T TS 466 N RANK 467 R RA
468 W W 469 P PY 470 M MLK 471 A VTAK 472 V VIL 473 F FLG 474 Y YFI 475 G RWNGY
476 M MILV 477 L LVI 478 N DNQ 479 M ITVM 480 A SA 481 F TGAFCLS 482 V VIYR 483
N N 484 S SA 485 Y HKYFC 486 I LIV 487 I IVL 488 Y YQW 489 C DMSCKRQ 490 H
ILEHAT 491 N HNIA 492 K HSNKV 493 I SQIPN 494 N DENSG 495 K WKP 496 Q AQG 497 E
DGKE 498 K KNVA 499 P TLPV 500 I TPYIQSV 501 S ETRNSYK 502 R RY 503 K GRKT 504
E MALEKY 505 F FQ 506 M LIM 507 K KERQ 508 K QKENIS 509 L L 510 S AGSYP 511 I
RMKTIAL 512 Q TSADQL 513 L LM 514 T VITF 515 T LATSGY 516 P PSGE 517 W

QHKHWFV 518 M MQIEL 519 Q KAQREH 520 E REKQS 521 R RT 522 L AKLVN 523 Q
LTRKQEP 524 A NPAEK 525 P ESAPMK 526 T RKTPN 527 L LISP 528 K PKS 529 R RVADTF
530 T ESATYNH 531 L LAVI 532 R RA 533 D LKVDRQ 534 N SRINL 535 I LIE 536 T
AGKTSE 537 N RSINK 538 V VHKIQ 539 L LF 540 K GRKPI 541 N PETNKD 542 V DSVETP
543 V MSNVLT 544 P PAQ 545 A FPAG 546 S VSATR 547 S PMSH 548 E DAEGV 549 N
PKND 550 I QIVNSM 551 S EPDSTR 552 N VNTE 553 E DE 554 P EP 555 E TGEV 556 P
FVPM 557 K KG 558 K TVKPR 559 R RKQY 560 R RKSTV 561 Y RYG 562 C C 563 G
HYQGTRK 564 V TIVFYDE 565 C C 566 S PSR PR 567 Y LVSYKN 568 K KR 569 K LKDI
570 R QDR 571 R RS 572 M KMD KD 573 T STA 574 K TKS NR 575 A HYTAR 576 Q TISQY
577 C CF 578 C YIVCKPN 579 K TSKA 580 C C 581 K KTPA 582 K KSNR 583 A HFAVNP 584
I VIL 585 C C 586 G LRGFM 587 E QEK 588 H CHP 589 N ATNC 590 I KNVIF 591 D QFTDE
592 V VFMIL 593 C CY 594 Q AEPQH 595 D DNST 596 C CQ 597 I VRGIFLA 598 G REGHD
(211) TABLE-US-00003 TABLE 3 A B C D AGROTIS_position agrotis Acceptable Beneficial_Ag
1 M M 2 E PSEAG 3 S SMK 4 R TARDN 5 Q SRQFI 6 R SGFRLYE 7 L
GLTDSRA 8 N RTANDQ 9 Q KDEQH DE 10 D RLEDHN 11 E SEAIR 12 I ILA 13 A
GNLASR 14 T NQLTAH 15 I VIFLCM 16 L HLFM 17 E NEDQA 18 N QLSNE 19 D
REDSV 20 D ADSLE 21 D AVEDST 22 Y KEYLD 23 S NESVFY 24 P RDIPGS 25 L
RVSLEGDY 26 D AIEDS 27 S VFISD 28 D VDES 29 S PGSVE 30 E GDEP 31 A
TEAKVP 32 E RSEAT 33 D DS 34 R FHNRC 35 V GVDSC 36 V TSIVD 37 E TIDES
38 D LRDSH 39 D TPDEN 40 V SVDE 41 W WEFQN 42 S LSQY 43 D DFW 44 N
NTSC 45 E EDS 46 D DESQ 47 A SVEA 48 M SEMIFTRA 49 I GIDV 50 D SPD 51 Y
ETYFASP 52 V VLD 53 E EHD 54 D DPSEVL 55 T ITVPL 56 S GSQ 57 R DRAPN 58
Q NQTS 59 E FESRG 60 D TDSQ 61 P PGE 62 D EDS 63 N DQVNE 64 N TQVNIME
65 I IGSLET 66 A SPADVY 67 S QSED 68 Q SDQRE 69 E ESNDP 70 S SENVD 71 A
ENLAPID 72 N EDGN 73 L QMLV 74 E VPE 75 V ALQVGD 76 T DSTQ 77 S HNSELA
78 L VLSA 79 T TAGR 80 S ERSDQ 81 H ERGHN 82 R HSTRAM 83 I NWIDSF 84 I
MAILC 85 S STA 86 L SKLAR 87 P DPGQ 88 Q RQG 89 R RPS 90 S SFTY 91 I FYI
92 C YKCRS 93 G GS 94 K K 95 N NDG 96 N RNKEG 97 H YTIHP 98 V KIVCA 99
W W 100 S ANSGY 101 T CRTPK 102 T QASTPN 103 K PKC 104 G LPHGSNQ 105 R
SNQRTF 106 T RPST 107 T ARTN 108 G VGSI 109 R R 110 T VTL 111 S PRS 112 A QSAE 113
I HESIL 114 N NP 115 I IP 116 I IVF 117 R QTKR 118 T RGMTSE 119 N TNQVR 120 R NPRA
121 G VGQL 122 P SVP 123 T NKT 124 R LRVNT 125 M TQFMDIG 126 C EACT 127 R
DKVRS 128 N DNT 129 I PAVI 130 V KLDVYFS 131 D DLET 132 P PEI 133 L FLQIYS 134 L
SDLNEK 135 C ICAF 136 F WF 137 Q NHQK 138 L KLI 139 F LF 140 I MVIF 141 T DNST
142 D DEQS 143 E ESPAD 144 I IM 145 I LEI 146 H QSRHD 147 E EVID 148 I TIM 149 V LV
150 K KEDLT 151 W WHMY 152 T T 153 N N 154 V EHLVAS 155 E KEYS 156 M IGMA IGA
157 I IRSE 158 V QSRVLH 159 K YEVKRS 160 R RFLQ 161 Q SQRTV 162 N KGNQ 163 L
FKLENTS 164 I SNIKPA 165 D DTMLA 166 I KSITPE 167 S DNSR 168 A PYEAHFV 169 S
EAKSTYH 170 Y LYWFMK 171 R RSKHQ 172 D NESDP 173 T LTI 174 N DTN 175 T
MLQTEI 176 M VTMDACS 177 E ED 178 M LMI LI 179 W HRNWK 180 A AR 181 L FVYL
182 V IVF 183 G GA 184 I LI 185 L LT 186 T LYITV 187 L FLMAI 188 T TAM 189 A AG 190 V
VL 191 M FYMRIT 192 K KR 193 D SDA SA 194 N NGK 195 H HRG 196 L EQLMS 197 S
NASL 198 T VLTE 199 D NQSDK 200 E YDSE 201 L LW 202 F FWD 203 D ANDTR 204 A
TSAR 205 T DETS 206 F GEFLV 207 S TNSL 208 G GS 209 T EPTMD 210 R IRV 211 Y FY F
212 V RPVS 213 S CMAST 214 V VT 215 M M 216 S S 217 R KLR 218 E NREDQ 219 R RT
220 F FY 221 E LAHEDQY 222 F VFML 223 L IL 224 I LVSIQ 225 R HNR 226 C CVFNS 227
M LIM LI 228 R RH 229 M FM 230 D DN 231 D ND 232 K PSRKT 233 T DTSA 234 L DTLIV
235 R RP 236 P EVPD 237 T EGTD 238 L RLQ 239 R RAPK 240 S EASGQK 241 D SIDNHT
242 D D 243 A KRAVN 244 F ILFM 245 I AILTH 246 P APK 247 V IVLF 248 R SR 249 K
YQDKPS 250 L IVLM 251 W FYWI 252 E TED 253 I KEILSQ 254 F FWL 255 I VIS 256 N

GKHNQ 257 Q NIQR 258 C CLF 259 R QKPRIA 260 L KDLQAN 261 N IVLNA 262 Y YH
263 V NTVS 264 P VP 265 G CYGS 266 G EPGSAQ 267 N YNHF 268 L ALVI 269 T TC 270 V
VI 271 D D 272 E E 273 Q MERQS 274 L L 275 L VL 276 G PAGLS 277 F F 278 R RK 279 G G
280 R R 281 C TCL 282 P HKPQL 283 F LF 284 R MR 285 M IQMV 286 Y Y 287 I MLI 288 P P
289 N MNS 290 K K 291 P PR 292 D ADS AS 293 K KR 294 Y Y 295 G G 296 I LI 297 R KR K
298 F LIF LI 299 P MIWPLYF 300 M CAMK 301 M LAM LA 302 C CV 303 D DAE E 304 A AS
305 A NYKAGS 306 T NTS 307 K GYSKF 308 Y Y 309 M FSAMTV 310 I YLWISV 311 D
NKDY 312 A CMAGFL 313 I YEQIML 314 P IVP 315 Y Y 316 L TALE 317 G G 318 K RDK
319 S GQSD 320 T SPT 321 K DGKQSL 322 T GTL 323 N APND 324 G GYP 325 L LKEVPA
326 P TVKPG 327 L VKLRT 328 G PGSA 329 E THMEGF 330 F QDRFYKE 331 Y SVYI 332 V
V 333 K IDLKWE 334 E HRED 335 L LIM 336 T AVSTI 337 K KQES 338 T PGT PG 339 V LIV
340 H FSQHLA 341 G GQR 342 T STQ 343 N NGCH 344 R RHF 345 N NH 346 V IVL 347 T
TY 348 C CMVF 349 D D 350 N N 351 W WF 352 F FY 353 T TS 354 S SG 355 I IY 356 P EPRT
357 L LT 358 A IYGAFM 359 K EAKTL 360 N YHENA 361 M LM L 362 L KLQY 363 Q KQCN
364 A KNRAEL 365 P P 366 Y HY 367 N GKNDR 368 L LT 369 T TP 370 I CAMIS 371 V VLCT
372 G G 373 T T 374 I MVI 375 R KRN 376 S KSR 377 N N 378 K KR 379 R RTPK 380 E
ECGQ 381 I ILM LM 382 P P 383 E KPSEED 384 E EKVAS 385 I FIL 386 K LRKIT 387 N
PEKNDR 388 S SRIKT 389 R KQRDG 390 S QGSL 391 R RN 392 P DEPRQ 393 V VIMGP 394
G GNHEA 395 S ST 396 S SY 397 M LIMAV 398 F YFL 399 C GACR 400 F YFK 401 D
AQTDN 402 G GDEK 403 P QDKPL 404 L NFALI 405 T TA 406 L IVL 407 V LVK 408 S SF
409 Y HYF 410 K VICKDA 411 P P 412 K K 413 P KRP 414 S NSAK 415 R KR K 416 M
ANMV 417 V V 418 F IFLYV 419 L LMVA 420 L LM 421 S ST 422 S ST 423 C MLCI 424 D
HD H 425 E HTED 426 N ADNE 427 A ESAN 428 V AESV 429 I VIL 430 N DSNR 431 E ESTQ
432 S TERSQ 433 N TDNR 434 G GDV 435 K K 436 P P 437 D ESQDL 438 M IMC 439 I IVS
440 L GTLMK 441 F FDYE 442 Y Y 443 N NS 444 Q KSQ 445 T TY 446 K KM 447 G GSA 448
G G 449 V V 450 D D 451 S ENSTRV 452 F IVLFT 453 D D 454 Q KQE 455 M KLVM 456 C
CITOS 457 K ARKSH 458 S ITSVYN 459 M YM Y 460 S TDSN 461 A SVCA 462 N SQNT 463
R R 464 K RNK 465 T TS 466 N RANK 467 R RA 468 W W 469 P PY 470 M MLK 471 A VTAK
472 V VIL 473 F FLG 474 Y YFI 475 G RWNGY 476 M MILV 477 L LVI 478 N DNQ 479 M
ITVM 480 A SA 481 F TGAFLS 482 V VIYR 483 N N 484 S SA 485 Y HKYFC 486 I LIV 487 I
IVL 488 Y YQW 489 C DMSCKRQ 490 H ILEHAT 491 N HNIA 492 K HSNKV 493 I SQIPN
494 N DENSG 495 K WKP 496 O AQG 497 K DGKE 498 K KNVA 499 P TLPV 500 I TPYIQSV
501 N ETRNSYK 502 R RY 503 K GRKT 504 E MALEKY 505 F FQ 506 M LIM 507 K KERQ
508 N QKENIS 509 L L 510 S AGSYP 511 T RMKTIAL 512 D TSADQL 513 L LM 514 T VITF
515 T LATSGY 516 P PSGE 517 W QHKWV 518 M MQIEL 519 Q KAQREH 520 E REKQS
521 R RT 522 L AKLVN 523 K LTRKQEP 524 A NPAEK 525 P ESAPMK 526 T RKTPTN 527 L
LISP 528 K PKS 529 R RVADTF 530 T ESATYNH 531 L LAVI 532 R RA 533 D LKVDRQ 534
N SRINL 535 I LIE 536 T AGKTSE 537 N RSINK 538 V VHKIQ 539 L LF 540 K GRKPI 541 N
PETNKD 542 V DSVETP 543 V MSNVLT 544 P PAQ 545 P FPAG 546 S VSATR 547 P PMSH
548 A DAEGV 549 N PKND 550 N QIVNSM 551 S EPDSTR 552 E VNTE 553 E DE 554 P EP
555 G TGEV 556 P FVPM 557 K KG 558 K TVKPR 559 R RKQY 560 S RKSTV 561 Y RYG
562 C C 563 G HYQGTRK 564 F TIVFYDE 565 C C 566 S PSR PR 567 Y LVSYKN 568 K KR
569 K LKDI 570 R QDR 571 R RS 572 M KMD KD 573 T STA 574 K TKSNR 575 T HYTR
576 Q TISQY 577 F CF C 578 Y YIVCKPN 579 K TSKA 580 C C 581 K KTPA 582 K KSNR 583
A HFAVNP 584 I VIL 585 C C 586 G LRGFM 587 E QEK 588 H CHP 589 N ATNC 590 I KNVIF
591 D QFTDE 592 V VFMIL 593 C CY 594 Q AEPQH 595 D DNST 596 C CQ 597 V VRGIFLA
598 G REGHD

(212) TABLE-US-00004 TABLE 4 A B C D E F G H Source Tposon left end Tposon right end
Tposon SEQ ID NO Tpose SEQ ID leu leu ura leu ura trp Heliothis 1 3 66 39 427 215 130
Spodoptera litura 91 92 71 46 >250 0 0 Pieris rapae 93 94 72 47 >250 0 0 Myzus persicae 95 96 73

48 >250 0 0 Onthophagus taurus 97 98 74 49 >250 0 0 Temnothorax curvispinosus 99 100 75 50
 >250 0 0 Agrilus planipenn 101 102 76 51 >250 0 0 Parasteatoda tepidariorum 103 104 77 52 >250
 0 0 Pectinophora gossypiella 105 106 78 53 >250 0 0 Ctenopusia agnata 107 108 79 54 >250 0 0
 Macrostomum lignano 109 110 80 55 >250 0 0 Orussus abietinus 111 112 81 56 >250 0 0 Eufriesea
 mexicana 113 114 82 57 323 0 0 Spodoptera litura 115 116 83 58 400 0 0 Vanessa tameamea 117
 118 84 59 389 0 0 Blattella germanica 119 120 85 60 248 0 0 Onthophagus taurus 121 122 86 61
 >250 0 0 Onthophagus taurus 123 124 87 62 >250 0 0 Onthophagus taurus 125 126 88 63 >250 0 0
 Megachile rotundata 127 128 89 64 >250 0 0 Xiphophorus maculatus 129 130 90 65 >250 0 0
 (213) TABLE-US-00005 TABLE 5 A B C Transposon Tposon Transposase D E F G H I J Source
 SEQ ID NO source Tphase SEQ ID Tphase DNA seq ID leu leu ura leu ura trp rel ex rel trans 1
 Agrotis 67 Agrotis 40 43 273 471 120 1.73 0.44 2 Agrotis 67 Helicoverpa 41 44 302 563 144 1.86
 0.48 3 Helicoverpa 68 Helicoverpa 41 44 412 547 211 1.33 0.51 4 Helicoverpa 68 Agrotis 40 43
 330 551 245 1.67 0.74
 (214) TABLE-US-00006 TABLE 6 A B C D E 1 Transposase none none 39 39 SEQ ID NO 2
 Day viability viable cells viability viable cells 3 1 94.12 1.03 93.10 0.87 4 3 92.15 0.55 91.01
 0.27 5 5 80.66 0.22 73.53 0.20 6 7 57.58 0.05 54.03 0.06 7 10 27.18 0.03 34.38 0.02 8 12
 27.05 0.04 32.48 0.04 9 14 31.88 0.04 21.48 0.03 10 17 41.46 0.04 50.98 0.06 11 19 no live cells
 no live cells 73.71 0.23 12 21 no live cells no live cells 94.96 1.03 13 24 no live cells no live cells
 98.38 4.14 14 26 no live cells no live cells 99.02 >4 15 27 no live cells no live cells 99.50 >4
 (215) TABLE-US-00007 TABLE 7 A B C D Day 7 Day 10 Day 12 Day 14 1,064 1,524 1,771
 2,053
 (216) TABLE-US-00008 TABLE 8 A B C 1 Days viability viable post-transfection cells 2 1
 94.82 1.07 3 2 93.91 0.10 4 5 84.29 0.11 5 7 48.39 0.03 6 9 37.89 0.04 7 14 33.00 0.03
 8 16 34.26 0.04 9 19 54.69 0.11 10 21 49.44 0.09 11 23 53.14 0.13 12 26 75.00 0.10 13 28 90.56
 0.70 14 30 95.69 2.16
 (217) TABLE-US-00009 TABLE 9 A B C D E F G H I J 1 Tphase no no yes yes no no yes yes no no
 2 Left 13 13 13 13 19 19 19 19 13 13 3 Right 14 14 14 14 14 14 14 14 20 20 4 -na- V VCD V VCD
 V VCD V VCD V VCD 5 -na- 96.3 1.24 95.4 1.32 95.2 0.85 93.2 1.04 96.5 0.29 6 -na- 94.9 0.54
 92.0 0.17 97.5 0.63 93.1 0.29 96.0 0.26 7 -na- 93.1 0.30 79.3 0.19 92.3 0.65 80.8 0.28 89.0 0.21 8 -
 na- 80.8 0.53 63.9 0.15 83.6 0.34 63.2 0.12 68.2 0.09 9 -na- 64.3 0.42 56.6 0.11 63.3 0.40 48.3 0.10
 37.1 0.07 10 -na- 47.4 0.30 71.1 0.27 47.6 0.17 44.0 0.08 28.4 0.03 11 -na- 29.3 0.15 83.1 0.83 32.4
 0.13 53.6 0.19 35.9 0.04 12 -na- 22.6 0.09 97.5 2.56 24.3 0.07 88.5 0.81 24.2 0.02 13 -na- 13.9
 0.07 N/D N/D 7.9 0.02 96.0 3.32 N/D N/D 14 -na- 18.0 0.04 99.0 5.34 N/D N/D N/D N/D N/D
 N/D K L M N O P Q R S T U 1 yes yes no no yes yes no no yes yes -na- 2 13 13 13 13 13 13 13 13
 13 13 -na- 3 20 20 21 21 21 21 22 22 22 22 -na- 4 V VCD V VCD V VCD V VCD V VCD Days
 post- transfection 5 94.9 1.16 95.1 1.14 91.0 0.50 95.4 1.45 94.1 0.58 1 6 94.6 0.32 96.0 0.31 94.4
 0.75 96.8 0.37 91.3 0.77 3 7 82.6 0.44 92.8 0.46 81.0 0.77 95.2 0.39 82.9 0.77 5 8 65.4 0.22 84.5
 0.33 68.7 0.52 84.2 0.23 74.3 0.69 7 9 59.1 0.22 68.1 0.42 64.1 0.67 58.2 0.17 70.1 0.86 10 10 74.6
 0.45 49.4 0.21 75.2 1.28 42.8 0.14 81.2 1.76 12 11 88.6 1.48 31.0 0.14 85.1 2.4 27.4 0.08 89.6 3.3
 14 12 97.8 2.75 21.8 0.05 97.3 6.9 23.5 0.03 98.3 9.9 17 13 N/D N/D N/D N/D N/D N/D N/D N/D
 N/D N/D 19 14 N/D N/D N/D N/D N/D N/D N/D N/D N/D 21
 (218) TABLE-US-00010 TABLE 10 A B C D E F seq id no live int ex int freq ex freq 39 363 56
 92 0.0006 0.0010 158 123 76 92 0.0025 0.0030 159 201 63 81 0.0013 0.0016 369 103 19 31
 0.0007 0.0012 370 145 40 41 0.0011 0.0011 371 107 17 29 0.0006 0.0011 372 180 45 80 0.0010
 0.0018 373 140 25 31 0.0007 0.0009 374 160 26 16 0.0007 0.0004 375 165 46 42 0.0011 0.0010
 376 188 57 62 0.0012 0.0013 401 275 31 57 0.0005 0.0008 402 294 23 18 0.0003 0.0002 403 149
 5 7 0.0001 0.0002 404 198 7 11 0.0001 0.0002 405 145 7 16 0.0002 0.0004 406 137 10 14 0.0003
 0.0004 407 121 8 7 0.0003 0.0002 408 131 17 30 0.0005 0.0009 409 154 5 12 0.0001 0.0003 410
 195 21 25 0.0004 0.0005 411 193 20 52 0.0004 0.0011 412 150 12 14 0.0003 0.0004 413 193 7 19
 0.0001 0.0004 414 172 13 12 0.0003 0.0003 415 135 8 7 0.0002 0.0002 416 234 12 19 0.0002

0.0003 417 151 23 34 0.0006 0.0009 418 131 20 34 0.0006 0.0010 419 161 19 29 0.0005 0.0007
420 249 9 17 0.0001 0.0003 421 211 8 13 0.0002 0.0002 422 240 35 42 0.0006 0.0007 423 240 33
49 0.0006 0.0008 424 196 7 7 0.0001 0.0001 425 172 11 12 0.0003 0.0003 426 292 12 31 0.0002
0.0004 427 185 22 27 0.0005 0.0006 428 145 11 23 0.0003 0.0006 429 310 38 54 0.0005 0.0007
430 163 12 20 0.0003 0.0005 431 94 7 12 0.0003 0.0005 432 212 19 31 0.0004 0.0006 433 213 15
14 0.0003 0.0003 434 189 8 4 0.0002 0.0001 435 165 13 16 0.0003 0.0004 436 311 22 50 0.0003
0.0006 437 118 8 7 0.0003 0.0002 438 124 13 15 0.0004 0.0005 439 169 8 5 0.0002 0.0001 440
101 8 16 0.0003 0.0006 441 111 6 3 0.0002 0.0001 442 163 5 5 0.0001 0.0001 443 160 7 7 0.0002
0.0002 444 87 3 10 0.0001 0.0005 445 63 4 7 0.0003 0.0004 466 280 2 7 0.0000 0.0001 467 129 2
4 0.0001 0.0001 468 132 2 10 0.0001 0.0003 469 178 4 12 0.0001 0.0003 470 206 2 5 0.0000
0.0001 471 245 1 2 0.0000 0.0000 472 165 2 1 0.0000 0.0000 473 266 6 9 0.0001 0.0001 474 195
5 7 0.0001 0.0001 475 239 2 6 0.0000 0.0001 476 182 3 10 0.0001 0.0002 477 117 2 10 0.0001
0.0003 478 170 3 11 0.0001 0.0003 479 138 2 9 0.0001 0.0003 480 148 1 2 0.0000 0.0001 481 152
2 3 0.0001 0.0001 482 211 3 8 0.0001 0.0002 483 202 2 4 0.0000 0.0001 484 102 1 3 0.0000
0.0001 485 95 2 5 0.0001 0.0002 486 95 2 1 0.0001 0.0000 487 94 2 5 0.0001 0.0002 488 68 1 7
0.0001 0.0004 489 111 3 3 0.0001 0.0001 491 209 0 0 0.0000 0.0000 492 252 0 2 0.0000 0.0000
493 217 0 1 0.0000 0.0000 494 183 0 2 0.0000 0.0000 495 127 0 0 0.0000 0.0000 496 210 1 0
0.0000 0.0000 497 101 0 0 0.0000 0.0000 498 293 0 0 0.0000 0.0000 499 216 0 6 0.0000 0.0001
500 300 0 1 0.0000 0.0000 501 55 0 2 0.0000 0.0001 502 127 0 0 0.0000 0.0000 503 79 0 0 0.0000
0.0000 504 333 0 0 0.0000 0.0000 505 147 0 0 0.0000 0.0000 506 208 0 0 0.0000 0.0000

(219) TABLE-US-00011 TABLE 11 E F G A B C D Int Weight Ex Ex Position From To Int Weight
Std Weight Weight Std 300 M R 0.620 0.085 0.746 0.064 43 L S 0.374 0.045 0.207 0.047 162 T I
0.306 0.066 0.018 0.061 239 I L 0.278 0.078 0.281 0.040 507 L I 0.268 0.055 0.193 0.063 421 Y F
0.234 0.063 0.278 0.064 603 Q K 0.222 0.051 0.471 0.070 305 M N 0.213 0.048 0.459 0.079 81 V
E 0.200 0.093 0.068 0.063 241 H N 0.192 0.052 0.036 0.048 450 V I 0.191 0.077 0.170 0.052 136
S V 0.177 0.043 0.293 0.049 155 D T 0.166 0.067 0.069 0.048 153 L F 0.157 0.060 -0.280 0.068
264 S R 0.149 0.046 0.256 0.067 152 K L 0.143 0.060 -0.061 0.065 312 L I 0.133 0.041 0.136
0.043 268 T D 0.122 0.080 0.116 0.066 280 V P 0.111 0.065 0.185 0.046 299 L F 0.106 0.060
0.016 0.064 430 H Y 0.081 0.072 -0.070 0.059 81 V P 0.069 0.057 0.027 0.042 238 V F 0.063
0.066 0.091 0.050 512 H N 0.062 0.073 0.136 0.066 495 Y F 0.060 0.068 0.208 0.090 297 T C
0.055 0.059 0.213 0.070 396 K R 0.050 0.071 -0.067 0.075 149 I C 0.023 0.078 0.055 0.069 268 T
E 0.014 0.049 0.069 0.049 41 S V 0.014 0.057 0.112 0.055 589 S T 0.003 0.069 0.067 0.040 600 V
I -0.021 0.065 -0.106 0.046 150 W F -0.025 0.051 0.080 0.054 604 C H -0.032 0.058 -0.207
0.085 428 L V -0.045 0.058 -0.119 0.058 233 K R -0.045 0.062 -0.122 0.054 459 I M -0.052
0.053 -0.126 0.062 188 L T -0.080 0.078 -0.320 0.076 419 Y F -0.084 0.041 -0.653 0.072 552 L
I -0.084 0.055 -0.079 0.063 603 Q E -0.087 0.094 -0.162 0.082 586 Q D -0.097 0.052 -0.128
0.049 137 N K -0.125 0.058 -0.022 0.066 137 N T -0.137 0.044 -0.022 0.050 136 S P -0.149
0.054 -0.143 0.063 501 S A -0.155 0.091 -0.123 0.075 378 I Y -0.219 0.106 -0.268 0.083 314 L
F -0.221 0.068 -0.009 0.067 74 Q S -0.225 0.080 0.074 0.043 499 D N -0.258 0.059 -0.656
0.090 303 M I -0.278 0.082 -0.251 0.065 241 H R -0.293 0.058 -0.311 0.100 485 R K -0.312
0.061 -0.343 0.055 586 Q R -0.312 0.071 -0.032 0.053 135 V G -0.336 0.079 -0.493 0.055 485
R N -0.359 0.121 -0.675 0.214 334 R K -0.396 0.073 -0.233 0.045 314 L I -0.424 0.070 -0.306
0.076 233 K L -0.426 0.087 -0.399 0.047 385 K L -0.583 0.049 -0.554 0.062

(220) TABLE-US-00012 TABLE 12 A B C D E F SEQ ID NO. live int ex int freq ex freq 39 604
360 308 0.0024 0.0020 369 612 260 424 0.0017 0.0028 158 512 668 1000 0.0052 0.0078 372 668
444 604 0.0027 0.0036 159 736 1004 840 0.0055 0.0046 375 568 428 512 0.0030 0.0036 376 616
264 416 0.0017 0.0027 377 888 844 1056 0.0038 0.0048 160 616 1892 1920 0.0123 0.0125 161
384 744 1048 0.0078 0.0109 162 404 832 828 0.0082 0.0082 163 576 1432 1292 0.0099 0.0090
164 648 880 936 0.0054 0.0058 165 576 848 1360 0.0059 0.0094 166 568 900 1200 0.0063 0.0085
378 1460 1100 1468 0.0030 0.0040 167 592 1400 1756 0.0095 0.0119 168 184 364 488 0.0079

0.0106 379 332 352 0.0039 0.0041 169 436 1056 968 0.0097 0.0089 170 560 1072 1188
0.0077 0.0085 171 220 1256 1256 0.0228 0.0228 172 328 688 1836 0.0084 0.0224 173 648 1016
1216 0.0063 0.0075 174 452 2144 2584 0.0190 0.0229 175 460 664 1280 0.0058 0.0111 176 320
588 656 0.0074 0.0082 177 380 808 1456 0.0085 0.0153 178 476 940 1160 0.0079 0.0097 380 648
528 1120 0.0033 0.0069 179 360 772 1232 0.0086 0.0137 180 664 948 1300 0.0057 0.0078 181
236 820 1084 0.0139 0.0184 182 396 916 960 0.0093 0.0097 381 440 320 496 0.0029 0.0045 382
376 392 560 0.0042 0.0060 183 384 572 1260 0.0060 0.0131 446 528 308 592 0.0023 0.0045 184
444 648 1076 0.0058 0.0097 185 316 860 1012 0.0109 0.0128 186 356 992 1184 0.0111 0.0133
187 312 500 740 0.0064 0.0095 188 424 636 1016 0.0060 0.0096 189 252 632 1080 0.0100 0.0171
190 296 748 1104 0.0101 0.0149 191 320 416 964 0.0052 0.0121 192 356 1444 1504 0.0162
0.0169 193 904 1500 2244 0.0066| 0.0099 194 332 1068 2152 0.0129 0.0259 195 360 632 924
0.0070 0.0103 383 488 420 312 0.0034 0.0026 196 528 992 676 0.0075 0.0051 197 440 692 656
0.0063 0.0060 198 612 960 1688 0.0063 0.0110 199 600 1680 2044 0.0112 0.0136 447 440 213
288 0.0019 0.0026 200 580 1064 864 0.0073 0.0060 201 544 780 696 0.0057 0.0051 202 216 295
380 0.0055 0.0070 203 312 473 508 0.0061 0.0065 204 308 477 416 0.0062 0.0054 448 484 283
348 0.0023 0.0029 205 396 702 696 0.0071 0.0070 206 572 1704 1104 0.0119 0.0077 207 516
1780 1140 0.0138 0.0088 208 508 868 968 0.0068 0.0076 209 300 712 720 0.0095 0.0096 210 492
800 996 0.0065 0.0081 211 328 676 804 0.0082 0.0098 384 564 360 564 0.0026 0.0040 212 424
720 796 0.0068 0.0075 213 544 664 656 0.0049 0.0048 214 540 1056 940 0.0078 0.0070 215 520
960 876 0.0074 0.0067 385 464 360 324 0.0031 0.0028 386 628 724 1196 0.0046 0.0076 216 472
728 572 0.0062 0.0048 387 576 600 700 0.0042 0.0049 217 428 1176 612 0.0110 0.0057 218 552
1080 744 0.0078 0.0054 388 492 496 696 0.0040 0.0057 219 276 532 640 0.0077 0.0093 389 508
600 500 0.0047 0.0039 220 404 736 612 0.0073 0.0061 390 576 384 512 0.0027 0.0036 391 388
432 796 0.0045 0.0082 221 480 636 820 0.0053 0.0068 222 388 560 560 0.0058 0.0058 223 452
1244 1180 0.0110 0.0104 224 384 848 880 0.0088 0.0092 225 236 340 516 0.0058 0.0087 226 548
1040 1060 0.0076 0.0077 227 260 816 852 0.0126 0.0131 228 492 672 848 0.0055 0.0069 229 288
1032 884 0.0143 0.0123 230 580 820 760 0.0057 0.0052 392 584 568 996 0.0039 0.0068 393 780
884 1380 0.0045 0.0071 231 340 628 796 0.0074 0.0094 449 788 380 600 0.0019 0.0030 394 532
400 724 0.0030 0.0054 232 512 756 900 0.0059 0.0070

(221) TABLE-US-00013 TABLE 13 A B C D E F SEQ ID NO. live int ex int freq ex freq 465 224
11 15 0.002 0.003 298 215 108 129 0.020 0.024 297 161 86 94 0.021 0.023 296 228 119 169 0.021
0.030 295 264 150 177 0.023 0.027 294 194 67 105 0.014 0.022 400 179 26 53 0.006 0.012 293
199 141 152 0.028 0.031 399 200 21 21 0.004 0.004 292 265 122 138 0.018 0.021 291 209 49 84
0.009 0.016 290 205 109 111 0.021 0.022 289 234 69 56 0.012 0.010 288 200 100 134 0.020 0.027
287 292 124 223 0.017 0.031 464 248 19 11 0.003 0.002 463 239 8 15 0.001 0.003 286 210 154
133 0.029 0.025 137 229 243 363 0.042 0.063 285 236 103 117 0.017 0.020 284 293 145 150
0.020 0.020 283 253 157 156 0.025 0.025 136 171 164 191 0.038 0.045 282 227 108 90 0.019
0.016 281 233 92 96 0.016 0.016 280 314 167 225 0.021 0.029 279 337 176 189 0.021 0.022 278
258 124 129 0.019 0.020 277 275 122 134 0.018 0.019 276 311 150 149 0.019 0.019 275 239 119
176 0.020 0.029 274 249 79 103 0.013 0.017 462 160 12 29 0.003 0.007 273 294 129 138 0.018
0.019 272 224 140 138 0.025 0.025 271 254 75 102 0.012 0.016 135 195 164 233 0.034 0.048 461
270 19 34 0.003 0.005 270 199 155 184 0.031 0.037 269 286 147 105 0.021 0.015 268 253 118
139 0.019 0.022 267 264 101 78 0.015 0.012 398 254 30 30 0.005 0.005 266 245 143 269 0.023
0.044 265 326 212 374 0.026 0.046 397 239 32 40 0.005 0.007 264 251 61 78 0.010 0.012 263 297
113 160 0.015 0.022 134 159 163 162 0.041 0.041 262 256 126 121 0.020 0.019 261 295 164 215
0.022 0.029 260 274 136 183 0.020 0.027 259 335 195 247 0.023 0.029 396 390 35 46 0.004 0.005
460 297 21 27 0.003 0.004 459 244 8 11 0.001 0.002 458 249 5 7 0.001 0.001 457 232 10 16 0.002
0.003 258 264 167 180 0.025 0.027 257 209 142 180 0.027 0.034 256 295 191 283 0.026 0.038
255 265 79 92 0.012 0.014 254 219 155 198 0.028 0.036 253 256 137 176 0.021 0.028 456 207 6 8
0.001 0.002 252 308 123 144 0.016 0.019 455 320 23 25 0.003 0.003 251 260 130 109 0.020 0.017

490 357 4 8 0.000 0.001 250 234 149 142 0.025 0.024 454 240 14 19 0.002 0.003 249 180 105 116
0.023 0.026 248 244 120 77 0.020 0.013 453 257 7 4 0.001 0.001 452 326 9 12 0.001 0.001 133
235 263 526 0.045 0.090 247 314 72 129 0.009 0.016 246 247 143 193 0.023 0.031 245 243 181
201 0.030 0.033 244 276 128 143 0.019 0.021 243 224 148 120 0.026 0.021 242 184 79 84 0.017
0.018 241 246 92 140 0.015 0.023 132 135 110 169 0.033 0.050 131 209 176 227 0.034 0.043 240
284 208 263 0.029 0.037 451 269 14 15 0.002 0.002 239 278 140 209 0.020 0.030 238 283 120
139 0.017 0.020 450 246 8 7 0.001 0.001 237 239 93 114 0.016 0.019 236 302 137 175 0.018
0.023 395 256 21 40 0.003 0.006 235 340 176 210 0.021 0.025 234 214 132 150 0.025 0.028 233
285 172 232 0.024 0.033 174 380 134 209 0.014 0.022 158 351 61 89 0.007 0.010 39 381 31 51
0.003 0.005

(222) TABLE-US-00014 TABLE 14 int ex A B C D E F G H Position From To Weight Weight Std
Weight Weight Std Number Of Occurences 41 S V 0.21 0.03 0.17 0.03 19 43 L S 0.00 0.00 0.00
0.00 96 81 V E 0.00 0.00 0.00 0.00 96 83 D S 0.28 0.05 0.22 0.05 5 85 V L 0.10 0.05 0.07 0.04 5
125 P S 0.01 0.05 0.08 0.04 5 126 Q S 0.23 0.06 0.21 0.07 5 131 Q R 0.38 0.05 0.43 0.04 5 131 Q
T 0.37 0.06 0.22 0.05 5 136 S V 0.24 0.02 0.16 0.02 19 140 E C 0.40 0.05 0.23 0.05 5 140 E A 0.24
0.05 0.20 0.04 5 151 N Q 0.00 0.06 -0.11 0.05 5 169 K E 0.43 0.06 0.48 0.05 5 212 N S 0.13 0.04
0.02 0.03 5 239 I L 0.17 0.03 0.12 0.02 19 268 T D 0.00 0.00 0.00 0.00 96 297 T C 0.21 0.03 0.20
0.04 9 300 M R 0.00 0.00 0.00 0.00 96 305 M N 0.00 0.00 0.00 0.00 96 312 L I 0.28 0.03 0.23 0.03
10 316 C A 0.22 0.04 0.23 0.03 5 322 N T 0.31 0.07 0.29 0.06 5 357 H R 0.32 0.06 0.21 0.05 5 360
K Q 0.14 0.05 0.08 0.04 5 397 K S 0.32 0.05 0.32 0.04 5 421 Y F 0.22 0.03 0.16 0.02 19 447 A N
0.18 0.06 0.11 0.06 5 447 A D 0.07 0.05 0.08 0.05 5 449 A S 0.22 0.03 0.07 0.03 5 449 A V 0.10
0.06 -0.08 0.06 5 450 V I 0.26 0.05 0.23 0.04 9 476 K L 0.06 0.03 -0.22 0.04 5 485 R K 0.11 0.05
0.06 0.04 5 492 V A 0.22 0.06 0.17 0.05 5 495 Y F 0.42 0.03 0.31 0.02 19 585 L K 0.46 0.08 0.27
0.06 5 595 T K 0.00 0.08 -0.08 0.05 5 241 H N -0.07 0.04 0.03 0.03 10 351 P G -0.19 0.08 -0.22
0.05 5 500 I M -0.52 0.07 -0.36 0.06 5 379 E P -1.13 0.10 -0.67 0.06 5 316 C M -1.74 0.10
-1.17 0.07 5 485 R N -1.77 0.08 -1.33 0.05 5 357 H A -1.90 0.05 -1.41 0.07 5

(223) TABLE-US-00015 TABLE 15 A B C D E F SEQ ID NO. live int ex int freq ex freq 368 288
117 220 0.016 0.031 157 268 280 452 0.042 0.067 367 160 98 146 0.025 0.037 366 296 127 232
0.017 0.031 365 67 15 24 0.009 0.014 364 248 151 224 0.024 0.036 156 300 212 316 0.028 0.042
363 528 320 464 0.024 0.035 155 236 172 292 0.029 0.049 154 268 220 348 0.033 0.052 362 288
169 292 0.023 0.041 361 268 132 252 0.020 0.038 360 180 49 97 0.011 0.022 359 376 162 296
0.017 0.031 358 420 170 404 0.016 0.038 357 652 324 576 0.020 0.035 153 312 218 36 0.028
0.005 356 404 187 276 0.019 0.027 355 360 119 207 0.013 0.023 354 340 150 316 0.018 0.037
353 220 60 90 0.011 0.016 352 216 106 173 0.020 0.032 351 348 172 356 0.020 0.041 350 400
209 400 0.021 0.040 349 284 107 276 0.015 0.039 152 292 218 376 0.030 0.052 348 211 104 188
0.020 0.036 151 220 266 476 0.048 0.087 347 222 83 133 0.015 0.024 150 313 214 324 0.027
0.041 149 262 223 500 0.034 0.076 346 452 193 328 0.017 0.029 345 300 187 336 0.025 0.045
344 360 142 332 0.016 0.037 148 283 248 296 0.035 0.042 343 195 68 113 0.014 0.023 342 201
76 133 0.015 0.026 341 266 138 248 0.021 0.037 340 338 204 280 0.024 0.033 339 355 241 364
0.027 0.041 338 276 181 280 0.026 0.041 147 305 385 824 0.050 0.108 337 156 83 192 0.021
0.049 336 206 113 124 0.022 0.024 335 224 132 220 0.024 0.039 334 252 93 184 0.015 0.029 333
319 166 320 0.021 0.040 332 710 469 1336 0.026 0.075 331 275 173 404 0.025 0.059 330 300 135
324 0.018 0.043 329 211 127 320 0.024 0.061 328 173 79 168 0.018 0.039 327 169 98 140 0.023
0.033 326 242 73 128 0.012 0.021 146 290 204 360 0.028 0.050 325 330 192 384 0.023 0.047 145
290 415 748 0.057 0.103 324 313 204 348 0.026 0.044 144 211 441 936 0.084 0.177 323 226 115
212 0.020 0.038 322 212 104 164 0.020 0.031 143 243 188 548 0.031 0.090 321 324 176 324
0.022 0.040 320 475 201 264 0.017 0.022 319 339 170 352 0.020 0.042 318 290 163 276 0.022
0.038 317 299 176 316 0.024 0.042 316 352 178 348 0.020 0.040 315 176 64 180 0.015 0.041 314
300 101 260 0.013 0.035 313 274 158 316 0.023 0.046 312 264 136 320 0.021 0.048 311 270 142
260 0.021 0.039 310 346 185 408 0.021 0.047 309 284 163 180 0.023 0.025 142 326 373 868

0.046 0.107 141 220 150 320 0.027 0.058 308 360 151 300 0.017 0.033 307 304 171 308 0.023
0.041 306 461 184 388 0.016 0.034 140 328 288 664 0.035 0.081 305 447 181 384 0.016 0.034
304 268 140 266 0.021 0.040 303 362 129 267 0.014 0.030 302 262 161 290 0.025 0.044 301 343
101 183 0.012 0.021 300 24 142 247 0.023 0.040 139 452 344 564 0.030 0.050 299 326 105 190
0.013 0.023 138 301 450 896 0.060 0.119 134 568 284 432 0.020 0.030 174 236 56 133 0.009
0.023 158 272 37 109 0.005 0.016 39 55 0 0 0.000 0.000

7. References

(224) All references cited herein are incorporated herein by reference in their entirety and for all purposes to the same extent as if each individual publication or patent or patent application was specifically and individually indicated to be incorporated by reference in its entirety for all purposes. To the extent the information associated with a citation may change with time, the version in effect at the effective filing date of this application is meant, the effective filing date being the filing date of the application or priority application in which the citation was first mentioned.

(225) Many modifications and variations of this invention can be made without departing from its spirit and scope, as will be apparent to those skilled in the art. The specific embodiments described herein are offered by way of example only, and the invention is to be limited only by the terms of the appended claims, along with the full scope of equivalents to which such claims are entitled. Unless otherwise apparent from the context, any embodiment, aspect, element, feature or step can be used in combination with any other.

Claims

1. A polynucleotide comprising an open reading frame encoding a transposase, the amino acid sequence of which is at least 90% identical to SEQ ID NO: 39, operably linked to a heterologous promoter, wherein the amino acid sequence comprises at least 2 mutations selected from S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, 1149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, L507I, H512N, L585K, S589T, T595K and Q603K, relative to the sequence of SEQ ID NO: 39.
2. A polynucleotide comprising an open reading frame encoding a transposase operably linked to a heterologous promoter, wherein the amino acid sequence of the transposase is selected from SEQ ID NO: 131-490.
3. The polynucleotide of claim 1, wherein the transposase can excise or transpose the transposon from SEQ ID NO: 66.
4. The polynucleotide of claim 3, wherein the excision activity or transposition activity of the transposase is at least 2-fold higher than the activity of the wild type transposase set forth by SEQ ID NO: 39.
5. The polynucleotide of claim 1, wherein the heterologous promoter is active in an in vitro transcription reaction.
6. The polynucleotide of claim 1, wherein the heterologous promoter is active in a eukaryotic cell.
7. The polynucleotide of claim 6, wherein the eukaryotic cell is a mammalian cell, optionally, wherein codons of the open reading frame are selected for expression in the mammalian cell.
8. The polynucleotide of claim 1, wherein the open reading frame further encodes a nuclear localization sequence fused to the transposase.
9. The polynucleotide of claim 1, wherein the open reading frame further encodes a heterologous DNA binding domain fused to the transposase.
10. The polynucleotide of claim 9, wherein the heterologous DNA binding domain is derived from a Crispr Cas system, or a zinc finger protein, or a TALE nuclease.

11. An isolated mRNA encoding a transposase, the amino acid sequence of which is at least 90% identical with SEQ ID NO: 39, and wherein the mRNA sequence comprises at least 10 synonymous codon differences relative to SEQ ID NO: 1181, optionally selected for mammalian expression, wherein the amino acid sequence comprises at least 2 mutations selected from S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, 1149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, L507I, H512N, L585K, S589T, T595K and Q603K, relative to the sequence of SEQ ID NO: 39.
 12. A non-naturally occurring polynucleotide encoding a transposase comprising an amino acid sequence at least 90% identical to SEQ ID NO: 39 wherein the amino acid sequence comprises at least 2 mutations selected from S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, 1149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, L507I, H512N, L585K, S589T, T595K and Q603K, relative to the sequence of SEQ ID NO: 39.
 13. A non-naturally occurring transposase encoded by the polynucleotide of claim 1.
 14. The polynucleotide of claim 4, wherein the excision activity or transposition activity of the transposase is 2 to 10 fold higher than the activity of the transposase of SEQ ID NO:39.
 15. The polynucleotide of claim 1, wherein the amino acid sequence comprises at least 4 mutations selected from S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, 1149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, L507I, H512N, L585K, S589T, T595K and Q603K, relative to the sequence of SEQ ID NO: 39.
 16. The isolated mRNA of claim 11, wherein the amino acid sequence comprises at least 4 mutations selected from S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, 1149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, L507I, H512N, L585K, S589T, T595K and Q603K, relative to the sequence of SEQ ID NO: 39.
 17. The non-naturally occurring polynucleotide of claim 12, wherein the amino acid sequence comprises at least 4 mutations selected from S41V, L43S, V81E, V81P, D83S, V85L, P125S, Q126S, Q131R, Q131T, S136V, E140C, E140A, 1149C, N151Q, K152L, L153F, D155T, T162I, K169E, N212S, V238F, I239L, H241N, S264R, T268D, T268E, V280P, T297C, L299F, M300R, M305N, L312I, C316A, N322T, H357R, K360Q, K396R, K397S, Y421F, H430Y, A447N, A447D, A449S, A449V, V450I, K476L, R485K, V492A, Y495F, L507I, H512N, L585K, S589T, T595K and Q603K, relative to the sequence of SEQ ID NO: 39.
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